

THE PRIMORDIAL EVENT AND THE ETERNAL PRESENT

BY

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Abstract

The appeal to and use of events in philosophical theorising, implicitly or explicitly, is utterly ubiquitous. However, relatively little attention has been given to events themselves, compared to other entities, like objects. This is unfortunate because events are as problematic, if not more so. This thesis has two aims. The first is a ‘negative’ aim. I argue that the two most natural construal of events, that they are entities that occur at some time or over some interval of time, are untenable posits in our ontology and that we should deny that they exist (except for one). This is because they run into problems to do with either Zeno’s paradoxes, the perception of change, or ineliminable arbitrariness in their individuation. The second is a positive aim. I argue that, in light of the negative conclusions reached, we are forced into certain views not only about events, but time as a whole. Namely, I argue that a view I call *event monism* is true. This is the view that there is single event constitutive of the entire duration of the history of the universe. I defend this view from objections and show that it has far-reaching consequences both for how we conceive of the entities that exist and our role as inquirers into the structure of this event.

Introduction

Philosophers and lay-people alike are always speaking of events. Going to a concert, taking out the rubbish, cooking a carbonara, going on a date: these are all events. Philosophers have technical use for and philosophical aspirations to use or describe such events.

For example, in the philosophy of action, one wants to be able to draw a line between an ordinary occurrence when it is an action taken by an intentional agent. Further, it seems that when we plan to achieve something, we do such planning in terms of events. For example, we might have some goal B and we might know that to bring it about, we believe we must first achieve A. Here A and B are events. Many philosophers claim that events (whether type or token) are the causal relata i.e., ‘Event A caused event B.’ Philosophers are presupposing events when they write things like ‘A has mental state *M* at time *t*’ or, ‘the world is in state *S* at time *t*.’ Even ethics is concerned with events. For example, consequentialists are concerned with the consequences of our actions. Both the action itself, and its consequences, are events. But what actually are events? The key thing ordinarily taken to differentiate an event from other kinds of entity, like objects, is that their existence is always indexed to some time or interval of time. My intention is to investigate precisely what events are or could be, especially in light of this feature.

My investigation into events will not be an investigation into the intricacies of how we use events in everyday ordinary language, nor will it investigate the logic or semantics of such talk. Nor am I interested in whether there are abstract facts about the world that describe it in ways that sound like they are events. If it turns out that this is all events are, then so be it. However, I don’t see how they could be this if there is no actual entity, namely, an event, to make such facts true. Rather, I am interested in whether events exist, in the sense that we think a concert is an event or in the sense that history is made up of events, and if they do exist, what they must be. Since our talk implicates them so heavily, there seems to be a *prima facie* assumption towards their existence, which I grant. However, it is precisely this assumption which I wish to overturn. Making the most minimal assumptions I can about such events, I seek to show that no such entities exist, at all.

The plan is as follows. Part I is the ‘negative’ aspect of the project, where I attempt to show that no theory of events is satisfactory and that we should reject all except one event. I do this by going through a number of different alternatives and rejecting all of them. In Chapter I, I begin by introducing something I call the *world-line*, which is an explanatory device I use to provide a taxonomy of events and frame the overarching discussion and arguments. Next, I provide a brief overview of what events are supposed to be, including a brief discussion of the intuitive notion and of what other philosophers have thought of them. Based on my taxonomy and the suggestions of other philosophers, I outline the possibilities for an ontology of events and the composition of the world-line. Following this, the work really begins. I go through each of these possibilities and reject them in turn. I begin with states-only views (Chapter II),

followed by states-and-events-views (Chapter III), and finally, with events-only views (Chapter IV and V). (I explain what I mean by these terms in chapter I.) Finally, in Chapter VI, I briefly summarise the ground covered, and lessons learned. It should be noted that this part is not entirely negative. Its arguments and conclusions will pose substantive constraints on our metaphysical theorising, most importantly, our conception of time. More specifically, they demand the reality of a primitive duration underlying the entire history of the universe, that is not composed of either states or events.

Part II is the ‘positive’ aspect of the project where I outline my own view based on the constraints developed in the previous part. My own view is the conjunction of two theses: *event monism* and *temporal monism*. Event monism is the view that reality is not divisible into states or events at all, but rather that it is a single whole, with a real duration. I call this reality *the primordial event*, as it is constitutive of the entire history of the universe. Temporal monism is the view that is entailed by event monism and reductionism about time. It says that there can be no times at which things can occur, even though there is duration, because there are no events that times can be indexed to.

Chapter VII is where I introduce these views, motivate them by reference to the conclusions of Part I, flesh out the details, and defend them from objections. I show that my conclusions are not only immune to my previous attacks, but theoretically preferable to the other views. In Chapter VIII, I relate my own view to the ongoing debates about time. I first situate it in relation to the debate between presentism and eternalism and show how it relates to the occurrence/existence distinction that is essential to them. Next, I situate it in relation to the debate between A- and B-theories of time, showing that the very idea of categorising theories of time in this way illegitimately presupposes the existence of events. Finally, in Chapter IX, I close by discussing how we are supposed to make sense of inquiry if my view is accepted.

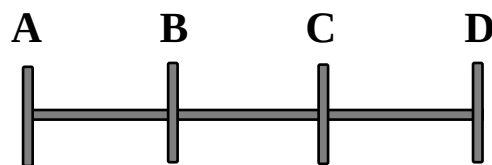
Part I:
Against Events

Chapter I: The Event Space

Introducing the World-Line

In this section and the next, I outline some conceptions of events as worldly entities. To begin with, what all accounts of events include is some reference to time; a single object would not be an event simply by its own existence. It only becomes an event once this object is in some static or evolving state, at some instant or interval of time (or, as related to other events in an order of occurrence). For example, a rock by itself is not an event, but if it is thrown, hurling through the air, this is an event. Even if a rock sits perfectly still over the course of a day, this still seems like an event, albeit a boring one.

Within this basic account, there seem to me to be two types of events. I represent these possibilities as follows. Take a line AD that represents the ordered history of the universe in time as a series of events. It is not important for now whether there is a ‘beginning’ or ‘end’ to this series, nor is it important for now whether there is a proper ‘direction’ to this sequence. It could be that we experience this event order one way but that is an illusion, and it could be that there is no beginning nor end. Nor does it matter right now whether all of them always exist, or just the present ones. Nor that many events will be co-occurring at any given time. Now, suppose we have an ordered series of points on this line, A, B, C, and D, that represent different times where some events will begin, end, or occur. There will of course be many other events populating the line, and overlapping with these events, but it will be easiest to talk about these few. This would look as follows:



This is the kind of thing I take philosophers to have in mind when they discuss events. For those who index events to time, the variables will be times. For those who index events to relations of order, the variables will be related to each other in “before” and “after” relations. Going forward, I call this the *world-line*. There are two kinds of events represented by this scheme: those that occur over an interval that are temporally extended, and those that occur at an instant that are not temporally extended.

Options for an Event Ontology

The first type of event is those that occur during the intervals between points, AB, BC, and CD. These are what we would ordinarily think of as events, they are occurrences that take place over some temporal extent. For example, cooking a carbonara or going to a concert. I call these,

simply, *events*. Events of this type are essentially temporally extended; they occur over some duration. In ordinary cases, a complete description of one these events would be impossible. We only have so much time to describe such things, so we normally just represent those aspects of the event that are salient to us. It would be unnecessary to speak of the chemical composition of the pots, and the direction of the wind entering the kitchen window, when explaining that we have made a carbonara, though presumably those things factor into it. Philosophers like Davidson think that we have good reason to think that such events constitute a fundamental ontological category.¹ He thinks this primarily because much of our talk seems to be irreducible to talk of objects alone, without events, thus implicating their existence. Furthermore, he thinks that the problems they face with regards to their individuation seem just as surmountable as those that objects face. Thus, “there is as good reason to believe events exist.”²

Another way of capturing events with temporal extension is to identify objects and events. This collapsing of events into objects is only a small step from four-dimensionalism. Four-dimensionalism is the view that objects do not just exist in space, but also in time. They have not only spatial parts, but temporal parts. This means all objects just are events in the sense that they occupy some definite duration. For example, on this view, I am not a spatial unity *now*, wholly present in each moment, I am extended in time, across my entire life. Each phase of my life is a temporal part. This simplifies the problem of events, because events, such as AB or cooking a carbonara, are just objects with spatial and temporal parts. This has the apparently strange consequence of making things we ordinarily take to be merely spatial objects, events. For example, it is not just the cooking itself that gains temporal parts, but things like the spaghetti and the individual pots too. Depending on the version of four-dimensionalism, the identification of objects and events does not have to be complete. It could be that all objects are events but not all events are objects. However, for our purposes only the former fact that all objects are events for the four-dimensionalist will be relevant to any future discussion of the view.

However, these objects are only events in the sense that interval AB is if they are not analysed to be the sum of an infinite number of temporally extensionless stages. They must either be whole and indivisible (the concert is its own whole, being a part of nothing), or only divisible into parts of finite length (the concert’s songs would be finitely extended parts of the whole concert). If the umbrella term is four-dimensionalism, this view would be ‘perdurantism,’ the view that temporal parts have duration. If they were analysed into temporally extensionless moments or stages, the view would be a ‘stage view,’ the view that temporal parts do not have duration, which would make them events in the next sense.

The second type of event occurs at the points themselves, A, B, C, and D. These events do not take place over any temporal extent. Rather, they occur at an instant, and can be thought of as what you were to get if you ‘paused’ time in some moment. I call these kinds of events, *states*. If they exist, and in spite of the fact that they are temporally unextended, they must still figure in the causal nexus of the universe. Further, at any given time, many such events will be

¹ Davidson, Donald. “The Individuation of Events.” In *Essays in Honor of Carl G. Hempel*, 216-234. Dordrecht: Springer, 1969. 307.

² Davidson, “The Individuation of Events”, 307.

co-occurring. For example, this is the language sometimes used when philosophers formulate determinism in the free will debate. Here is an influential example from Peter Van Inwagen:

Our definition of ‘determinism’ will also involve the notion of ‘the state of the entire physical world’ (hereinafter, ‘the state of the world’) at an instant...Our concept of ‘state’ must be such that, given that the world is in a certain state at a certain time, nothing follows logically about its states at other times...[and] If there is some observable change in the way things are (e.g., if a white cloth becomes blue, a warm liquid cold, or if a man raises his hand), this change must entail some change in the state of the world.³

A slightly more sophisticated formulation of determinism is given by Carl Hoefer in the *Stanford Encyclopedia of Philosophy* article on the matter. He defines it as follows:

Determinism: The world is governed by (or is under the sway of) determinism if and only if, given a specified way things are at a time t , the way things go thereafter is fixed as a matter of natural law.⁴

The idea that there can be “a way things are at a time t ”, and implicitly that there can be later ways things can be, is to presuppose that there are states in this sense.

Some, such as Jaegwon Kim and Jonathan Bennett,⁵ go further and suppose that everything that involves an object, property, and a time is also an event of this kind. Thus, Kim writes:

According to this account, each individual event has three unique constituents: a substance (the “constitutive object” of the event), a property it exemplifies (the “constitutive property” or “generic event”), and a time. An event is a complex of these three and I have used the notation $[x, P, t]$, or variants thereof, as a canonical notation for events in general... Event $[x, P, t]$ exists just in case substance x has property P at time t .⁶

Kim calls this the property exemplification theory of events. Thus, on this account, it is enough that if some ball was red at time t , then there was a red ball event.

Both of these views also entail that events are utterly ubiquitous. If there is some change in the way this substance is, there exists a new event. For example, if the red ball is painted green, there is now a green ball event. In van Inwagen’s formulation, there is a “state of the world” event at any time t , no matter how arbitrarily close to other times this state is because a change in time is a change in state (because the world is an ordered series of states). In Kim’s view, an event exists insofar as some substance and a property of it exists at some time. Since

³ Van Inwagen, Peter. “The Incompatibility of Free Will and Determinism.” *Philosophical Studies* 27, no. 3 (1975): 185-186.

⁴ Hoefer, Carl. “Causal Determinism.” *The Stanford Encyclopedia of Philosophy*, Spring 2016. <https://plato.stanford.edu/archives/spr2016/entries/determinism-causal/>.

⁵ Bennett, Jonathan. “What Events Are.” In *The Blackwell Guide to Metaphysics*, edited by Richard Gale, 43-65. Oxford: Blackwell Publishers, 2002.

⁶ Kim, Jaegwon. “Events as Property Exemplifications.” In *Supervenience and Mind*, 33–52. Cambridge: Cambridge University Press, 1993. 35.

there is almost constant change in the world, and since some property or other of some particular substance almost always changes over any arbitrarily small duration, there is ubiquity here too. Thus, this kind of event, if it exists, is incredibly profligate. Especially due to the fact that many states will be co-occurrent in any state of the world, with an extremely large number of other states.

As I noted earlier, I am not interested in the intricacies of how these things figure in our talk; I am interested in their ontological nature, in whether they exist, and if they do, how it is possible that they could constitute the world in time. Philosophers make prodigious use of both types of events in their theorising, as if they were some harmless theoretical posit. However, few stop to question whether we are metaphysically warranted to make such assertions. I take it we are warranted in speaking of events and states if, and only if, they exist. However, if it turns out that such constructions are only fictions erected in being in order to navigate its contours, then we must radically reconfigure our ontology. I focus on three choices for an ontology of events:

- (1) States and Events: The universe is made up of states and events.
- (2) States-Only: The universe is made up of states and no events.
- (3) Events-Only: The universe is made up of events and no states.

The first view seems to be David Lewis' view. He is a four-dimensionalist who thinks that events have a beginning and end in time, but also that they have temporal parts that could be both temporally extended (events) and unextended (states). He calls what I am calling events (inclusive of both kinds), "stages".⁷ While Lewis' stages could be both temporally extended or extensionless, he says of the unextended stages that they are unproblematic but also that "I do not really need them."⁸ It is plausible that most philosophers have a view such as this because most will help themselves to both kinds of events in their theorising.

It is unclear how popular the second view is. However, this view is held by anyone who believes time to have the structure of a dense set of points where duration is derived from duration-less states. For example, Adolf Grünbaum holds such a view, which he thinks is the proper analysis of the special theory of relativity.⁹

The final view, that there are only events, is held, for example, by David Hume. He writes that "every moment must be distinct from, and posterior or antecedent to another...time, as it exists, must be composed of indivisible moments."¹⁰ For Hume these 'moments' are what all intervals are ultimately decomposable into. The view is also held by William James, who writes that: "The quickest way to avoid the contradiction would be...to treat real processes of change no longer as being continuous, but as taking place by finite not infinitesimal steps, like the successive drops by which a cask of water is filled, when whole drops fall into it at once or

⁷ Lewis, David. *On the Plurality of Worlds*. Oxford: Blackwell, 1986. 202.

⁸ Lewis, David. "Survival and Identity." In *Philosophical Papers Volume I*, 55-77. New York: Oxford University Press, 1983. 76.

⁹ Grünbaum, Adolf. "Relativity and the Atomicity of Becoming." *The Review of Metaphysics* 4, no. 2 (1950): 143-186.

¹⁰ Hume, David. "A Treatise of Human Nature, 1739." In *Hume: The Essential Philosophical Works*, edited by Tom Griffith, 1-552. Ware: Wordsworth Editions Limited. 35.

nothing.”¹¹ Both thinkers advocate for the existence of *atomic events*, events which are not decomposable into any further elements.

I see each of them as having the task of explaining, if these events are the stuff of reality, in what way do they make up the universe as we know and experience it? In what way is the world line constituted? Presumably, without any such events at all, there would be nothing at all, as there would not even be a present. Thus, the question is one of *composition*: what composes the world-line, and how does this support our observations of it? I investigate whether each of these proposals is plausible or not. However, it is worth noting that some philosophers – presentists – are not concerned with this problem of composition because they only think a single time exists (at which many states are occurring). One could be either a state-presentist (all that exists is the present state(s)) or an event-presentist (all that exists is the present event(s)). I leave out discussion of these views due to constraints of space and because the resources needed to reject them will be implicit in my rejection of the non-presentist alternatives.¹² Onto those alternatives.

¹¹ James, William. *Some Problems of Philosophy*. London: Longmans, Green, and Co., 1911. 172.

¹² However, see Appendix I for a discussion and rejection of these presentist views.

Chapter II: The States-Only Proposal

The first route one could take for an ontology of events is a states-only view, where events are reduced to states. There are a few different ways this could go. First, (and this is not Kim's view) instead of saying that my cooking a carbonara occurs over some finite duration, we would say that the event of my cooking a carbonara is set of property exemplification events. If we wanted to speak of the universe on this view, we might talk about the collection of all substances at a specific time. Each event has a location in space determined by the location of the substances and their respective properties at this time. For the carbonara, we would expect each event to be located in the kitchen. Alternatively, we could say that the entire world line in time, as formulated in van Inwagen's version of determinism, is constituted by an ordering of states, where time just indexes a particular state making up the line. Finally, we could take a four-dimensionalist view of states and say that the entire concrete universe is made up of an infinite series of states. Here, such states and their characteristics are the basic constituents of the universe.

Since time is dense, and on all of these views a change in time entails a change in state, the world-line will be dense with states. This is because we can take any arbitrarily small interval, which is just any two states on this view, and find an infinite number of states between them. Therefore, between any two points in time there are an infinite number of states between them. It would still rightly be represented as a line because it is dense with points, but it appearing to be a line is partly an illusion, because it is actually an infinity of points. The line is discrete. My target in this chapter will be all of these positions that wish to build reality out of states. While it is not clear that anyone actually thinks that this view is right, this is the way a lot of philosophers talk. Even if no one believes this view, my arguments here are necessary to reject the other views, so it will be worthwhile, nonetheless. There are two major problems for the states-only view that I will outline in turn. First, Zeno's dichotomy shows that motion and change would be impossible. While it seems like this could be escaped by embracing a kind of eternalism about events the next problem shows that this too is untenable on a states-only view. Thus, it cannot account for the perception of change without relegating it to another world. To Zeno.

Zeno's Dichotomy

The first problem with the states-only view is that motion and change become impossible. Why would this be the case? We can see this through a study of one of Zeno's paradoxes of motion, the dichotomy. Of course, most philosophers think Zeno's arguments are unsound and have been many times refuted. However, I do not myself think this is the case, which puts me in the minority. I think at least some of them are sound, but only if they are properly formulated and target the right views. The argument I discuss here is sound only when it targets a view that countenances no real duration (i.e., any states-only view), which is precisely what I will level

it against. Even though you will likely disagree with my discussion here, bear with me at least to the next section, which will not rely on such an argument. Anyway, onto the dichotomy. One might formulate it as follows:

The Dichotomy: Suppose Rowan wants to walk from A to B. If he completed his walk, the path traversed would be made up of a dense set of states, the journey being the set of those states. Between any two points there is an infinite number of other points. In order to even begin the journey by travelling any arbitrarily small distance, Rowan would have to travel through an infinite number of states. It is impossible to pass through an infinite number of states. Therefore, it is impossible for Rowan to travel from A to B. On the states-only view, all motion and change is like this. Therefore, if the states-only view is true, all motion and change is impossible.

But motion and change are possible, Rowan can travel from A to B. Therefore, the states-only view must be false. The key premise I need to defend here is that on the states-only view it is impossible to pass through an infinite number of states.

Two things about this paradox to note before I defend it. First, it applies to all of the states-only views of events because the way each theory must describe a completed motion requires that the journey is composed of an infinite number of states. Second, it is worth reflecting briefly on how temporally extended events, rather than duration-less states, would solve these problems. If we had a theory of events say, like Davidson, we might individuate certain occurrences into single whole events that take place over some finite duration. In this case, we would have the event: “Rowan travelled from A to B”, where indivisible motion, rather than motion made up of an infinite number of states, occurred during the event. Alternatively, if we supposed that there were such a thing as atomic events, a smallest possible physical space and time, there would be a determinate finite number of events such that only a finite number of tasks are required to complete any given motion, solving the paradox.

There are many responses to these paradoxes. One popular response is what I call the *standard mathematical objection*. This is supposed to work by showing that the sum of a convergent series of acts is finite and thus that an infinite culmination of states is possible to complete within some finite time. Whitehead, for example, writes: “Zeno...illegitimately assumes this infinite series of acts of becoming can never be exhausted. But there is no need to assume that an infinite series of acts of becoming, with a first act, and each act with an immediate successor, is inexhaustible in the process of becoming. Simple arithmetic assures us that the series just indicated will be exhausted in the period of one second...Thus this paradox of Zeno is based upon a mathematical fallacy.”¹³ The idea is that if you take a journey that is one second long, it does not matter if you matter if you subdivide it into an infinite series of segments, its divisions only get smaller and always only sum to 1.

However, this response is inadequate. Suppose that the distance I attempt to traverse between A and B is 1 and it takes 1 second for me to complete it. If I begin by traversing half

¹³ Whitehead, Alfred North. *Process and Reality*. Edited by David Ray Griffin and Donald W. Sherburn. New York: The Free Press, 1978. 69.

of this distance, then half of that distance, then half of the next distance and so on, we can represent the time it takes me to travel by the following function:

$$Z = 1/2 + 1/4 + 1/8 + 1/16 + 1/32... + 1/2^n$$

The limit of the function Z is 1 because the difference between it and 1, $(1/2)^n$, becomes arbitrarily small as n increases indefinitely. We can always find a finite number of terms in the series that sum to a number greater than n , but less than 1, because these finite number of terms approach 1 as n tends towards infinity. In our case, as each state is added to the next, the distance between Rowan and point B is supposed to become smaller and smaller. Rowan approaches the limit 1 as the number of states tend toward infinity. The standard mathematical objection points out that this is a mathematically unproblematic summation that never exceeds 1 and thus that the action is simply completed after one second. However, it makes no sense, on the assumption of a states-only view, to say that the function Z sums to 1 when n is of infinite value because it implies that Rowan actually must pass through an infinite number of states. No matter how many states Rowan passes through he can never pass through an infinite number of them because such a process can never terminate. No matter how many steps are made in the series, it never actually reaches 1. Asking the function to sum to 1 is like asking someone to count all of the natural numbers. It is an impossible task.¹⁴

It will not do to reply that this is nonsensical because we have already stipulated that the sum is 1 second and therefore it will only take 1 second. It must be remembered that we begin from point A and duration-less states are the only thing on this view that could get us to B. *If the journey were completed*, it would take a finite time, one second. The problem is that for the journey to be completed on the states-only view, Rowan must pass through an infinite number of states to get there. Indeed, the problem is worse: no finite time could ever even pass in the first place because every finite interval, no matter how small, is made up of an infinite number of states that must come to pass for it to have occurred. This means that no journey can even get started. No matter how small the interval of time, a process that must undergo an infinite number of operations can never terminate, because after any amount of time (like one second), only a finite number of steps could possibly be completed. We might be able to accurately describe my journey from A to B after the fact by using a convergent series that sums to 1, but that can never tell us how we got started moving in the first place. This solution conflates a completed definition of some already existing series of actions with the actual ongoing processes of the world that starts out incomplete. It presupposes a journey already completed.

William James supposes that this error comes from confusing what he calls the “growing infinite” with the “standing infinite.”¹⁵ I think he is right. This distinction is similar, but different, to Aristotle's distinction between the actual and potential infinite.¹⁶ The standing

¹⁴ See Black (1951) for the classic discussion of impossible tasks or ‘supertasks.’

¹⁵ James, *Some Problems of Philosophy*, 167-172.

¹⁶ This distinction is what Aristotle uses in his response to Zeno’s paradoxes. In this case, a change that is actually infinite is some change that has an infinite number of parts. These changes are actually constituted by an infinite number of smaller changes. Conversely, a change that is only potentially infinite is some change that is ‘continuous.’ What Aristotle means by continuous is some entity whose extremities (in this case their start-state and end-state), which are different, nonetheless form a whole: “continuity belongs to things that naturally

infinite is the complete description of some collection, like the set of all natural numbers. For example, suppose there were an infinite number of stars existing spatially in some world state. For every number n there is a corresponding star. Similarly, for my completed journey AB, the states-only view will have it that there is an infinite number of states that are constitutive of it and that it could be divided into. For every t , where t is infinitely granular, there is a corresponding state in the journey. In the standing infinite, we are dealing with static forms of being, things completed that can be chopped up in whichever way we so wish. The standing infinite is something already completed, by definition. Note that this is how infinite numbers are defined, as they cannot be defined by the process of counting. The growing infinite is different, however.

The growing infinite is a process that carries on infinitely, namely, without end. This process is not given in advance beforehand but must be successively counted through some process. For example, imagine trying to count all the natural numbers. Undertaking these processes would be a growing infinite and are not merely “medically impossible” as Russell would have you believe, but logically impossible. Even ‘after’ an infinite amount of time, one would still be counting! While we can easily define the limit that such an operation tends toward, infinity in the counting case and 1 in function Z , there is no finite or infinite time that could elapse such that a growing infinite could reach its end. At every point in this process, the number would still be finite. Note that my objection here is not that changes take some finite amount of time and on the states-only assumption they are made up of an infinite number of steps, therefore changes must take an infinite amount of time, contradicting our assumption. I assume only that each of the infinite terms in any supposed change occurs during a durationless state and that an infinite number of those states could never come to pass, making change impossible, just as counting all the natural numbers would be. An infinite summation cannot actually be completed on this view.

Those who offer this solution, such as Russell¹⁷ and Salmon,¹⁸ assume that the dichotomy can be solved in an ontology of instants because all motion is adequately captured by the standing infinite. However, this begs the question against the argument as stated. The identification of motion with the standing infinite presupposes that some motion or time could be completed in the first place, since the standing infinite only deals with functions already defined and completed. But this is precisely the thing at issue: whether I could complete (or

in virtue of their mutual contact form a unity.” Such changes are still considered infinite because they could potentially be ‘stopped’ at any of an infinite number of different points during the change. However, they are not actually stopped at any point, other than where the change actually does end (even though they could be). Thus, such a unity is not actually made up of an infinite number of parts. If some continuous change were to stop at every point it possibly could, actualising every potential change, the change would no longer be continuous, it would be discontinuous, and the changes making it up would no longer form a unity. Rather, it would form an actual infinity of changes that are supposed to be constitutive of the entire change. This, according to Aristotle, would make the change impossible: “to the question whether it is possible to pass through an infinite number of units either of time or of distance, we must reply that...If the units are actual, it is not possible; if they are potential, it is possible.” It should be clear that the account of change that the states-only view must provide is that changes are actual infinities. Thus, on Aristotle’s view, they could not get around Zeno’s paradoxes.

¹⁷ Russell, Bertrand. *Our Knowledge of the External World*. London: George Allen & Unwin Ltd, 1914. See chapters 6 and 7.

¹⁸ Salmon, Wesley. *Space, Time, and Motion*. Minneapolis: University of Minnesota Press, 1980. See chapter 2.

even start) my journey. It is a crucial premise that we assume the journey is not yet completed, but if we begin with the assumption that time is an infinite series of states, no journey could ever be completed. The problem is that motion on this states-only view is a growing infinite. This is what makes motion and change impossible. In terms of distance, it is impossible to occupy every state in a growing infinite, no matter how much time passes. And in terms of time, it is impossible that any finite time passes in the first place, because an infinite number of temporal states of a growing infinite must first terminate, which cannot happen. Russell even grants this: “If you set to work to count the terms of an infinite collection, you will never have completed your task”¹⁹, recognising the distinction between the standing and the growing.

One way of getting around this is by saying that all changes are, in fact, already ‘completed’ and that is why we can assume it in our refutation of the argument. This is what eternalism says about time and events. Eternalism is the view that all events and their relations, what we normally think of as past, present, and future events, all always exist. Each event exists in a ‘permanent’ or ‘timeless’ relation to all other events. However, this does not mean that all events are occurring all the time. Rather, eternalism makes a distinction between occurrence and existence. Each event is indexed to the world-line at a specific time, which is when it temporarily occurs in time and is when we experience it, even though they all exist eternally. It is this time-indexing that orders the events as before and after one another. In sum, eternalism argues that events eternally *exist* as determinate entities, but do not eternally *occur*, to account for their order and our experience of them. Through the essay I refer to this as the occurrence/existence distinction. Thus, it might be objected that eternalism is true, and that, therefore, motion and change is rightly considered to be a standing infinite rather than a growing infinite. Indeed, it seems to me that the standard mathematical objection presupposes, for its answer, that eternalism is true.

This would avoid the problems raised by Zeno’s paradoxes as any motion or change could be treated as an existing event made up of an infinite collection of states, therefore making the assumption that the journey is in some sense already completed, unproblematic. Such an account does not have to explain how any motion can get through the infinite number of points that must make it up, or how any finite time interval can be completed by the infinite succession of states in an ongoing process because each event exists, already ‘completed.’ Event AB would take one second to fully occur, even if it is composed of an infinite number of states, because one of the eternal properties of the event is that its occurrence will have a duration of one second. Thus, if the states-only view retreats to eternalism to avoid Zeno’s paradoxes, then the event AB just is the co-existence of each of its stages in time occurring in order. I call this view *states-only eternalism*. This seems to be the strongest response to the argument. Indeed, it seems the only way of making sense of the purported import of the standard mathematical solution, as it otherwise seems to completely miss the point of Zeno’s paradox by presupposing that all changes are already complete. I now move on to a further argument that puts pressure on this states-only response.

¹⁹ Russell, *Our Knowledge of the External World*, 181.

States-only Eternalism & The Perception of Change

The second problem with the states-only view is that it cannot account for the perception of change. I outline the argument I intend to defend in full here, before going on to defend its truth. The full argument, including last section's conclusion, goes as follows. On the states-only view, change must be either described by a standing infinite or a growing infinite. Suppose change was described by a growing infinite, like the process of counting all of the natural numbers (as it is in the statement of the dichotomy). If this were right, Rowan can never begin the journey AB. But Rowan can begin, and complete, the journey AB. Thus, change cannot be described by a growing infinity of states coming-to-be. This, I have already established in the previous section. Now suppose motion was described by a standing infinite, like the collection of all natural numbers, as the states-only eternalist would like to do. However, if this is right, and the experience of change is a part of this world like anything else is, *we would never experience any change at all*. But we obviously do experience change and there is a good case for that experience being part of the world. Therefore, change cannot be described by the standing infinite.

The key thing to be explained here is exactly why we should think that the truth of states-only eternalism and our perception being part of this world would mean that we cannot perceive change. This is quite a radical claim, so it will need a strong argument. However, such an argument is at hand. I will uncover it in what will at first feel rather peculiar, but piece by piece, if you bear with me, you will see my point.

First, we perceive change. This should be obvious to all of us. In order to be perceiving anything intelligible at all, we must be experiencing some change. Now, this premise does not rely on there being actual things out there that we perceive to be changing. That is, it does not require there to be, say, a person crossing from one side of a road to the other side for us to perceive what seems like a person crossing from one side of the road to the other. It could be that neither this person, nor the road, exist at all and we would still be experiencing some change. Namely, we would be experiencing change such that it seems like there is someone crossing the road. All this point requires is that there is some kind of qualitative *succession* that is perceived, not that there is some particular object undergoing some change. Denial of this claim would amount to saying that our seeming awareness of the world is completely static, like a single photograph that never changes, which is just false. In order to even be reading this sentence, you need to be successively perceiving its parts. There needs to be no analysis of such change for us to recognise its genuine occurrence in perception. Nor does this presuppose that there is a perceiver to perceive this change. Such perception need not 'belong' to a subject, it just needs occur. In a phrase: *there is the perception of change* (or, there is changing perception).

Second, the perception of change, qualitative succession, necessarily occurs over some duration. This premise is about the conditions for the possibility of such perception at all. In the language of events that I have been using until now, the perception of change cannot be a state (cannot occur at a single time, instantaneously), it must be an event (it must occur over some duration). Why should we think this? In the case of a single state or event this is obvious. If all we perceived were a single duration-less state at a single time, all we would perceive

would be a single unchanging sensation. On the other hand, if all we perceived were a single event that occurred over some duration, we would experience the succession of its internal changes as it unfolds over its duration. If the event were my crossing the road, my starting on one side and getting to the other would be successively perceived as the event unfolds in its duration. And once again, there could be no such things as roads or persons crossing them and there would still have to be a qualitative succession supporting the seeming entities, that still must be accounted for.

Now, any states-only view (not just states-only eternalism) would disagree with this. They must reject that the perception of change is due to a real duration because states are duration-less. They would instead have to say that the perception of change is not constituted by the successive unfolding of some event in duration, but the successive occupation of (an infinite number of) different times, at which a duration-less state (that does not itself change) occurs. Thus, for states-only eternalism, perceiving my crossing the road, while appearing to be an event which occurs over some duration, is actually just the existence of an infinite number of states that occur at different times. Does this mean that my premise is false, that the perception of change *must* occur over a real duration, rather than the successive occupation of an infinite number of states? I will now argue in the negative, and that this explanation does not, and cannot, actually account for our perception of change.

The problem with this is that the idea that we could ‘successively occupy’ different states cannot actually be secured on any states-only view, because for the states to exist in a succession, they must be temporally posterior to one another, thereby giving them a real order. However, to be temporally posterior to one another, they must exist over some duration. Without such a duration, there can be no real succession and thus no perception of change. This point, which I will now argue for, will require some spelling out. However, the idea is rather simple.

What characterises the perception of change is that things appear to be one way *and then* they appear to be another way. I appear to be on one side of the road *and then* I appear to be on the other. This is what I mean by *succession*. The essential feature of this succession is that it must be constituted by some temporal order, where the particular character of succession at an *earlier* time is different from that at a *later* time. That is to say, the qualities are temporally posterior to one another. However, it is here where the presuppositions of any states-only view come back to bite. In order for these qualities to be temporally posterior to one another, they must be separated by some duration. To see this, consider the following argument.

Suppose that you perceive my crossing the road. According to states-only eternalism this perception is made up of an infinite number of states A, B, C, D..., jointly referred to as event AD. Either AD, the perceptual event, happens over some duration or it does not. It cannot happen over some duration as states are duration-less. No matter how many states are posited to make up the perception of change, they cannot compose to constitute an event with a real duration. This is because states are essentially temporally unextended, and duration is temporal extension. We can see this point analogously in the spatial case. Suppose I had an unending supply of two-dimensional pieces of paper. They have a length and width but have no height. Now imagine I am to place them on a desk on top of each other, at an infinite speed. There is

no amount of time that could pass, in principle, that would mean the stack of paper gains any height whatsoever. The stack would always remain completely two-dimensional. The same is true for duration-less states. While they are made up of their objects and their respective properties, they are not extended in time and thus have no duration. No matter how many of them there are, you cannot get duration from the duration-less. Thus, the states-only eternalist must say that the perceptual event AD has no duration.

However, if the perceptual event AD has no duration, then the states constituting it cannot succeed one another in a way that supports the successive character of our perception. We saw there that what was essential to our perception of change is that the particular character of it at an earlier time is different from that at a later time. You see me on one side of the road, then you see me on the other side. If such events do not occur over some duration, then there can be no such thing as earlier and later. This is because, what it is for some entity to be earlier or later than something else is that there is some temporal ‘distance’ between that entity and another, such that one can be said to really succeed the other. In order for your perception of my crossing the road to occur the way it does, you must have first perceived me on one side of it, then on the other. What is denoted by ‘then’ in the last sentence is the fact that the qualities had to be temporally posterior to one another for you to experience my crossing the way that you did. They cannot have occurred at the same time; they must have occurred over some duration.

If the temporal constituents of that perception are duration-less states, as states-only eternalism would have it, then *such an event cannot take any non-zero time to elapse and thus cannot bear qualities that are temporally posterior to one another*. Rather, every constituent of the perceptual event must occur simultaneously as each of its infinite states elapse instantaneously. This makes the actual perception of change impossible. Therefore, on the states-only view, the perceptual event AD cannot occur over some duration or over no duration. Given we actually perceive change, the states-only view must be false, or perception must occur in a world different to the states-only world.

This can perhaps be made clearer by pointing out its spatial analogue. Suppose you have a bookshelf where one of its shelves is full, from one side to the other, packed tight with books. For imaginations sake, suppose they are of the same height, depth, and thickness and that we have numbered each book from left to right such that they are each assigned a number corresponding to the actual ordered spatial relations holding between them. What makes possible the fact that these books (1) fill up the entire shelf, (2), are each fully spatially posterior to one another and (3), can be given a meaningful numbering corresponding to their actual ordered locations in space is the fact that they have a real thickness, the fact that they are actually extended in that dimension.

To see this, suppose now that each of these books was somehow made two-dimensional, instead of three, losing their thickness. If we pack them all tightly together again, and we place them, one by one, next to each other, in the same order we had them in, (1) to (3) no longer hold. They would instead (4) fill up *none* of the shelf from left to right because they have no thickness. Even if we put an infinite number of such books on the shelf, they would not fill it, they would not fill any of it (in that dimension). Nor can they (5) be fully spatially posterior to

one another. Rather, given they have no thickness whatsoever, they will be entirely spatially co-incident with one another, sitting ‘on top of’ one another, co-located. Finally, and most importantly, they (6) cannot be meaningfully ordered by our numbering system anymore, given that their order corresponded to the fact that they were actually differentiated along a dimension, by their thickness. Now that they are not posterior to one another in that dimension, their ordering is meaningless as they are entirely spatially co-incident in that dimension.

It does not help to say that they still have their numbering anyway, because that numbering does not correspond to the real phenomenon of being ordered along the bookshelf. It no longer corresponds to anything and is therefore a pseudo-order. That is to say, it does not actually inform us about the way the concrete world actually is. We can see this because scrambling the order and assigning each book a new number at random and saying it corresponds to their actual order would be equally as informative about reality as the original order is once the dimension that initially allowed us to establish that order is abolished.

The case is exactly analogous in the temporal dimension, too. Once duration is abolished from one’s ontology and only states are admitted to exist, three things follow: (1) there can be no duration; (2) no occurrence can be temporally posterior to any other occurrence and; (3) the notion of there being a real order or succession of occurrences in time becomes meaningless. It becomes meaningless for the same reason the ordering of two-dimensional books becomes meaningless: the dimension that their ordering corresponded to has been abolished.

In fact, the temporal case is more damning than the case of the books, because the books on the shelf could still be located in the missing dimension without merging entirely with the others. However, in the temporal case, a whole dimension is under threat because states-only eternalism countenances *none of it whatsoever*. A better analogy would be to imagine that one of the spatial dimensions were abolished entirely and we were still trying to differentiate objects along it. It just does not make sense to say one thing has greater depth than another, if there just is no depth. Thus, whatever states are supposed to exist timelessly on states-only eternalism, will all exist, *at* the same time. That is to say, they will not just exist timelessly (as the eternalist will usually have it), they will also all *occur* at a single time, simultaneously, because they cannot be made to be temporally posterior to one another. This is just the same as the two-dimensional books. They could not be made to be spatially posterior to one another and instead ended up being spatially co-located. But if all states that are supposed to occur in some successive order actually just occur simultaneously, then any order applied to those states is a pseudo-order. By which I mean that it is meaningless, because it does not correspond to the real differentiation of some entities along a dimension, and because the order could now be arbitrarily rearranged with no loss (or gain) of information.

However, and here is the primary issue, if the perceptual event AD is also made up of such duration-less states, then no successive perception could have occurred in the first place. For there to have been such perception, you had to perceive me on one side of the road, *and then* the other. But there is no meaning to such a ‘then’ when it comes to a universe of states because it presupposes a successive order that cannot really exist. If there are only states, there cannot exist the essential character of perceived change, namely, the succession of some

qualities into others. Succession is incompatible with no duration. Thus, the perception of change, succession, necessarily occurs over some duration.

One objection to this premise is that it is not an undeniable fact that the perception of change occurs over some duration, as one instant of memory experienced concurrently with a perception could constitute the experience of change.²⁰ The idea here would be that the temporal contents of experience are not necessarily of some duration. Rather, they could be duration-less, where the appearance that it is not is due to our memory (or some other experiential mechanism) that makes it appear as if we are perceiving temporally non-coincident contents (because there will be no change within a single duration-less experience). Indeed, compelling suggestions have been made along these lines by Chuard (2011) and Arstila (2018).

However, I could accept such a view with no damage whatsoever to my argument. I am not claiming that all perceptual occurrences have content that is necessarily *of* some non-instantaneous duration.²¹ My argument is operating at a level beneath the debate about what the temporal structure of the contents of experience are. My argument is about what must exist for the experience of change to occur at all. Whether one thinks the contents of temporal experience are punctual or specious, you still must think there is the experience of some succession, which is to say, the experience of one thing *and then* the experience of another thing. Whether or not the ‘individual’ experiences of *this* and now *this* is duration-less is irrelevant here. It only matters that there is content at one time, and then content at a different time. For there to be such a succession of content, there must be some real duration over which it must occur.²² If there is no duration underlying these successive perceptions, which is the contention of the states-only view, there is no perception of change because each perceptual content would occur simultaneously rather than successively. This is true even if the content of perception is of duration-less states, supplemented with memory (or some other experiential

²⁰ Thanks to Cei Maslen for this objection.

²¹ That is, the claim of ‘retentionalist’ and ‘extensionalist’ models of temporal experience. See Dainton (2022) for an overview.

²² An interesting thing to note here is that this is Thomas Reid’s view. He thinks both that duration is prior to the experience of succession and that the contents of experience are duration-less but supplemented with memory. Reid is often taken to be the representative of the ‘cinematic’ model of temporal experience. That is, any view which argues that temporal experience is always duration-less. This would seem to align him with a states-only view. However, this is not the case. Like myself, he also thinks that the succession of temporal experience is parasitic upon a real duration, and he thinks this even though the content of experience has no duration. He writes: “If it be meant that the idea of succession is prior to that of duration, either in time or in the order of nature, this, I think, is impossible, because succession...*presupposes* duration, and can in no sense be prior to it; and therefore it would be more proper to derive the idea of succession from that of duration.” He argues as follows: “suppose a succession of as many ideas as you please, if none of these ideas have duration, nor any interval of duration between one and another, then it is perfectly evident there can be no interval of duration between the first and the last, howsoever great their number be. I conclude, therefore, that there must be duration in every single interval or element of which the whole duration is made up. (*Essays on the Intellectual Powers of Man*, Essay III, chapter II.3.)” In effect, he is arguing that our present perception cannot be *of* a duration (the cinematic model) yet still must be *part of* a duration, for succession to be possible. He goes on to say that we come to know this more basic duration through succession, which is just the present perception of different parts of duration, plus memory of its previous parts. Thus, Reid thinks both that temporal contents are cinematic and that duration is presupposed by succession. I am arguing for the latter in this section, and that the former is compatible with it, in this paragraph. If Reid’s view is correct, then my argument is sound.

mechanism).²³ However, one would just have to say there is some duration between each duration-less episode of perception. Otherwise, no number of instantaneous individual perceptions could add up to the non-zero interval required to make succession possible. There is no such duration on the states-only view. Onto the next premise.

Third, change in perception is change in the world. This premise sounds strange, or even trivial, not the least because this notion of ‘world’ will be unclear. But hold off on any objections until the end. I lay it out like this to show both that the argument does not presuppose a theory of mind and that I am not making a purely phenomenological point about the perception of change. It is a claim about the *immanence* of perceptual change with the rest of the world. Or, put another way, a claim that the world must be constituted in time by a real duration, and not merely by states, for there to be the perception of change. This is because the experience of change either (1) is that change or (2) changes along with that change, but is a different kind of entity or property, depending on one’s theory of mind. This premise will need some more defence and clarification, but I think such a defence is ready to hand. Though its plausibility can be better approached by considering its negation, that change in perception is not change in the world, but instead, change in some other world.

If change in perception is not change in the world but change in some other world of another character, three issues arise. First, some explanation must be given for *how* we are perceiving anything at all. If perception and the perceived are in the same world, this is not an issue because perception is the immediate relation or arrangement between things in the world. However, if perception is in a wholly different world than ‘this’ one, then an ontological impasse opens between them. Some explanation of how they interact must be given. Indeed, it is stronger than this. If perception were in a different world from this one and there is no plausible account linking worlds, it seems like we would not be perceiving anything at all. For perception to be possible (without some interaction principle) it must be situated in a world that acts as its object (and that’s even if one thinks that what is delivered in perception radically differs from what the world is ‘really like’). By positing two separate worlds, this seems to be made impossible or at least inexplicable. This is the *interaction* requirement of any two-world theory.

Second, some explanation of *what* we are perceiving must be given. If our perception is in another world and is therefore not perceiving the non-perceptual world, an account must be given for what it is we must instead be perceiving, without also collapsing the distinction between worlds (presumably into some kind of idealism). That is to say, it does not help to say that we perceive representations because regardless of whether the content delivered to us in those representations accurately represents the world, most representationalists would presumably still think that the representations are part of the world. If this is the case then the distinction between worlds collapses, and the perception of change, even if it is a mere representation, it is still a change in the world after all. If not, you would have to say that the

²³ Although one would have to accept that there are durations between these duration-less experiences that we do not perceive, otherwise such experience could never ‘add up’ to the non-zero temporal interval required to make succession possible. Chuard (2011) suggests that such a view is possible in passing on page 11. This also seems to be Reid’s view (see previous footnote).

thing doing the representing resides in a different world from its representations.²⁴ And if we accept this, the first worry about their interaction would have to be answered. This is the *content* requirement of any two-world theory.

Further, this brings us to the third problem with the idea that the perception of change is not in this world. If we accept it, there seems to be no explanation for *why* we are perceiving anything at all in the first place. If our perception somehow resides in a different world from everything else, where did perception come from in the first place? There cannot be an obvious worldly explanation for its genesis because it is not part of *this* world. Once again, some principle would have to somehow show that perception first emerges from and then transcends this world, into another. Alternatively, one could explain the relation in reverse. They would have to say that the world first emerges from and then transcends perception, into another. However, both explanations require some plausible explanation of this transcendence. (This is the role normally played by the notion of ‘emergence.’) Of course, one could reject the need for providing a genetic explanation for either world by saying that both worlds always already existed. (This is the strategy of any dualist panpsychism.) However, this still faces the initial two problems. This is the *genetic* requirement of any two-world theory.

Now, none of these points are arguments against any particular views, they are just an abstract presentation of the explanatory burdens that a view must meet if they are to argue for a plausible dualism of worlds that would be able to block my third premise. First, they must provide some principle of interaction between worlds. Second, they must provide some explanation of whether and how the content of perception relates to the other world. Third, they must explain the genesis of either perception or the world that shows how one could first emerge out of, then (metaphysically) transcend the other, or how both worlds were always already existing and related to one another. If you are to deny the premise that a perception of change is a change in the world, these are the gaps that must be closed for one to also have a plausible metaphysics.

It is now worth pointing that there is a perfectly reasonable theory of mind that meets each of these explanatory problems *and* posits two worlds. Namely, substance and property dualism. Substance dualism posits a different kind of entity, where property dualism posits a different kind of property. It is these different kinds that play the role of the separate ‘world’ in my presentation. On these views, the mental is related to the physical either by causal interaction or some metaphysical relation like supervenience (interaction), you are perceiving ideas, mental properties, or the non-perceptual world itself (content), and usually, the mental emerges out of physical somehow (genetic), though there are of course a multitude of other

²⁴ This is perhaps the view of some dualists and of Kant. In the case of Kant, there is debate on whether he really countenance two worlds or just one, where two-world interpretations have it that appearances and things-in-themselves are not one entity seen from a certain aspect, but numerically different entities, where one-world interpretations have it that they are the same entity. I do not have a view on the matter, but the subsequent argument will have consequences for what view is actually tenable for him to hold. In my view, the whole argument shows his one world position to be untenable in principle, because the noumenal self is non-temporal while the empirical self is temporal. This leaves open only the two-world view, which I think is much weaker in the details, for the same reasons I reject the idea that a change in experience is not a change in the world.

views that will not argue down these particular lines. Are these counterexamples to my premise that a change in perception is a change in the world? No. Here is why.

Whatever kind of dualism one might accept, substance or property, the only way to bridge the three explanatory gaps is by postulating some metaphysical relation between worlds. For example, dualists often think that the mind covaries with, or supervenes on, the physical. Alternatively, one might think that minds can even affect the body directly, between worlds. However, it is not important for my discussion here what these relations are, just that they are generally posited, usually in order to close the three explanatory gaps I outline above. However, once such a connection is posited, once it is admitted that the mental supervenes on, or covaries with, the physical, even in the most minimal way, it means that whenever we experience change as the succession of mental properties (or otherwise), there will be some corresponding succession in the physical. This is because, for most theories, a change in the mental will always correspond to some change in the physical. This is especially true for those who advocate supervenience because the mental supervenes on the physical, but not vice versa. That is, for the mental to have changed from A to B, the corresponding physical states had to change from C to D, but not always vice versa. Denying this would amount to a view where the entity supervening upon another somehow changes without the entity supervened upon changing, contradicting the theory. Thus, any two-world theory with some relation connecting worlds (the first explanatory gap) will inevitably have to admit that a change in experience is a change in the world, after all.

The same is true for the content requirement. Either the content of perception is about the other world or its own. If it is about the other world, the world must be changing because we are constantly perceiving change, securing the premise. If it is about its own world only, and you deny that the other world is really changing successively, perception would be utterly uninformative about the that other world. But even worse, taking this route would require one to say that no change in perception ever changes anything in the other world. However, if the substrate of that perceiving, the thing(s) doing the perceiving or representing, is in that world, then this means arguing that no perception would ever change it. If perception never changes the conditions for its possibility, why would perception ever exist in the first place? At minimum, we surely learn things by perceiving, which is a change in the very constitution of that perception. Alternatively, if you think both the perception and perceiver is in the same perceptual world, the impetus for positing a second, non-perceptual world is lost. Why posit this other duration-less, unchanging world when everything that is cared about is in the changing one? Thus, as before, any two-world theory that wishes to explain the content requirement will also inevitably admit that a change in experience is a change in the world, after all.

And finally, the genetic requirement. Some account must be given that explains why there was one world *before* and another world *after* its emergence from the initial world (unless both worlds were always already co-existent), which presupposes a successive order of occurrence. As established by the previous premise, a successive order of occurrence presupposes a real duration. If you have a view where both worlds always already existed but wish to deny the premise, one world, the perceptual world, must have duration and the other

world must have no duration. If this is the case, an account would have to be given for how and why they are supposed to relate to each other without a principle that necessitates positing a duration in both. No plausible account seems to be forthcoming.²⁵ Thus, any two-world theory that wishes to explain the genetic requirement will also require granting this point.

I think that the general lesson with regards to any attempt to bridge these gaps is that any plausible two-world theory will always have to posit some principle of interaction or systematic relation between worlds for them to make sense. The problem is that once this is posited, my premise must be accepted. If you deny that there is such a connection which links the two worlds, you end up with a theory that cannot account for the interaction, content, or genesis of perception, with the rest of the world, at all. This is because, without the interaction, a satisfying account cannot be given for either the content or genesis of perception. On an account that denies the interaction, and thus that change in perception is change in the world, it must have the perception of change as occurring in a completely free-floating world that connects in no way with the change-less one. If change in perception is *not* a change in this world in this way, and there is no metaphysical explanation joining the two worlds, I call this perceived change *illusory*. I call it this not merely because our perception will be misleading about the nature of the other world (like a regular illusion which *is* part of the world), but because this change *is not part of the world at all* (and stands in a mysterious relation to it). Any future use of the term ‘illusory’ will me meant in this technical sense (as otherworldly, rather than merely misleading).

The last thing to note on this premise is that any monistic view about the kinds of concrete entity, such as physicalism, idealism, and neutral monism, is straightforwardly committed to the truth of it. This is because every entity in one of these monistic worlds is of the same kind and can be unproblematically taken to reside in the same world. This is the great benefit of such views, that these explanatory gaps do not have to be closed. Thus, any change in perception is always a change in the world because all entities are of the same type and are usually considered to be part of the same world. For a physicalist, a change in perception must be just another physical change in the world, because everything is part of the physical world. For an idealist, a change in perception must be a change in the world, because our perception just is the world, or is an aspect of it. If you deny this, you would be re-opening the explanatory gaps closed by being a monist and covertly taking up an untenable dualism without interaction. Therefore, given all of this, we should say that a change in perception is a change in the world.

The preceding discussion may appear to be rather off topic. However, the benefit of this discussion is that shows that this premise (and the first) does not presuppose a theory of mind, or rely on some merely phenomenological point, and that it is true according to all but the most implausible theories of mind. Namely, those that countenance two worlds with no real connection between them, those that close none of the explanatory gaps. Furthermore, and this is even more important, it rules out responses which say that even though *it seems like* there is a real succession and therefore a real duration, that is merely our experience of the world, not

²⁵ The only view I can think of that has something like this is Plotinus’ system of hierarchical emanation. Here, the eternal, duration-less forms of intellection (and the One) exist logically prior to, and explain, through a relationship of emanation, soul and matter, which do have duration (and are less perfect).

what the world is actually like, meaning we should not actually be troubled by the argument and can go on advocating a duration-less states-only view. On this view, the perception of change and order would be a kind of secondary quality, like colour is sometime supposed to be. This response is ruled out because it amounts to the saying that perception is not actually immanent in the world but is in some other world, making it illusory in the sense above. My point is that the perceived change *cannot* be a secondary quality or illusion if we are to have a plausible ontology.

Now to bring all of these threads together. The argument thus far has been as follows. There is the perception of change. The perception of change necessarily occurs over some duration. And finally, that a change in perception is either a change in the world or it is an illusion in the otherworldly sense. Given we would have to accept an implausible theory of mind that closes none of the explanatory gaps outlined above, I propose that the perception of change is not an illusion and is in fact part of this world. Thus, a change in perception is a change in the world. Here is the conclusion we can now draw: if there were no real duration, the experience of change (being in this world) would be impossible. There is the perception of change. Therefore, there is a real duration. Since states-only eternalism cannot countenance a real duration, it cannot explain the perception of change, and it cannot therefore be true. The only way out that it has is to affirm a most implausible two-world ontology with no relation between worlds. Change cannot be described by the growing infinite of states coming-to-be, per the last section, or the standing infinite of an infinite number of states being successively occupied, per this section. Thus, change is not possible at all for the states-only view, which means it must be false.

To conclude this section, even though it seemed like all changes could maybe be explained by the appeal to eternalism and the successive occupation of an infinite number of states, we saw that it could not. We saw this through a study of the perception of change. We saw that it must occur over some duration and that it could not realistically be doubted or relegated to another world. This meant that for there to be the perception of change, there must be a real duration. Given that states-only eternalism does not countenance such a duration, it is incompatible with the possibility of perceiving change at all, which means that we should reject it.

States-Only Views & Time

The overall argument against the states-only proposal went as follows. Zeno's dichotomy showed that the states-only view is untenable because it implied that all changes were the coming-to-be of an infinite number of entities, which is impossible. However, it seemed that a states-only view might have recourse to eternalism and time-less relations to get oneself out of this bind. What the next section showed is that states-only eternalism cannot account for the perception of change which is undeniable on all but most implausible assumptions. Thus, neither a generic states-only view, nor a states-only eternalism can be true, unless one insists on an implausible two-world ontology. In both cases, the problems come from not accepting

the existence of events with a real duration. A real duration is a condition for the possibility of immanent perception and real change.

The implication of this is that it shows not only that reality cannot be composed merely of states, but also that *time cannot merely be a dense collection of instants*, where an instant is a duration-less instant of time. This is because, whether time is something over and above the events that make it up or whether it is merely the measurement of the actual order of events that make it up, the only events that can be indexed to a dense set of instants (rather than both instants and intervals of time) are states, because instants are duration-less and could only thus support the existence of duration-less states. Thus, if time were merely a dense collection of instants, the states-only view would be true, and perception would be impossible. Therefore, we do not have valid recourse to a description of the universe in time based on any dense set of temporal points precisely because we do not have valid recourse to a description of the world from discontinuous states to events. Time must, at minimum, be made up of intervals. The error in supposing that we could construct a line out of instants always begins by taking some determinate reality, already existing, and retroactively imagining we can construct it out of the convenient fictions we derive from it. Onto the next proposal.

Chapter III: The States & Events Proposal

States & Overdetermination

The second, and perhaps the most intuitive proposal for composing the world-line, is that the universe is made up of both states and events. On our world line, we would have both intervals, like AB, and points, like C and D. This would mean there may be events, such as Rowan cooking a carbonara, and of states, such as Rowan's mental state or the state of the world at time t . This also includes views that take the world-line to have a real duration but be infinitely divisible. This view would avoid the problems raised in the last chapter, because events have duration, which means we can account for the successive perception of change. However, in this chapter, I will provide reasons for rejecting this proposal too.

There is one major problem with the proposal that there are states and events. Any states that are also parts of events overdetermine reality. We can see this with an example. Suppose I am kicking a football around with my friend Lachlan. At some point, he kicks the ball over the goal and through the window of a nearby house, smashing it. Now we have an event, "Lachlan smashed the window." Let's call it event AB. Suppose the event starts when Lachlan started taking a run-up to kick the ball and ended with us inspecting the window. All of this occurs over some definite interval of time, some duration. Here is the problem with the states and events proposal: we can fully characterise this event, and any such event, without recourse to duration-less states. In other words: what further thing is gained by supposing that, on top of the real temporal extent and its accompanying arrangement of objects, already well defined, that there are an infinite number of states populating them as well? Remember, one cannot say that the interval is made up of such states (making them constitutively necessary), as I have rejected an account that supposes duration to be made up of purely duration-less states. Thus, something that is not a state has to exist, namely, something with duration, an event. But as soon as we admit that, states become superfluous in explanation.

To see this, take two worlds containing the event "Lachlan smashed the window", one with states populating it and one without. Would things be any different, causally, empirically – in any meaningful way? The answer is no. The window still breaks, we still feel guilt and so on. The event occurs as if the states were not there. Thus, they overdetermine reality and play no ultimate explanatory role. We should reject entities that play no explanatory role from our ontology.

One might object that states are actually indispensable to practices, both within science and within ordinary language. Such talk is utterly ubiquitous and we often never, even in principle, have the tools to pursue descriptions at the level of events, rather than of states. My answer is that I am not interested in what it is useful for us to wield in our practices of navigating the world, I am interested in what the world actually is. We can certainly use the language of states for purposes of inquiry. But that does make them necessary posits, because, on this hypothesis, they are merely parts of events that already fully explain them. I do not

think, absent some positive metaphysical (non-pragmatic) reason or argument, that because something is useful, that it also exists in a fundamental sense.²⁶

Ockham's Razor & Arbitrariness

Just to reinforce this point, states overdetermine reality in precisely this sense: no explanatory needs can be met by the existence of states that are not already met by the existence of events. Further, I am claiming that if there are no explanatory needs met by postulating particular entities or kinds of entities, we should not do so. This is because, I think some theory has an evidential edge over another theory if it is identical with another one, but it does not postulate entities without reason. Thus, in this case, I am claiming that due to the structure of the world-line proposed by the states and events view, events already explain everything states could possibly explain, making them explanatorily superfluous.

In other words, I am just invoking Ockham's razor. One formulation of the razor can be found in the *Stanford Encyclopedia*:

(OR₁) Other things being equal, if T_1 is more ontologically parsimonious than T_2 then it is rational to prefer T_1 to T_2 .²⁷

The author supposes that this formulation is the natural paraphrase of the most common formulation of the razor:

(OR₂) Entities are not to be multiplied beyond necessity.

This is right, as long as we read "Other things being equal" in the paraphrase strictly, as implying that the theories actually do explain the exact same phenomena, but that the latter theory posits further entities that are unnecessary for this task. That is to say, in situations where T_2 is just $T_1 + X$, and X is some entity or type of entity playing no explanatory role, it is rational to prefer T_1 . While this is a rather narrow scope for the principle, I think this is its proper place.²⁸ A more literal rendering of what the razor means would then be as follows:

(OR₃) Entities are not to be postulated to exist (multiplied) if they are not necessary to explain anything (beyond necessity).

It simply means that, given the choice between two theories that are exactly the same, except that one posits some further entity that plays no explanatory role, we should accept the theory

²⁶ This is not to say I am not open to indispensability arguments, arguments that are trying to meet some necessary explanatory burden, or transcendental arguments for necessary metaphysical structures that attempt to meet some explanatory burden. These are of a different structure to the kind posit I am considering here. These kinds of argument *do* give a positive explanatory reason why they are to be taken seriously. For example, Plato thought that his Forms made possible what he took to be the undeniable intelligibility of the world. In the previous chapter I make a transcendental argument for the existence of a real duration to explain something I take to be undeniable: the perception of change.

²⁷ Baker, Alan. "Simplicity." *The Stanford Encyclopedia of Philosophy*, Summer 2022.
<https://plato.stanford.edu/archives/spr2016/entries/determinism-causal/>.

²⁸ This is because I do not think some theory is automatically superior to some other theory just because it has less entities. What matters is whether those entities are doing explanatory work. Ontological profligacy is entirely justified insofar as that profligacy is taken to be necessary to explain some feature of the world.

that does not posit these further entities. It says that we should not postulate entities unnecessarily, that is, *for no reason*. If an entity is postulated for no reason, we are under no obligation to accept it. But how can I justify this principle?

We can see the appeal of such a principle with an example. Take the event of Lachlan smashing the window. Suppose I argued that the natural explanation we would give of the window being smashed was all correct. That Lachlan kicking the ball and it hitting the window was sufficient to break the window. Now suppose I argue that simultaneous with the football smashing the window, a God decided to smash the window as well. No explanatory demand is fulfilled by such an appeal because everything would have happened as it did without God, especially given I am not replacing the causal power of the ball as the thing smashing the window with God smashing the window but instead supplementing it with God. In this case I would surely be wrong to postulate further entities because no phenomena are accounted for that were not accounted for before. In other words, I am postulating an entity for no explanatory reason. I am just arbitrarily supposing there is something else there. The case is the same if we supplement events with duration-less states.

Arbitrariness is objectionable in metaphysics because it amounts to arguing that we should postulate entities for no reason, which would trivialise the whole endeavour. Suppose we were to ask someone who postulates such a God, or any other entity doing no explanatory work, why they do so. Given they play no explanatory role, could a satisfactory reason be given? They cannot say “because it explains so and so phenomena”, because so and so phenomena have already been explained in their ontology. They cannot say “because it is useful to suppose that it exists” because just because something is useful does not mean it exists. And surely, if one cannot give a reason for why we should accept a posit, then we should not accept it.

If some stranger came up to you and asked if you could give them \$100, and when you asked why, they reply “oh, for no reason” (and you somehow knew this was true), should the onus be on you to give a reason why you should not give them the money? No, the onus is on them, and if they proffer no reason (and you come up with no reason yourself), then you have no reason to give them money. The same burden holds for arbitrary metaphysics. If someone postulates some entity, the burden is on them to show why it is necessary to explain some phenomenon. Once again, denying this principle would amount to arguing that we should postulate entities for no reason, which surely trivialises the whole endeavour of metaphysics, because then, anything goes.

No Grounds for Schaffer’s Laser

Unlike myself, some philosophers are not opposed to certain kinds of explanatory redundancy in metaphysics that would be ruled out by Ockham’s razor, given the right conditions hold. Jonathan Schaffer holds such a view. He explicitly rejects Ockham’s razor and instead accepts

what he calls “The Laser.”²⁹ The idea is that if we accept the view that reality is structured according to hierarchical grounding relations³⁰ then it is not the proliferation of any old entities that are objectionable, but only those that are fundamental. The point is that so-called ‘derivative’ entities are fine to be multiplied. But what is this idea that reality is structured according to hierarchical grounding relations?

Grounding is a supposed metaphysical relation whereby some existent entities are dependent on other entities in a relationship of descending fundamentality. For example, take a modified version of the classical doctrine of atomism, that all there is, is atoms and void *plus* some entities that they ground. Here, one might suppose that atoms and void are the most fundamental existents, and that all other ordinary existents, such as tables, chairs, and organisms, are derivative entities grounded by those existents. There are two important aspects of Schaffer’s grounding relationship to note for my purposes.

First, entities that are not grounded in anything are called *fundamental* entities, while all other entities that are grounded in things, are called *derivative* entities. Fundamental entities are necessary to posit because they explain the ‘base’ reality, while derivative entities are explained by their being grounded in fundamental entities. It is not important what kinds of entities are fundamental and which kinds are derivative, as this will be theory dependent. For example, for atomism, atoms and void are fundamental, while everything else is derivative. Second, the grounding relation is asymmetrical. Namely, that the grounded depends on the grounds, but not the other way round. If the grounded ceased to exist, the grounds would remain. While if the grounds ceased to exist, the grounded would go with it. We can see this with atomism and a chair. The chair could be destroyed while the atoms that make it up remain, but not vice versa. The putative benefit of such an approach is that we are still right to speak of the grounded entities as existents, rather than reducing them to the fundamental entities, or eliminating them outright.

We could even perhaps use this scheme to recover states by supposing that states are grounded in events. That is, that they are dependent on events, but still have existence, albeit a derivative one. While this seems natural enough, I have a major problem with this approach.

The problem with this explanation of states and events is that it does that not succeed in overcoming Ockham’s razor, as I have formulated it. As I mentioned earlier, there are no explanatory needs to be met with regards to positing states, over just events. Why bother further positing a primitive metaphysical relationship to join the two? Especially considering I am not worried about saving our talk. Interestingly, Schaffer actually partially agrees with this assessment. He thinks that whatever one’s ontology is, their fundamental entities *fully explain reality*. However, he also thinks that instead of following Ockham’s dictum not to multiply entities beyond explanatory necessity, we ought to follow another principle, The Laser:

²⁹ Schaffer, Jonathan. “What Not to Multiply Without Necessity.” *Australasian Journal of Philosophy* 93, no. 4 (2015): 647.

³⁰ Schaffer, Jonathan. “On What Grounds What.” In *Metametaphysics: New Essays on the Foundations of Ontology*, Edited by Edited by Chalmers, David, Manley, David, and Wasserman, Ryan, 347-383. Oxford: Oxford University Press, 2009.

“*The Laser*: Do not multiply fundamental entities beyond necessity!”³¹

The intuition that this is a good replacement can be made clear with a simple story. I adapt his own:³²

Rowan comes up with a theory of reality that posits 100 types of fundamental thing, with seemingly perfect predictive adequacy and tight argumentation. But Will comes along and posits a theory of only 10 fundamental types of things, which, in varying ways, combine to make those 100 things posited by Rowan’s theory (and those 100 things only), including themselves. Will’s theory has the same predictive adequacy and arguments.

In this case Schaffer rightly points out that we would think that Will has the better theory. He thinks this story, and our preference for Will’s theory, supports his *Laser* because according to him, Ockham’s razor, rather than rightly lining up with our intuitions, deems both theories equally parsimonious. Both account for and believe in the existence of a total of 100 entities. The reason we think that Will’s theory is better is that we have an intuition that favours ‘*The Laser*’, namely, that it is fundamental entities we object to being multiplied beyond necessity, not derivative ones. Therefore, derivative entities should be viewed as “ontological free lunch”³³ in the sense that their existence simply follows unproblematically from their fundamental entities through the grounding relation.

Putting aside whether all things are equal (in the strict sense given above) between the theories in such a way that Ockham’s razor is applicable, I have a more general concern with Schaffer’s whole approach. That is, with *The Laser*, grounding, and fundamentality more generally. By Schaffer’s own lights, derivative entities are always explanatorily redundant. This means that in any case where we posit derivative entities, *we never actually have reason to do so*. This is the kind of arbitrariness I was trying to capture, and do away with, by my earlier formulation of the razor.³⁴

Once it is admitted that basic entities already explain everything there is to be explained, there can be no reason to ever postulate more entities. Take Schaffer’s own theory as an example. He thinks that we can explain everything in the universe by positing only one entity: the cosmos. However, he also thinks, and this is *The Laser*, that we can get every single other kind of entity we might ordinarily think exists from this for free, through the grounding relation. This is with the caveat, however, that ultimately, they are explanatorily redundant. He admits this redundancy when he writes that “There is redundancy in what exists, but such redundancy is mitigated. For there is a hierarchy – some things are basic, and some things are merely derivative. Redundancy is tolerable provided the redundant entities are properly grounded in what is basic.” However, if the cosmos is supposed to explain everything there is to explain, what reason does Schaffer have to postulate these further derivative entities? The answer

³¹ Schaffer, Jonathan. “What Not to Multiply Without Necessity.” *Australasian Journal of Philosophy* 93, no. 4 (2015): 647.

³² Schaffer, “What Not to Multiply Without Necessity”, 648.

³³ Schaffer, “What Not to Multiply Without Necessity”, 647.

³⁴ Michael Della Rocca makes something like this criticism incisively in chapter 8 (‘Metaphysical Explanation’) of his great book *The Parmenidean Ascent*. I owe some of following discussion to that chapter.

cannot be that “they explain so and so unaccounted for phenomena” because they do not explain anything that is not already accounted for by the fundamental entities.

Thus, it is a consequence of his view that he accepts this thesis: *no derivative entities are ever necessary to posit*. But if we are never required to postulate any derivative entities, there is neither constraint on (or even impetus for) entity postulation, and anything goes. But this is what the original razor was supposed to cut against: arbitrarily multiplying entities beyond necessity. If everything is already explained by your ontology, not only is there no reason you should ever postulate further entities, but also anything you do posit over and above the fundamental existents have the explanatory status of the window-smashing god from earlier. And for that reason, we should reject them.

Schaffer could have two responses to this charge. The first is that we might have practical (or other non-explanatory) reasons to posit derivative entities, even though they play no explanatory role. However, I do not think such an argument would work. We can engage in all kinds of inquiry assuming that certain entities exist, and assuming they have certain properties, for practical purposes, without thinking that they actually do. (I argue for something like this in Chapter IX.) The second thing he could say is that he can avoid the specific charge of arbitrariness by saying that he is actually okay with postulating *every* set of possible non-fundamental entities, rather than some arbitrary subset of those entities. This move could avoid the charge of arbitrariness because it does not stop at this or that set of derivative entities, but instead posits *all* of them. Rather than “anything goes”, it is “everything goes”, and everything is not arbitrary in the same way that anything is.

Indeed, it is precisely this view that Schaffer actually takes. In “On What Grounds What”, he calls his view “existence permissivism.”³⁵ This is the view that the question of whether something exists is trivial because everything exists (even if it is ‘merely’ a fiction). The caveat is that they all exist somewhere in the hierarchy of existing things, and it is figuring out the structure of this hierarchy, and importantly, what the fundamental entities are and how they give rise to derivative entities, that is the non-trivial question. How this hierarchy will be ordered is theory dependent. On his own theory, everything is explained by one fundamental entity: the cosmos, in which everything else is grounded in. Does he get around the charge of arbitrariness through this move?

I think not. First, arguably it is still arbitrary to posit *any* entities whatsoever, including all of them, given they still play no explanatory role. Second, and more fatal for the view, is that it also follows from the fact that no derivative entities are ever necessary to posit, that *the grounding relation itself is never necessary to posit*. This is because the only reason the grounding relation is there is to relate fundamental to derivative entities. But if derivative entities are never necessary to posit, then neither is the grounding relation that is supposed to ground them. Thus, even if accepting the set of all derivative entities means that the choice of entities admitted into existence are not decided on arbitrarily, the grounding relation itself still is because we seem to have no reason for postulating it in the first place. Suppose I agree with his monism and do not presently accept the grounding relation exists but am open to changing

³⁵ Schaffer, “On What Grounds What”, 356-62

my mind. What can Schaffer offer as a reason for me to accept it? He cannot say that I should accept it “because you need grounding to explain the structure of the universe”, because according to him, we do not. Nor can he say that I should accept it “because it is useful to talk about entities that are not fundamental” because it being useful to talk about non-fundamental entities entails neither that they exist, nor, therefore, that I should accept that a grounding relation exists.³⁶

In sum, there can be no reason we should accept that there even is a grounding relation in the first place. This is because, on Schaffer’s account, the only reason we would ever need it is to relate fundamental to derivative entities, of which there is also no reason to ever do. But if there can be no reason to accept the grounding relation, then we should reject it (or merely refuse to accept it). But if we reject the grounding relation, we must also reject the framework of grounding and fundamentality more generally because the relation is necessary to secure the structure between levels. But if we do that, we must then reject The Laser, because it only makes sense on the grounds that there really is a distinction between fundamental and derivative entities. Hence, I think that Schaffer’s attempt to block the razor fails and that we should accept it after all, only postulating entities that we deem to play a necessary role in explaining some phenomenon.³⁷

To return to my own project, the grounding relation and The Laser will not be a route to saving states and blocking Ockham’s razor, after all. Thus, to conclude, the proposal of a world made up of events and states collapses into a view such that there are only events. We saw this because once we countenance events with a real duration, states become explanatorily superfluous, meaning we should reject them. They are unnecessary in our final metaphysical theorising about the structure of the world line and insisting that they exist is to insist on an unacceptable arbitrariness. Onto the next proposal.

³⁶ Nor can it be replied that the choice between a theory without derivative entities and a theory with them is *also* arbitrary given that they have the same explanatory power. This reply does not work because the theory without derivative entities has reasons for accepting every entity in their ontology but the theory with them only has reasons for accepting some (the fundamental entities). Every derivative entity and grounding relation will necessarily be posited without reason, and therefore arbitrarily.

³⁷ If such phenomenon includes a relationship of descending fundamentality, then so be it. But that relationship should not be conceived of as already entirely explained by its fundamental entities. It should be conceived of as a necessary explanatory posit, doing some of its own work, and thus providing a reason for accepting it into our ontology. I suspect that the reason Schaffer does not take this route with grounding is because of the notorious difficulties with notions like strong emergence that argue for something like this.

Chapter IV: Common-Sense Events (Events-Only)

The third proposal is that the universe is made up only of events. This is the event-only proposal. On our world line, this would be the intervals AB, BC, CD. At this point, talk of states has been shown to be inadequate, so we have to return to events.

First, we could have a theory of events that picks out things like cooking a carbonara and going to a concert. I call these *common-sense events*. While they constitute the world-line in a manner like the intervals indicated above, they will also overlap in a great many cases. For example, the event “the first song of a concert” completely overlaps “the concert.” The goal of having a theory of common-sense events would be to have an account of where the appropriate temporal boundaries can be drawn such that the events we speak of can be said actually to exist, in the same way that having a theory of objects would give an account of how we can rightly say that the world is full of real objects with spatial boundaries. Second, we could have a theory of *atomic events*. The notion of an atomic event is the temporal equivalent of the spatial atom. Atoms (not the theoretical posits of atomic theory, but the conceptual notion) are indivisible substances of a finite size that make up all spatial phenomena by particular arrangements. The temporal equivalent would be finite intervals of time that combine to compose all more complex temporal phenomena. Depending on the theory, these atomic events may or may not overlap on the world-line, but they are, in all cases, the fundamental constituents of it. I discuss these two events-only proposals over this chapter and the next, and critique them in turn.³⁸

In this chapter, I outline some problems with common-sense events. My arguments here are either influenced by or come specifically from the literature around the elimination of ordinary objects, but I adapt them for use against events instead. In short, Davidson was right when he said that “The conclusion to be drawn, I think, is that the individuation of events poses no problems worse in principle than the problems posed by individuation of material objects”.³⁹ I just think, contrary to Davidson, that these problems cannot be overcome in principle.

Vagueness and Events

The first problem I have with common-sense events is that their boundaries are always arbitrary. This means that, in the last analysis, any specification of some event will be arbitrary in a way that justifies their elimination. For example, take the event “Rowan cooking a carbonara.” What temporal boundaries are we to draw around this event? We could suppose that it begins once I have actually started cooking, once the pasta is in a pot, and some bacon is on the stove. Or we could suppose that it started once I started getting the ingredients and

³⁸ As before, if you are a four-dimensionalist and advocate either view, the events in question would be space-time regions or spatiotemporal quanta, respectively. These would build up not just the temporal order, but the entire concrete universe.

³⁹ Davidson, Donald. “The Individuation of Events.” In *Essays in Honor of Carl G. Hempel*, 216-234. Dordrecht: Springer 1969. 307.

utensils out, perhaps when I walked into the kitchen. Even along this short timeframe of five minutes or so there are infinitely many temporal points at which the event could have begun. And until now, if it started at *any* of these points, we would still want to affirm that it is the same event. That is, we would not say that there were two events “Rowan cooking a carbonara”, but one that started at a single determinate point.

But we can go further. Maybe it started when I purchased the ingredients from the supermarket, or when I first formed the intention to cook it. Maybe it started at my birth and my first bite of my own home-cooked carbonara was the termination of a lifelong quest begun from childhood. From here we may as well continue this process back to the start of the universe and everything until then was the long arc of history bending towards carbonara-cooking. Here we are not so happy to call these the same event, especially and most obviously in the case of the whole universe. However, it is not clear where we can draw the line.

In the next section I will argue that if there are events at all, every candidate proffered here will be an event too, but here I argue that any individual common-sense event is itself vague in this way. *We can always expand or contract the description of any event such that it includes either wider or more narrow boundaries without seemingly losing its identity.* Onto the argument.

Take any generic event AB. Suppose we have specified the objects in the particular case, and we are only dealing with the temporal aspect of it. This event must have some temporal duration, represented by the interval AB. However, as we have just noted less formally, it is we are not able to know where those boundaries really lie. That is, if we added or took away a finite duration of arbitrarily small length, we would still like to call the event AB, that same event. I use nanoseconds (one billionth of a second) in the argument, but any even smaller arbitrary duration could be used instead. We have the beginning of an argument:

- (1) There are common-sense events, like AB.
- (2) If there are common-sense events, they must exist over some finite duration t .
- (3) If the event lost one nano-second, it would still be the same event.

This is what I was getting at earlier. We could say that the carbonara began when I walked through the door, when I got the utensils out, and similarly we could say that the event begun one nano-second before or after either of these events. Further we would want to say that it retains its identity as an event. Thus, premise (3) is plausible.⁴⁰ This is just a standard sorites-style argument. But before I go on, I consider a couple of objections.

It may be objected that (3) is not true for all common-sense events. For example, some events seem to have sharp boundaries that make this not the case. Take a game of football, for example. For it to be a football game, it must last until the referee’s whistle at the end. It simply

⁴⁰ If this still seems like a strange premise, there are a couple ways of thinking it through. First, note that we ordinarily think that common sense events are modally robust. For example, suppose a band I really like held a concert in Wellington. Now suppose they didn’t play one of their songs. We would still think of it as one and the same concert. That is, the concert of the band I like, in Wellington. Second, you could think of it in terms of objects instead. If some object, such as a table, lost one of its constituent atoms, we would still like to call it that table. I think this intuition ought to carry across to events as well. The point here is that some tiny change in temporal constitution should not affect our judgement that it is the same numerical event.

would not be a football game if it fell a nano-second short of that whistle. This objection does not land. The whistle is not a separate thing that must be reached in order for the happenings to count as an event. Rather, the whistle is constitutive of the end of the event. Thus, I do not mean that if some event, minus one of the very things making it that event, lost one nano-second then it would still be that event. What I mean, in this case, is that if the ref blew the whistle one nano-second earlier than he did we would still call it the same football game. And indeed, football refs do blow their whistles all the time in a rather arbitrary manner, especially when analysed at the level of nanoseconds. They could blow their whistle at any of an infinite number of different times and still end the game.⁴¹ It is also true that a concert cannot end unless the band plays their last song, but that does not mean that if they stopped the last song short by a nano-second that it would not have been a concert. The cases are no different, whether the boundaries seem sharp or not.

Perhaps it could further be objected that there are events that are precise by human stipulation that would not fall prey to premise (3). For example, take a school-day, which is specified to start at exactly 9am and end at exactly 3pm.⁴² If this event, if it is so specified, lost a nano-second at either the beginning or end of it, it would no longer be the same event. Thus, it will not be vulnerable to premise (3).

It is hard to know what to say about such stipulated events. On the one hand, there seems to be good reason to think this is an existing common-sense event. On the other hand, whether we would like to accept entities we stipulate to be there as existing can end up becoming rather arbitrary. For example, I could name any arbitrary duration AB “Rowan’s x -minute coffee time” where x is some perfectly precise amount of time. It would not even matter for my ability to do so whether I actually drink coffee at that time or not. Would all durations I dub “Rowan’s coffee time” exist as events? It could be replied that only a certain amount of community acceptance of some stipulated event is necessary to institute it, in fact. However, then one would have to specify how many people, or to what extent people do accept something before it becomes instituted in fact.⁴³ I think any specification of this would be arbitrary and if one person cannot do it, I do not see why one million people could. Though perhaps it could be admitted that these events do exist in some sense, regardless of these problems.

However, I do not think that would be right. My own view about the matter, though I do not have room to defend it at any length here, is that such stipulated events should not be treated as telling us anything about whether the world is actually carved up into such events, i.e., whether they actually ‘exist.’ Rather, they should only be treated as ways in which we could *regard* the world as being carved up, in order to act or achieve some goal more easily, even though it is not actually that way. For example, treating the world as if it were carved up into “school-days” is an extremely useful way to co-ordinate our efforts of education. We can do this without committing ourselves to their existence in fact. The final chapter of the present

⁴¹ Of course, if there were two events, the football game as a whole and the football game minus one nano-second (and thus minus the whistle blowing), then the second event would merely be a part of the football game. But this is not what I mean, I am talking about a single event that could have been different.

⁴² Thanks to Cei Maslen for raising these counterexamples.

⁴³ It would also have to do it in a way that does not identify the instituted event with states of persons as, e.g., Searle (2010) does, because then it would not establish what the objection needs it to.

essay (Chapter IX) goes into more depth on this matter. For now, let us merely discuss what seem to be stipulation independent events. Onto the next premise.

By repeatedly applying the reasoning of (3), we get absurd conclusions. We can continually remove further and further nanoseconds such that we end up with a single nano-second left:

- (4) Therefore, something is still common-sense event AB, even if it loses all of its duration but one nano-second.

However, this seems false. Whatever we could possibly mean by cooking a carbonara or going to a concert, even if we have specified all the right objects, it seems impossible that either of them could be only one nano-second long. I leave it open that there could be common-sense events that are one nano-second long, for example, if we were timing something where we need an extremely fine-grained measurement. However, almost no events normally considered to be common-sense events will be of this length. Furthermore, the argument can make the duration arbitrarily shorter to a point in which no event can even be measure by existing methods. Thus, I doubt this would be a very strong defence. Common sense events are ubiquitously longer than that. Plus, we would get this conclusion:

- (5) For every nano-second that involves the same objects as AB, there exists another event of the same kind.

We get this result because if any old nano-second (with the right objects involved) can be considered an event, then every other nano-second like it can be. Given that the nearby nanoseconds for any nano-second involving the right objects would be virtually identical, we seem to be rationally required to accept that all object-involving nano-seconds are also events. Thus, going to a concert would not be a single unified event, rather it would involve a tightly packed series of trillions of nano-second long events. This is an absurd result. Even if were tempted to bite the bullet here and accept a proliferation of such events, it saves us almost nothing theoretically, because such entities are useless to common sense, science, and metaphysics.

Perhaps we can turn the argument by rejecting this conclusion and positing the relatively plausible negation of (5):

- (6) It is impossible for something of one nano-second duration to constitute a common-sense event.

However, if we do this, the argument can run in the other direction:

- (7) Something that is not an event is still not an event if it gains a single extra nano-second of duration.

This premise, like premise (3), is plausible enough. Two billionths of a second will be just as arbitrary a duration to suppose that there is an event. Plus, rejecting this premise leads to a corollary of premise (5), that for any two nanoseconds with the right objects, there is an event. While we would be cutting the number of events in half it is effectively as useless a conclusion to accept. However, if we repeat the reasoning by continually adding nanoseconds to this event, we get these results in the other direction:

- (8) Therefore, there is no finite duration t that constitutes an event.
- (9) There are no common-sense events.

(8) follows from (7) for the same reason that (4) followed from (3), namely, that we tolerate that there could be some miniscule difference in the constitution of some common-sense event without it losing its identity or gaining some new identity. If no determinate number of nanoseconds can be combined and non-arbitrarily identified as the point where some event AB and its objects become a common-sense event proper, we get conclusion (9), that there are no such events. Thus, we have a contradiction between premise (1) and (9), and between (2) and (8).

The options are as follows. Given that the conclusion contradicts premise (1), one must discard it or one of the premises regarding change. If we reject (2) and say that common-sense events exist but do not have duration, we end up with a states-only view, which have already been rejected. If we reject (3) or (7) by positing a stopping point where each particular event either is or is not itself, we can always ask why we did not stop one nano-second earlier or later, making the choice arbitrary. For example, if we insist that the carbonara cooking started precisely when I entered the room to begin cooking, we can always ask why it was not some other minutely different duration involving the same objects.

If one bites the bullet by accepting (5) and rejecting (6), you end up with infinitely many of arbitrarily small and unnecessary entities in your ontology that do not even line up with common sense in the first place (which is what these events were trying to capture). In fact, we can go further. Given that the argument can be reformulated with arbitrarily smaller and smaller durations (unless you believe in a minimum duration of time, which few do), *ad infinitum*, then there will be an infinite number of arbitrarily small events. Further, this solution trivialises the explanatory role these events might have played. If any duration whatsoever that includes the right objects can be arbitrarily individuated into a multitude of different durations limited only by our powers of imagination, then I fail to see how affirming their existence in reality, rather than in language, would actually help us. Nor is there good reason to suppose that existence is also actually carved according to the bounds of our imagination. Finally, one could accept the conclusions (8) and (9) by rejecting (1) and (2), that common-sense events exist at all. I believe this is what the argument suggests.⁴⁴

One must think this talk actually carves nature's joints to resist my suggestion that they do not really exist. This is because, if you are merely speaking of facts about how we use language then we are talking about different things. I am discussing whether events, as we speak of them, are actually there to be spoken of. But it is absurd to think there is such an

⁴⁴ It is worth noting here that there are many great replies to these kinds of paradox that I unfortunately do not have room to discuss. Most notably, supervaluationism which says that there are many acceptable sharpening's of any given vague event and epistemicism which says that each event has a single sharp boundary that is determinate in each case but that we can never know. I reject the former (supervaluationism) because it is usually merely a linguistic thesis about how vague language works (unless one is also a universalist about mereology, which I reject below) and because it gives rise to a second order vagueness at the boundaries of the acceptable sharpenings. I reject the latter (epistemicism) because the most prominent defender of such a view, Timothy Williamson, also makes what the boundary is dependent on language use rather than on the way the world is.

extensive proliferation of objects and events. However, I do not want my rejection of such a proliferation to rest merely on accusations of absurdity. Things can be absurd and true. Rather, I would like to reject an account that bites the bullet because it is post-hoc.

The dialectic of the preceding discussion goes as follows. My imagined opponent defends common sense events on the grounds that it seems reasonable to suppose that our spatiotemporal universe can be carved into common-sense seeming events, that is, into those things that we talk about in our everyday life and in the way carve up the world for scientific inquiry. There are things such as concerts, storms, road crossings, lightning strikes, running races, and so on. Then, through the induction step of a sorites series I show that we would be required, if we are to call a typical instance of an event an event, to call every arbitrarily small duration involving the same objects, an event as well. This means that we would have to end up saying that any duration involving the right objects would also be an event, which means that for any common-sense event, there must be an infinite number of events present, many of them in no way resembling the original one that are yet still that kind of event. A nano-second of a band tuning their instruments is still somehow a concert and there are trillions of them. But suppose my imagined opponent, in order to save common sense events, accepts the conclusion.

Doing this is objectionably post-hoc because the original justification for accepting common-sense events in the first place is that there were natural seeming spatiotemporal boundaries by which they could they be individuated. To accept the conclusion as a response to such an argument is to completely throw away that justification. For example, you must deny premises like (6), that it is impossible for something of one-nano second duration to constitute a common-sense event. Starting from our original justification this would seem to be almost certainly false, at least for most ordinary events.⁴⁵ No longer are these boundaries natural seeming and commensurate with common sense, they are infinitely abundant, and they are seemingly arbitrary, given that all kinds of different arrangements, of all kinds of different durations, earn the same name. And there seems to be no principled line we can draw between the single appropriate event and all of these other candidates.

I am not saying we cannot add new hypotheses to theories in order to save them, I am just saying that accepting the conclusion here is in no way in the spirit of the original view and in fact goes against it. Such a conclusion, that events are infinitely profligate, would never have been affirmed independently of these arguments, which is not even true for my own eliminativism about events.

The Problem of the Many

The second problem I have with common-sense events is that there is no way to non-arbitrarily individuate such events in the first place. It may seem that I have already shown this with the vagueness section, but I am making a slightly different point here. There I showed that once

⁴⁵ If this seems wrong, perhaps because we might measure something of a nano-second length, just imagine a duration one hundred times shorter (and so on).

we identify an event and suppose that it has identity conditions that permit some difference, paradox follows. Here I intend to show that identifying an event in the first place is paradoxical too. I show this by appealing to Peter Unger's 'problem of the many'⁴⁶ and showing how it applies not just to objects (as he has it) but to events too. I first outline the problem in the case of objects before showing how it applies to events.

The problem of the many purports to show that in a typical case of some object's existence, there are actually a great number of objects present. Take a cloud, for example. Suppose like our events, that it is of finite spatial extension, has a natural boundary, but that it is not divisible into further parts. It covers a perfectly determinate region of space. For simplicities sake, let us suppose it is made up of 'cloud-stuff' which is a homogenous substance. What is notable about clouds is that they seem to have indeterminate boundaries. It is vague where they begin and end due to the varied distribution of cloud-stuff. A great number of tiny regions of cloud-stuff can reasonably be included or excluded within the original boundary of the cloud without any absurdity. This is where the problem comes in.

In any typical case of a cloud, there are (at least) millions of other candidate regions that could also be recognised as a cloud. To see this, let us call the original cloud, cloud₁. Now consider another cloud, cloud₂. This cloud completely overlaps cloud₁ except for a miniscule region of cloud-stuff right on its boundary. If we accept cloud₁ as an object, why should we not also accept cloud₂ as an object? It has all the features typical of a cloud, indeed, it is essentially as typical a cloud as our typical cloud, cloud₁ is. But if we accept cloud₂ as an object then we would be committed to accepting millions of other clouds as present in any given cloud situation. This is because there are (at least) millions of other (mostly overlapping) regions that are just as permissible candidates for objecthood as cloud 1 or 2. I reproduce Unger's argument in full:⁴⁷

- (1) If there are clouds, then there are typical clouds.
- (2) If something is a (typical) cloud, then there is something that limits or bounds the cloud, that is, something that is the *boundary* of the cloud. [Otherwise, the cloud cannot be individuated as something against the rest of the universe.]
- (3) If something is the boundary of a typical cloud, then there are, in a *bounding envelope* centred on the boundary of any typical cloud, millions of other boundaries of clouds each of which limits, or bounds, a cloud that is different from any cloud thus bounded by any of the others.

The use of the term 'bounding envelope' is supposed to evoke an image of a kind of 'thick' region or boundary on the outside of our cloud that includes all of the permissible permutations of potential cloud boundaries. Some of them will differ in only the most minute way, while others might completely envelop those smaller than it.⁴⁸ Regardless, the possibilities here are

⁴⁶ Unger, Peter. "The Problem of the Many." *Midwest Studies in Philosophy* 5 (1980): 411-467.

⁴⁷ Unger, "The Problem of the Many", 419-422.

⁴⁸ You can read Unger's justification of his choice of words on pages 420-1. Nothing of substance hangs on it, it is only supposed to help one visualise the problem.

endless because in any region of cloud stuff we can always imagine a region smaller than any finite region considered, which can thus be taken to constitute a new boundary. Onwards:

- (4) If there are, in a bounding envelope centered on the boundary of any typical cloud, millions of other boundaries of clouds each of which limits, or bounds, a cloud that is different from any cloud thus bound by any of the others, then there are situations involving typical clouds, and in any of these there are millions of clouds.
- (5) If there are clouds, then there are situations involving typical clouds, and, in any of these, there are millions of clouds.

In disjunctive form:

- (6) Either there are no clouds at all, or else there are situations involving typical clouds, and, in any of these, there are millions of clouds.

The dilemma is as follows: either deny that there are clouds at all, or accept the absurd conclusion that in any given case of some typical cloud, there are millions of them. Two further things to note here. First, this works not just for clouds, but for all objects. This is because no thing whose constitution changes so minutely from another thing that is considered an object can rationally be held not to be an object. I will not defend the extension of this argument to all objects because it seems to be a losing battle to fight it. One would have to argue, for example, that clouds have indeterminate regions that admit of this problem, but pebbles or cars do not. This seems implausible as there are always miniscule regions of these objects that we take to be part of them where, without them, there is everything that is needed to constitute another typical instance of that object. Second, if you believe in atom-less gunk, that every object has proper parts (for example, everything could have half of itself as a part), since we can take smaller and smaller regions of whatever-object-stuff being excluded from the typical instance of that object as parts, *ad infinitum*, there would be an infinite rather than merely finite number of every object in any typical object-situation, if they existed. This will not be the case for atomists who think everything is made up of a finite number of simples, but the number will still be absurdly large.

The problem is that we are under considerable theoretical pressure once we accept any cloud whatsoever as an entity, to also accept those other entities that differ only minutely from it. Even if one insists that there is only one cloud there, the choice of regions will always be arbitrary between at least a million, and possibly an infinite number, of different choices. There is no principled way of selecting one rather than the other. But, as noted earlier in the case of vagueness, if we accept this arbitrary proliferation of common-sense objects, which is impossible to avoid on some plausible assumptions, almost all of them become both theoretically and explanatorily useless. Before I say more about why we should not accept such a view, let's move over to events.

As required by common sense events we have intervals like AB, BC, and CD of our world line. These events are of finite duration, and there are typical instances of them, such as cooking a carbonara or going to a concert. Each of them must have a boundary, a determinate beginning and end, and its correlative bounding envelope, but they will be temporal instead of spatial boundaries. It is easy to see how Unger's problem can be extended to such events. Once

again, we can assume that the objects are specified and that we are only dealing with a temporal problem (but it can easily be seen how the argument extends both to objects within a view like this, and to the four-dimensionalist view). I adapt the argument as follows:

- (1) If there are concerts, then there are typical concerts.
- (2) If something is a (typical) concert, then there is something that limits or bounds the concert, that is, something that is the *boundary* of the concert, the beginning and the end.
- (3) If there were no boundaries, an event cannot be individuated as something against the rest of the world-line, the entire history of the universe.
- (4) If something is the boundary of a typical concert, then there are, in a *bounding envelope* centred on the boundary of any typical concert, millions of other boundaries of concerts each of which limits, or bounds, a concert that is different from any concert thus bounded by any of the others.
- (5) If there are, in a bounding envelope centred on the boundary of any typical concert, millions of other boundaries of concerts each of which limits, or bounds, a concert that is different from any concert thus bound by any of the others, then there are situations involving typical concerts, and in any of these there are millions of concerts.
- (6) If there are concerts, then there are situations involving typical concerts, and, in any of these, there are millions of concerts.

In disjunctive form:

- (7) Either there are no concerts at all, or else there are situations involving typical concerts, and, in any of these, there are millions of concerts.

Thus, we are faced with the same dilemma – not only the same dilemma faced in the case of clouds, but that of which we faced in the case of vagueness: the endless proliferation of events, or their elimination. Further, just like in the case of objects, it can be shown that there are an infinite number of events in any situation, rather than merely millions. More on that below. First, why we should accept that there are no concerts, rather than that there are millions of them.

Once again, the dialectic is as follows. The defender of common-sense events assumes that there are typical cases of some event. Based on this assumption we can construct an argument, the problem of the many, showing that simply by positing that event, we are rationally required to posit at least a million, and possibly an infinite number of other events. To accept this conclusion would be post-hoc because the initial justification for accepting common-sense events in the first place is that there were natural seeming spatiotemporal boundaries by which they could be individuated. This is no longer the case as it is not natural at all to think that in any given situation involving an event, there are an infinite number of that kind of event overlapping with one another. And indeed, there is no reason, independent of the fact that we run into things like Unger's problem of the many that anyone would have ever formulated this idea. Once again, this is not to say that we should not respond to arguments by adjusting our theories, it just seems to me that once we adjust our theory in this way, the initial justification for going down the path of thinking these things exist at all, has slipped away. And

this suggests that we should reject our initial assumption that we could secure such entities at all, not that they must therefore be proliferated infinitely and arbitrarily. The response is once again objectionably post-hoc.

Perhaps it might be objected here that these events are posits that are necessary to explain different phenomena in all different kinds of inquiry and our everyday language use because they are the truth makers for the sentences making up theories and our everyday utterances. For example, cosmologists talk about events like ‘the big bang’, chemists talk about chemical reactions, and historians talk about events like ‘the French Revolution.’ Further, we surely know that sentences like “Rowan climbed Mount Taranaki” are true because there was an event wherein Rowan climbed Mount Taranaki. Without the actual event there to make such an utterance true, these theories and sentences become meaningless.

However, in all of these cases, I simply do not think that it is necessary that we must think there exists a common-sense event or mereological fusion over and above the given structure of the world for us to explain how these things constitute sensible utterances that are genuinely about the world, that can be understood by others, and even understood as being ‘true.’ I will not argue this point here as I do so in the final chapter (Chapter IX) of this essay. However, I take it that if my account there is successful, we have good reason here to go with no events, rather than many, as the many would not only undermine the original justification for picking out a typical case of some object in the first place, but also each and every one of them would be explanatorily superfluous. If this is also the case, then I can appeal to Ockham’s razor, as I discussed it in the previous chapter, as further justification for elimination over proliferation.

The moral of these arguments is that we should think that a kind of conventionalism is true and that the whole edifice of common-sense entities is an artifact of practical and not speculative life. I am prepared to accept what I have said so far as a successful attack on the existence of common-sense events (when combined with what I say later). However, the temporal version of this problem invites another issue that makes it fatal for the friend of events, that the spatial version never produced for objects. I will now spell this out in detail, including the derivation of an infinite rather than finite number of events in any given situation.

Each common-sense event must have determinate boundaries. That is, there must be a point where it begins and ends. These boundaries, where each event begins and ends, must occur at duration-less points. This is because, if there were no such points the events would never begin or end, and if the boundaries had some temporal extension, they would not actually bound the event. To see this, suppose there were no points where any given event begins and ends. If this were the case, the event could never start because there is no time at which it does, and even if it could, it would then never end, because there is no time at which it does. But common-sense events, if they exist, must at least begin, and will usually end (though it is possible that a common-sense event could not end). If they did not, we could not individuate them against the rest of the world-line, against the rest of the history of the universe.

Now suppose some event has a boundary with duration, rather than one without duration. Call the event A, and the boundary BC, which has a finite but miniscule duration.

There are two alternatives. Either the boundary BC is part of event A or it is not. If the boundary BC is part of event A, then it does not truly bound event A because point BC is not the end of event A, point C is, because everything up until that point is part of event A. All of the boundary BC, except its own boundary C exists before the end of both it and event AC. Thus, in this case, the boundary of A is just duration-less point C. If the boundary BC is not part of event A, then the event does not really end at the boundary BC, it ends at point B because no part of it is extended further in time. Thus, in this case, event A never had boundary BC in the first place, it just had duration-less boundary B. Thus, in all cases, events must have boundaries with no duration. That is, there is an instant of time at which every event begins and ends (if they begin and end).

However, this leads to a problem. Recall that I am supposing our world-line is made up of finite events only and no states. That is, the interval within any two boundaries is not made up of states but just overlapping events. However, if the events are overlapping, then there is an *infinite* (rather than merely finite) number of overlapping possible boundary points between any two other boundary points. This follows from the fact that they have no duration and that smaller durations can always be more finely discriminated over a continuous duration. Thus, Unger's problem produces not merely a great but finite number of candidate events, it produces an infinite number of candidate events in any given situation. This is because any allowance that there could be even one minutely alternative duration constituting a separate event, entails that there are an infinite number of almost exactly similar events nestled within the two boundary points of temporal difference. Thus, the disjunction is more severe, I conclude instead that:

- (8) Either there are no concerts at all, or else there are situations involving typical concerts, and, in any of these, there are an infinite number of concerts.

Obviously, once again, this argument extends to all candidate events and requires the defender of events to posit an infinite number of events in any given event situation. But importantly, this is not only an absurd and post-hoc conclusion, but it also invites Zeno back. Consider the following argument.

On the common-sense view of events, event AB occurs over some finite duration and has temporal boundaries A and B. But due to the lessons of Unger's problem of the many, there must be an infinite number of event boundaries between these points. Suppose that I wish to cross the road and that, if I were to succeed, AB would represent my journey. However, within interval AB there are an infinite number of other, smaller but almost exactly identical events. Thus, in order for me to complete my journey, I must pass through an infinite number of events. But as we established earlier, it is impossible to complete such a journey because, assuming it is not already existing as completed, a growing infinite has no end, by definition. Therefore, if this view is true, then I cannot cross the road. But I can cross the road. Therefore, (as before) either common-sense events do not exist, or all change is an illusion. Since change cannot be an illusion unless we relegate it to another world, we should say that common-sense events do not exist.

One could object that there is nothing objectionable about passing through so many events because, for the most part, they do overlap. Thus, when I begin completing my journey, it is not like I must complete them one by one. Rather, almost all of them are being completed simultaneously where an infinite number of such events will terminate once the boundary B is reached. Further, it will be said, on this view events are of a finite size. Thus, even if the world line were infinitely divisible as Unger's problem has led us to believe, there is nothing contradictory in supposing that infinitely many non-overlapping temporal parts add up in an unproblematic way to a finite size. On this view, the world-line would be like atom-less gunk. Thus, it is a fallacy to suppose that such a journey cannot be completed in a finite time. We can see this more easily in the case of a line segment. No matter how many times it is divided, even be it an infinite number of times, the sum of the lengths that it is divided into can never equal more than what we begun with. Thus, the event will have a certain duration AB and completing it will terminate, at the same time, an infinite number of other events.

However, this solution will not succeed. Careful attention to the case at hand will show this. Suppose I begin at point A. There are an infinite number of events that begin after point A but that also end at point B. I must pass every one of these points, infinite in number, on my first step. However, what does it mean to pass through an infinite number of things? This is the same as asking one to finish counting all the natural numbers, it is impossible. The simplicity of the line segment in the spatial case is what is tricking us here. We imagine the line segment to already be drawn, to already have happened, and thus that the sum of its infinitely divided segments unproblematically add up to the finite duration. However, as noted earlier, this begs the question against Zeno. It is to mistake a growing infinite, which is the consequence of a false theory of reality, with a standing infinite, which is an excellent tool for measuring empirical reality. It is to mistake the map for the territory.

If you suppose that some motion was already completed in some finite time, it is trivial that you can arbitrarily break it down into infinitely smaller segments and add it back up to the same duration. What I am saying is that we cannot presuppose that there already exists some ready-made and complete event AB, in answering the question of whether the task AB can ever be completed. We must start from the standpoint of an event yet to be fulfilled. But if we do that, then any task whatsoever becomes impossible to complete on this scheme of infinitely divisible finite events, because for any change to be completed, something must appear at an infinite number of different times. But to appear at an infinite number of points in a finite time is a conceptual impossibility because there is no end to a growing infinite. Thus, the common-sense theory of events, made problematic and infinitely ubiquitous by vagueness and Unger's problem of the many, contradicts something we know to be true: that there is change and motion at all. Therefore, it must be false.

In this case, one is forced back to eternalism. An eternalist could consistently reject my argument from Zeno's paradox and maintain the existence of common-sense events because they think all motion and change just are already completed relations between events. Since these events do have duration, unlike on the states-only view, they do not fail to derive duration from the duration-less. However, there is still considerable theoretical pressure to reject any

theory of common-sense events due to the two problems I have outlined above that are independent of this argument: vagueness and the problem of the many.

This concludes my discussion of the proposal that the world-line is composed of common-sense events, and no states. I have argued that we should reject that such entities exist at all. Given that I have already rejected the proposals whereby there are only states and events and states, we get the conclusion that *there are no states or events at all*. If this is right, we are left only with *the whole world-line itself*, where any carving up of it into intervals (events) or points (states) is an unreal abstraction from it. In Part II, I call this view *event monism*, the view that there is only a single undivided event making up the entire history of the universe. However, before I go forward, defend this view, and spell out its consequences, let's first see if atomic events can get around the problems raised.

Chapter V: Atomic Events (Events-Only)

We can avoid the conclusions in the previous chapter if we can show that instead of there being one big thing that everything else is an imagined fragment of (as I will advocate), there are many basic things that everything else is built out of.

I outline two theories of such events, as endorsed by David Hume and William James respectively. As I noted earlier, their view of events is that they are atomic. They think the world line is composed of fundamental temporal atoms. They are the substance of time. Both thinkers are similar in that they approach the problem as empiricists who have certain epistemological concerns regarding the nature of events. However, they also take somewhat different approaches to each other so it will be worth going over them in turn. It is worth noting that there are some going physical theories that advocate for temporal quanta that would be considered atomic events, but they are controversial. For this reason, few hold such a view. However, my approach here is to cover all the available conceptual ground in order to be exhaustive, to show what I think must be case. Thus, it is to see whether such quanta or atomic events are conceptually abhorrent, and thus inadequate for metaphysical theory in the first place. My discussion hopefully harbours some objections against such views as well. Onto Hume.

Hume's View

Hume's view of events is that they occur in a discrete sequence composed temporally of indivisible units of time.⁴⁹ What interests me here is not so much his discussion of the corresponding atomicity of space (that together jointly constitute events), but his discussion of time and how he derives its atomicity. Once we ascertain this, events for Hume are just the spatial configuration of objects plus their place in time. His view of time in *A Treatise of Human Nature* (henceforth the *Treatise*), has two main components: succession and indivisibility. The former is unique to time, while the latter applies to space as well.

Succession can be explained together by three points. First, as I noted earlier for atomic theories of events, times cannot overlap. He writes: "It is a property inseparable from time...that each of its parts succeeds another, and that none of them, however contiguous, can ever be co-existent. For the same reason, that the year 1737 cannot concur with the present year 1738, every moment must be distinct from, and posterior or antecedent to another."⁵⁰ Time unfolds in a discrete and sequential manner, where each event is contiguous. Between times A, B, C, and D, there is nothing else.

⁴⁹ I owe much of this section to the excellent discussions of Hume's theories in Baxter (1988), Jacquette (1996), and Esposito (2010).

⁵⁰ Hume, "A Treatise of Human Nature", 35.

Second, the idea of time is derived from the experience of the perceptible succession of objects or spatial states. To see this, imagine that you are being held completely still and a homogenous blue screen covers your entire visual field. Whether or not anything is changing in the object, no notion of time can be derived from this experience because there is no perceptible change. Now imagine a red ball being thrown through the centre and across your visual field. It is the perceptible change given by the successive states of the ball against the background that the notion of time is derived from. “Wherever we have no successive perceptions, we have no notion of time, even though there be a real succession in the objects...time cannot make its appearance to the mind, either alone, or attended with a steady unchangeable object, but is always discovered by some *perceivable* succession of changeable objects.”⁵¹

Third, and this is entailed by the second point, time is not conceivable without some real change. Hume thinks we cannot conceive of what it would be to have time pass without some real change – we cannot separate time from our successive perceptions. “Five notes played on a flute give us the impression and idea of time; though time be not a sixth impression”.⁵² This succession is one of the few things for Hume that we do not have to be sceptical about. It should be clear that this resembles my earlier discussion of succession in relation to the states-only view. Onto indivisibility.

Hume argues for the indivisibility of time in two ways. First, he argues that there is a phenomenological minimum of duration that the human understanding can experience. For example, we know that a film is given its appearance by a discrete juxtaposition of images, yet we experience it as continuous and smooth. The rate by which the images are projected onto the screen is too fast for human conception. Hume thinks that all of time and reality is like a film in this way. Obviously, this does not, by itself, prove that time must be indivisible. It only proves that the human mind has certain cognitive limitations. Thus, second, he argues for the logical impossibility of the infinite divisibility of time. However, Hume does use the former in his arguments for the latter. He constructs three arguments for his thesis of indivisibility.

Each of Hume’s arguments are in section 1.2.2 of the *Treatise*, I paraphrase them from there. His first argument attempts to show that if time is infinitely divisible there is actually no difference between a finite duration with infinite parts and an infinite duration with infinite parts. That is, both, owing to their infinite parts, would actually just have an infinite duration, which would be absurd. This is because any infinite sum of finite parts is also infinite. This is obviously fallacious. As discussed earlier, the sum of the parts of a line segment sub-divided infinitely never exceeds the initial line segment. Hume is only entitled to the conclusion based on his dubious phenomenological considerations. I will try to spell out his thinking though.

Here is the argument as he states it. Suppose time is infinitely divisible. Therefore, any duration must contain an infinite number of parts, otherwise the division would terminate somewhere. Thus, any finite duration must have an infinite number of parts. But if I take the smallest conceivable part of time (which is of some finite temporal extension) and assume that

⁵¹ Hume, “A Treatise of Human Nature”, 39.

⁵² Hume, “A Treatise of Human Nature”, 40.

howsoever I combine it with other parts, it must be a real quality of time, continually adding this smallest conceivable part of time to others *ad infinitum* would result in an infinite duration. This means that anything with an infinite number of parts has infinite length. Thus, if time is infinitely divisible, there is no difference between a finite and infinite duration, because they both have an infinite number of parts. This means that all finite durations are also infinite durations, which is absurd. If we follow Hume in supposing that there is a minimum temporal extent, we avoid this conclusion because any finite duration is simply composed of a finite number of parts.

The obvious fallacy seems to be that this argument posits the minimum phenomenological duration as a real constituent of time rather than an artifact of our cognitive apparatus. The response is that we may not be able to conceive of times shorter than this distance, but we can imagine that there are such parts, and that duration can undergo a process of infinite subdivision without therefore having parts that add up to an infinite duration, they would only add up to the same finite duration. Thus, Hume needs an argument that shows the logical necessity of indivisible rather than infinitely divisible temporal parts. Luckily, he has one at his disposal.

The second argument Hume offers is for the logical (and not merely cognitive) necessity of positing the existence of minimum indivisible moments or atomic events, in my terminology. The argument intends to show that if we supposed that time is infinitely divisible it would violate the succession of time, that event A must be entirely complete before event B comes about. Hume makes this point rather elliptically as follows:

It is certain then, that time, as it exists, must be composed of indivisible moments. For in time we could never arrive at an end of division, and if each moment, as it succeeds another, were not perfectly single and indivisible, there would be an infinite number of co-existent moments, or parts of time; which I believe will be allowed to be an arrant contradiction.

It is not clear from this passage why Hume thinks that the infinite divisibility of time leads to the co-existence of its parts rather than succession. However, to see why he might think this, we need only reflect on why infinite divisibility is incompatible with succession.

Remember, it is essential for succession that there is always a next moment of time e.g. A occurs, then B, then C, and so on. However, if time is infinitely divisible, it will be dense. And if time is dense, *there is never a next moment of time*. This is because, between any two points A and B, there is an infinite number of other points. This means that moments of time *cannot* exist in a succession because when events are dense, there is no succession. Assuming that time can only exist successively, as co-existent, or not at all, we must conclude that if time exists and is infinitely divisible, it must be in a state of co-existence rather than succession.⁵³

⁵³ It is remarkably interesting that once again we end up, from an entirely different direction, showing that eternalism is entailed by treating time as a geometrical line segment. The fact that co-existence contradicts the notion of time as Hume has defined it is precisely the argument McTaggart makes for its elimination in his classic paper “The Unreality of Time” (in the argument against the B-series, not the A-series). Hume rightly shows that we derived our notion of time from the change we actually perceive, and that co-existence is contrary

But this contradicts Hume's initial assumption of succession. Therefore, time is not infinitely divisible.

Perhaps this is fine. One might say: "So what if it turns out that time does not succeed in the way Hume thought it did, it is an outdated way of thinking about time anyway." However, Hume does have recourse to resources made available to himself through his phenomenological views about time. He could attempt to show that the experience of change would not be possible on the infinite divisibility view. From this, since we obviously experience change (once again, regardless of the mechanism or theory of mind), it must not be infinitely divisible. Let's see how this could work.

As I noted earlier, Hume thinks that in order to come to the notion of time we must experience some perceptible change between spatial states, for example, moments A, B, C, and D. This is what he calls succession. Now imagine again the blue wall with the red ball thrown into your phenomenal field. For us to experience the ball going from one side to the other, we must experience it as a sequence of moments succeeding each other. But if time is infinitely divisible, there is no such succession, because all states must co-exist. Therefore, if time is infinitely divisible it is impossible that we could either come to the idea of time or even experience change at all. Further, for there to be succession on this view, the ball would have to pass through an infinite number of states to even pass from A to B. But it is impossible to pass through an infinite number of cognitively minimum states, because it would take an infinite amount of time. Therefore, unless change and succession are illusions, time cannot be infinitely divisible. But the perception of change cannot be an illusion (and therefore neither can change itself) and we do have the idea of duration. Therefore, time is not infinitely divisible.

It is worth noting here that Hume's argument does not hinge on the cognitively limited minima of time constituting the world-line. Even if our cognitive apparatus had processing speeds that are a trillion times faster than they are now, the experience of succession would still be impossible if time is infinitely divisible, because no matter how fast our processing speeds are, there cannot be succession in an infinite series of events. Therefore, for Hume, it is a condition for the possibility of experience at all that there are indivisible temporal moments, because if there were no such moments, succession would be impossible. Thus, even if they are much smaller than our cognitive minimum, there must still be indivisible temporal moments.

I will omit discussion of the third argument regarding the reality of pluralities because it is compatible both with event atomism and event monism.⁵⁴ The important lesson is that Hume holds that all pluralities derive their existence from the parts in which they are composed. Thus, the world-line is composed of a discrete succession of moments. These moments and

to time. McTaggart agrees but wrongly takes that experience to be illusory. Both get it wrong by not taking duration (rather than succession or a particular analysis of change) as basic.

⁵⁴ Hume, "A Treatise of Human Nature", 34-35. This is the argument he attributes to "Mons. Malezieu." It is compatible with both because the main premise of the argument is that only indivisible things and not their collections or "unities" exist. This is precisely what the atomist and monist agree about. For the former, everything is resolvable into arrangements of indivisible atoms, for the latter, everything is resolvable into an indivisible whole.

their corresponding spatial arrangements constitute the events and thus the world-line for Hume. A further point to note is that putting forward a view that posits an atomistic discrete succession of events like this avoids almost all of the problems raised so far. It avoids Zeno's paradoxes since all durations have a finite number of indivisible parts, it accounts for change and motion through a real succession, and the spatiotemporal atoms are necessary posits of the theory because otherwise succession would be impossible. However, in spite of Hume's insistence that succession is accounted for in this theory, objections already raised will show that his account will not work.

One thing I haven't said about Hume's theory of time is that, while the parts are indivisible, they do not themselves have any duration. He does not say this explicitly, but it follows from his account of space, which is supposed to be directly analogous to time. For Hume, space is made up of extensionless points distributed and compounded in certain arrangements to form extension. He thinks this because if the ultimate parts of space had any extension, they would themselves have parts that are different from each other. For example, we could always imagine them halved.⁵⁵ Thus the parts of space have no extension. He is committed to the analogous position for time. He would argue as follows. Time is composed of parts. The ultimate parts must be indivisible (per the previous arguments), but they also cannot have any duration. If they did have some small but finite duration, they would be further divisible into parts. Therefore, the parts of time have no duration. This further means they have no internal differentiation; they are like a still of a film. Hume thinks these indivisible duration-less atoms of time are a "medium betwixt...infinite divisibility...and the non-entity of mathematical points"⁵⁶ that avoids the issues of both.

However, the problem with this is immediately obvious, as evidenced by the states-only chapter. It is impossible to derive duration from the duration-less. No matter how many atomic events succeed one another, if they have no duration, they cannot ever compose any duration. Of course, Hume could reply that all he means by duration is the succession of these duration-less states. However, there is a further problem that we could make by using Hume's own argument about succession. If all events are duration-less, and there are no intervals of duration between them, then there can be no duration between the occurrences of each event. But if there is no duration between the occurrence of each event, then they would all occur simultaneously and, in a single duration-less state of time. But if everything occurred simultaneously, the events would co-exist, rather than succeed. But, according to his own lights, there must be succession for us to experience change. Positing events with no duration, once again, makes succession impossible. Therefore, time cannot be made up of Hume's indivisible, duration-less events. And as it turns out, Hume didn't really hold a view of atomic events at all, he held something much closer to the states-only view. For the same reason we rejected that earlier, we must reject Hume's view now.

That being said, the discussion has been deeply illuminating, especially his second argument against the infinite divisibility of time, to the effect that if time were not indivisible, experience would not be possible at all because there could be no succession. This is very

⁵⁵ Hume, "A Treatise of Human Nature", 43.

⁵⁶ Hume, "A Treatise of Human Nature", 43.

similar to the argument I made in my rejection of the states-only view. Except I took a real duration to be the basic element that makes succession possible, rather than a real sequence of states.⁵⁷ I now move onto a better developed theory of atomic events.

James' View

William James holds a very similar view to Hume in that they are both sensitive to matters of experience. He wants, like Hume, to be able to resolve all of our conceptions back into elements of our experience. However, James will solve some of the problems generated by Hume's account by supposing that atomic events do have duration. This section will exposit and then evaluate his theory.

At the time of his writing, James thought that rationalistic metaphysics, especially absolute idealism, failed to explain being as it really must be. He thought this because any view that attempts to posit all of being as existing at once runs into insuperable difficulties, namely, that somehow everything we experience is an illusion. While he is speaking specifically of monism in this passage, it applies just as much (he thinks) to eternalism about time:

It contradicts the character of reality as perceptually experienced. Of our world, change seems an essential ingredient. There is history. There are novelties, struggles, losses, gains. But the world of the Absolute is represented as unchanging, eternal, or 'out of time,' and is foreign to our powers either of apprehension or of appreciation. Monism usually treats the sense-world as a mirage of illusion.⁵⁸

Whether this critique holds of eternalism I discuss later on. For now, it is enough to motivate his views. James also thinks that the mathematization or conceptualisation of reality, and especially of time, leads to all sorts of contradictions.

The common thread of James' critiques is always the same: that when we abstract away from concrete experience, from the interpenetrating flux we are the most intimately acquainted with, we argue ourselves into contradiction. He calls this tendency "conceptual translation", the attempt to reintegrate the static concepts we have formed from our flowing concrete experience, back into the real dynamic thing. It is "the vain attempt to reconvert the manifold into which our conception has resolved things, back into the continuum out of which it came."⁵⁹ The paradigmatic case of this is Zeno's paradox which, from the very beginning of philosophy, problematised the integration of our geometrical abstractions we take from experience, back into concrete reality. James sums up his view as follows:

If the aim of philosophy were taking full possession of all reality by the mind, then nothing short of the whole immediate perceptual experience could be the subject-matter

⁵⁷ This is also Reid's view, see footnote 20.

⁵⁸ James, *Some Problems of Philosophy*, 139.

⁵⁹ James, *Some Problems of Philosophy*, 91.

It is important to note here that James does not mean "continuum" in the mathematical sense. The mathematical continuum applied to temporal reality for him was utterly inadequate, as shown in the discussion of Zeno's paradoxes above. He means continuum in the intuitive sense of continually flowing.

of philosophy, for only in such experience is reality intimately and concretely found. But the philosopher although he is unable as a finite being to compass more than a few passing moments of such experience, is yet able to extend his knowledge beyond such moments by the ideal symbol of the other moments. He thus commands vicariously innumerable perceptions that are out of range. But the concepts by which he does this, being thin extracts from perception, are always insufficient representatives thereof; and, although they yield wider information, must never be treated after the rationalistic fashion, as if they gave a deeper quality of truth. The deeper features of reality are found only in perceptual experience.⁶⁰

Thus, we can think of his position as an inversion of Plato's view. Plato agrees that we perceive a world of becoming, but elevates the immobile concepts drawn out of this becoming into the ultimate reality, being. James does the reverse. He supposes the world of becoming we perceive is primary, while the concepts we derive from it are merely derivative realities used to describe actual, past, or possible perception, unfit to ever be fully re-inserted back into the world with complete adequacy. He writes:

[R]eality cannot be thus confined by a conceptual ringfence. It overflows, exceeds, and alters. It may turn into novelties, and can be known adequately only by following its singularities from moment to moment as our experience grows... We shall insist that as reality is created temporally day by day, concepts, although a magnificent sketch-map for showing us our bearings, can never fitly supersede perception, and that the 'eternal' systems which they form should least of all be regarded as realms of being to know which is a kind of knowing that casts the knowledge of particulars altogether into the shade."⁶¹

Of course, Plato agrees that the forms are unfit to precisely capture the world of becoming, but not because they are imperfect approximations as James thinks, but because they are perfect idealisations of the elements of that becoming. For each thinker, the others' direction of explanation is reversed. For James, all the concepts we form merely resemble and strive towards perceptual particularity. For Plato, all particulars in the world of becoming resemble and strive towards the eternal forms.

I do not have space to defend James' view at any length. However, as my forgoing discussion has shown, if we wish to save events and circumvent the issues I have raised so far, this seems to be a promising metaphysics. So, what does this point of view lead James to say about events? How would he suppose the world-line is composed?

From the forgoing discussion we can see James holds four views that relate to the following discussion of his events. First, temporal monism, the view that all of reality in time already exists, must be false. This is because if everything is given at once we cannot explain the perception of change and the reality of motion. Second, any view that treats time as infinitely divisible falls prey to Zeno's paradox and also cannot explain the perception of change or the reality of motion. These two views jointly entail an atomic theory of events since

⁶⁰ James, *Some Problems of Philosophy*, 97.

⁶¹ James, *Some Problems of Philosophy*, 99-100.

fundamental occurrences must be plural and indivisible. Third, his diagnosis of what went wrong in these considerations led him to affirm the ontological priority of percepts to concepts, where concepts are only generalisations that describe actual, past, or possible perceptions or perceived events. Thus, the primary reality for William James is a series of indivisible atomic events. However, these events, contrary to Hume, do have some duration. Thus, he has no problem composing the world-line or explaining succession. The world-line would be composed of events like AB, BC, and CD.

However, and this is the fourth, due to his rejection of monism and eternalism, it follows that the world line is always ‘incomplete.’ This is because, in order to explain the reality of motion and change, reality cannot be given all at once, but must be something that constantly grows. It must exist in time as we ordinarily conceive of it: as having a past that is past, a present which is now, and a future that does not exist but emerges from the present, in a piecemeal, atomic fashion. Thus, James affirms that there is *novelty*. Indeed, every occurrence is novel in this way, some being more predictable than others.⁶² “Time keeps budding into new moments, every one of which presents a content which in its individuality never was before and will never be again. Of no concrete bit of experience was an exact duplicate ever framed.”⁶³ To defend James on this point, experience does attest to this. Perception can never repeat itself. No matter how artificially similar two perceptions are, it will always be different because the perceiver must necessarily have changed since the previous perception, resulting in a different state of affairs. This is true even staring at an unchanging object over some arbitrarily small duration. Thus, all perceptions are novel in the sense that nothing exactly like it has happened or ever will happen again. If the world-line is incomplete the same thing can be said for non-perceptual events because they always necessarily have a different set of events preceding them.

The way this novelty comes about and emerges as the newest additions to the world-line is precisely in the form of atomic events. He describes these events as follows:

On the discontinuity-theory, time, change, etc., would grow by finite buds or drops, either nothing coming at all, or certain units of amount bursting into being ‘at a stroke.’ Every feature of the universe would on this view have a finite numerical constitution, Just as atoms, not half- or quarter-atoms are the minimum of matter that can be, and every finite amount of matter contains a finite number of atoms, so any amounts of time, space, change, etc., which we might assume would be composed of a finite number of minimal amounts of time, space, and change.⁶⁴

⁶² It is not clear that Hume followed James down this path. On the one hand, his rejection of necessary connection and his broad acceptance of a Newtonian succession of time seems to entail such a view. That is, owing to the looseness and separateness of Humean events, it follows that anything, and indeed things that are new, *could* happen – that’s the whole point of the discussion. However, many interpreters read him as a determinist. I personally don’t think he should be a determinist as his discussion seems to preclude any kind of determinism from being true. I cannot say myself, but I direct the reader to Harris (2005, 69) who writes “Hume...does not subscribe to determinism of any kind, whether Hobbesian or merely nomological.” And for a response, to Millican (2010).

⁶³ James, *Some Problems of Philosophy*, 148.

⁶⁴ James, *Some Problems of Philosophy*, 154.

Do not let his language fool you, he does not mean to say that we experience things literally as if it were a lethargic dripping from a just rained on wooden beam. Rather, it is more like an unbroken stream whose heterogeneous undulations bleed into each other, while nevertheless having real and distinct discrete phases.

Like Hume, James supposes that such atoms correspond to or are individuated in experience. We might be tempted to accuse him of transposing a mere cognitive function onto a wholly different underlying reality. However, he has a defence against this charge. As we saw, percepts are ontologically prior to concepts. Thus, the correct direction of explanation of reality for James is from experience to concepts, not the other way round. Concepts forced us into antinomies that led us to deny motion, change, and efficacy of experience altogether. Thus, it would be us who are wrong to object to that our scientific concepts are more real than our percepts. And it is these percepts that are the model for his piecemeal theory of reality. We might not accept these terms, but he is not making a simple fallacy. He writes:

Such a discrete composition is what actually obtains in our perceptual experience. We either perceive nothing, or something already there in sensible amount. This fact is what in psychology known as the law of the ‘threshold.’ Either your experience is of no content, of no change, or it is of a perceptible amount of content or change. Your acquaintance with reality grows literally by buds or drops of perception. Intellectually and on reflection, you can divide these into components, but as immediately given, they come totally or not at all.⁶⁵

The “threshold” he speaks of here is the “specious present.” The specious present is the variable duration which one can be aware of in time-consciousness. (James supposes it could last anywhere between a few milliseconds or as long or over 12 seconds. However, more realistically, we should only suppose that it is around a few milliseconds to a second long, maximum.) It differs from the duration-less present of Augustine, in that it consists of some objective duration. It is *both* present and temporally extended. Within this duration you have the immediate past, and the anticipation of the future at once. Further, each ‘moment’ within this duration (which is an unreal abstraction for James) melts into one another, it is utterly continuous within itself, breaking only at its edges, where it fades into memory or reaches beyond anticipation and into a non-existent future. Thus, unlike Hume’s indivisibly but static and internally undifferentiated atoms, James’ events are indivisible but qualitatively heterogeneous.

To conceive of this, think of a melody to a song. You do not seem to experience it in a duration-less, infinite discontinuity of notes. Rather, the last few notes played remain in your immediate awareness as new one’s push them out of your experience. Even though there is an earlier and later half of the specious present, it is not divisible, it remains discrete. If experience is discrete like this, since it is both an event amongst other events in the world and the model of all other events in the world, then all events are discontinuous but indivisible in this way.

⁶⁵ James, *Some Problems of Philosophy*, 154-55.

Obviously, a lot more would have to be worked out for this to become a satisfactory theory, however, the bare bones of it are there. I have hopefully showed how they would compose the world-line, and this is all I need to criticise the theory. Further, it should be obvious how this solves all of the problems so far. Motion and change occur during the finite durations of reality's drip. Novelty, the world-line's fundamental incompleteness, facilitates the drip and explains how we can experience this change. Finally, the atomicity of the events is necessary to explain how there can be novelty, and thus cannot be reduced to a more basic duration. Onto some criticisms.

There are a couple of problems with James' atomic understanding of events. First, they surreptitiously reintroduce instants into the world-line in a way that invites some of the problems faced by common sense-events. Second, even if they are superable, there is a kind of phenomenological vagueness to James' account.

The first problem with atomic theories of events is that they contradict themselves by reintroducing states into their ontology. This is something I have already noted in the case of common-sense events. It is entailed by the existence of some finite temporal interval that it has boundaries. That is to say, there must be a point of time where the event begins and where it ends. These boundaries cannot be of any duration, because if they were, something without duration must bound them too. States and events are logically co-emergent. That is, to posit one is to posit the other. Thus, like with common-sense events, we elicit a contradiction between these two theses:

- (1) There are only events are of finite duration.
- (2) If events of finite duration exist, duration-less events (states) must exist to bound them.

Of course, the friend of atomic events could accept (2) and reply by amending (1): "only those instants which bound the atoms necessary to make change and its experience possible." However, the sorts of problems raised regarding the vagueness of individuation come back here because there are an infinite number of imperceptibly different boundaries the atomic events could take on. We can see this if we consider in turn atomic events that can and then cannot overlap.

Atomic events must either sometimes overlap or never overlap. If one allows that atomic events can overlap *at all* then we would be under considerable theoretical pressure, due to the problems raised by Unger's problem of the many, to accept that there are an infinite number of atomic events overlapping one another in each case. This is because there will be an infinite number of minutely different events almost completely overlapping one another. But this means that if some novel event AB comes into being, then an infinite number of events would have to be created that differ only minutely from it, which is impossible. It would be impossible because the process of any event coming-to-be would be a growing infinite which James himself repudiates. But events do come to be. Thus, James would have to say that events never overlap and that instead they "tile" perfectly. But this invites a kind arbitrariness to where the atomic events are bounded.

If we suppose that atomic events do not overlap at all they must have a specific and sharp cut-off point that completely excludes all other events. Of course, on James' view, it is

perfectly natural that there is a non-arbitrary cut-off point where the present is gnawing into the future, but is there one in the past? Is it phenomenologically accurate to suppose that our experience drops a boundary down in the past such that no event ever overlaps? I do not think there is support for this conclusion empirically or phenomenologically. As an empirical matter, it is not enough merely to say that the boundaries are sharp because there is a cognitively minimum duration determined by the specious present, because it does not follow from the fact that we cannot experience some briefer duration that it cannot exist as an atomic event. It will always be plausible that there can exist some event briefer (or longer) than any so called 'minimum' by an arbitrarily small amount of time. Even if we broadly accepted James' approach to metaphysics and his view that we perceive a specious present, atomic events do not follow.

Phenomenologically, it seems to me the experience of change is one of a deep continuity without such stops. Sure, there are sharp qualitative transitions over some duration that seem to mark a sort of 'ending' to some phenomenological 'episode.' For example, my finishing writing this sentence and standing up to make a coffee. However, this relative qualitative discontinuity does not seem to mark some ontological dividing point, it is merely the internal diversification of an unbroken flow of awareness. It seems more likely from this point of view that there is no such experiential event structure, there is just the continuous interpenetration of change. When we listen to a song, we do not experience it in atomic phrases but as a continuous movement beginning before it starts and ending well into the silence that accompanies its termination. It is impossible to say precisely when some sound passes from specious present to past owing to the deep continuity between the two. The picture of drops does not seem compelling here either.

Put another way, if James allows that atomic events are of some non-overlapping duration, we can always ask of a particular event, both empirically and phenomenologically: "why not include one more or one less arbitrarily small finite segment of time to the present duration?" Like Hume, James thinks atomic events are conditions for the possibility of experience, but unlike Hume, James does not limit his atoms to a necessary and uniform length. His boundaries are contingent and elastic. Thus, this is an entirely appropriate question to ask him. And there is no satisfactory answer he can give.

Even if we supposed events were fixed unproblematically in this way, would there not still be a connection between any two succeeding events of a few milliseconds' duration. For example, take two events AB and BC. Suppose we are one third of the way through experiencing BC. Can we still retain the latter half or at least some of AB? What is their phenomenal connection? Surely some mediating specious present must hook into both events such that during some of BC, we are aware of some of AB. But that would mean specious presents overlap. But as noted earlier, if we allow that events overlap, we are forced to posit an infinite number of events being present in any finite duration, which leads to real growing infinities. But if no specious present mediates between them, then where each event begins the previous event is absolutely cut-off from present awareness. Right after present AB ends, even after an arbitrarily small duration at the beginning of BC, it is impossible that there could be any awareness of any of AB without overlap, which is absurd. (This is especially absurd for

James' since he firmly believes that the notion of a specious present is theoretically sound and that our 'stream of consciousness' is basically continuous.) Indeed, each event would be entirely separate streams of consciousness, inaccessible to one another.⁶⁶ Therefore, since James' atomic events can neither exclude each other nor overlap without collapsing into impossibility, they must not exist at all. And this is all according to James' own reasoning.

Even though I reject atomic events for the reasons given above, they do come close to solving some of the problems raised earlier. Furthermore, both Hume and James raise a problem for the view I arrived at earlier, event monism. Hume showed that without some real succession in time, the experience of change would not be possible. Indeed, this is what led him to posit his atomic theory in the first place. Similarly, James posed a challenge against monism and eternalism that must be answered. He argued that since these views suppose all of reality to exist at once, they cannot account for motion, change, and its perception. This is what led both thinkers to advance an atomic theory of events in the first place, that they play a transcendental role in making experience possible. Indeed, it was these considerations that led me to reject the states-only view right at the beginning of this investigation. Therefore, even if I reject atomic events, I must somehow incorporate these insights into the theory. Before going into more detail on such issues, I will take stock of the lessons and then draw some conclusions based on the entire preceding discussion.

⁶⁶ This argument combining the problem of the many and temporal consciousness undercuts the proposal in Dainton (2008) for an 'overlap model' of extensionalist temporal consciousness.

Chapter VI: Event Post-Mortem

So far, I have discussed four possible views of events and how they come together to compose reality in time, as represented by the world line. I began by discussing a view that supposed that the world-line is made up of merely discrete duration-less states. I marshalled two arguments against it: Zeno's paradox of The Dichotomy and its inability to explain the perception of change without relegating it to another world. Thus, I argued that the world-line is not composed of only states. The lesson of this chapter was that whatever time is, it must have some duration, otherwise there can be no experience of succession.

Next, I discussed a view that the world-line is composed of states and events. Here, I showed that any complete positive description of an event within the world-line would include everything that could be known at any instant within it. Thus, I concluded, through an appeal to a modified Ockham's razor, that all states in the world-line should be reduced to events, because they are explanatorily superfluous and overdetermine reality. The lesson of this chapter was that unless some entity does genuine explanatory work in our fundamental theory of reality, there is no non-arbitrary reason to suppose they exist, which is no reason at all.

Following this, I discussed a view that the world-line is composed of common-sense events, such as cooking a carbonara or going to a concert. I showed that acceptance of such events led to paradoxical and metaphysically suspect conclusions, including an infinite proliferation of events and the impossibility of change (for non-eternalists). The lesson of this chapter was that the way that we carve up events in time is conventional rather than metaphysical, unless one is willing to proliferate entities to absurd lengths and without the justification that motivated us to think they were there in the first place.

Finally, I discussed two views that world-line is made up of atomic events, events of a finite but indivisible spatiotemporal extent. I showed how certain aspects of them are deeply unsatisfactory or fail to avoid problems raised before that. However, what these accounts did reveal is that there are certain considerations of experience that must be taken aboard any theory of events. I tackle this problem head on after expounding my own view.

Part II:

The Primordial Event & The Eternal Present

Chapter VII: The Primordial Event

There is room...for a doctrine which regards homogeneous space and time as principles of division and of solidification introduced into the real with a view to action and not with a view to knowledge, which attributes to things a real duration and a real extensity, and which, in the end, sees the source of all difficulty no longer in that duration and in that extensity...but in the homogeneous space and time which we stretch out beneath them in order to divide the continuous, to fix the becoming, and provide our activity with points to which it can be applied.

– Henri Bergson, *Matter and Memory*

This part of the essay will outline my own view about how to construct the world-line. In this chapter, I introduce the views that are suggested by the discussion in Part I. I flesh out their details and defend some of their consequences, including a discussion of how I can have structure and order without division and how I explain the perception of change. By showing that my own view is immune to my previous attacks, I show that it is theoretically preferable to the other views discussed.

Event Monism: Introducing the Primordial Event

The considerations of Part I lead me to two conclusions. The first only has to do with the structure of the world-line, while the second is what this structure entails about time. The first conclusion is what I am calling *event monism*, which states that the world-line is whole and indivisible. It has no parts, while also having a real duration. Another way of saying this is that the entire history of the universe is a single indivisible event. I will be calling this event, which is constitutive of the entire history of the universe, *the primordial event*. I call it ‘primordial,’ because it is the duration necessarily prior to the temporally extended entities we think we are carving out of it.

This view follows from my discussion for three reasons. First, because a world-line constructed merely out of states means there would be no duration, which makes perception impossible. Second, because we found that we are never warranted in breaking up the order of events further into instantaneous states on top of that, because they overdetermine the world-line. Third, we found that carving the world-line up into events at all seems to commit us to premises that make individuation either arbitrary or infinitely profligate and post-hoc. I argued that this is true whether the events are common-sense or atomic and thus that they should be rejected. Thus, event monism and the primordial event is the only view that fulfils the conditions demanded by my discussion so far. It (1) countenances a real duration, (2) does not overdetermine the world-line and (3) has individuation that is neither arbitrary nor post-hoc (because there is none).

There are a few notable consequences of this. First, since I think that relationism about time is true,⁶⁷ it entails a further thesis about time. Namely, it entails what I am calling *temporal monism* (as opposed to temporal atomism). For a relationist about time, the order of time and its measure is indexed to the order of events. However, since there is only a single event on my view, there must therefore only be a single indivisible Time that corresponds to the world-line. This means that time is indivisible in the same sense that the primordial event is. Another way of saying this is that all of time is a single indivisible interval (in the same sense that the concept of the temporal atom is indivisible). Whether this means time exists at all is up for question. I think that on my view it still does, because the primordial event has a real duration which ‘temporal monism’ is supposed to capture, it is just that this time is unlike what we ordinarily call time, because it is actually indivisible. That is, there are no times, points at which things occur, but there is Time, which just is the universe’s duration.

Second, if objects exist, this event includes within it their entire existence. Indeed, if we are to be four-dimensionalists about objects, this straightforwardly entails existence monism, the view that there is only one concrete existent, the entire universe. Third, even though the world-line and time is undivided on this view and therefore that temporal entities to be ordered are, in some sense, unreal, there still must be a *qualitative* order within time that gives the particular character to its duration. As we will see, this is a feature of our experience and inquiry that cannot be denied. The primordial event is the entire, internally heterogeneous, undivided history of the universe. And it is this undivided unity that we conventionally carve our temporally extended entities out of.

For example, even though there is no ontological division between today and yesterday as independent temporal entities, it can still be said that there is a real qualitative succession or order between the features of the world we call yesterday and the features of the world we call today. This is intersubjectively verifiable in that people roughly agree that the same changes occur over some duration, we are not running around in complete confusion, and we are able to communicate successfully with each other by grouping these occurrences into pragmatic bundles within language and inquiry. This is compatible with some of our best scientific theories because this temporal order of qualitative becoming within the primordial event is absolute, that is, the same in all frames of reference, even though these features may appear to happen at different ‘times’ from different perspectives.⁶⁸ My contention is just that this structure cannot be actually divided in this dimension, it only *differs* along it. Along these lines of temporal difference, we graft onto reality our languages, theories, models, maps, and descriptions, to varying degrees of success. We are the cartographers of this difference.

⁶⁷ Unfortunately, I will have to assume that relationism is true for the rest of the essay. The section where I argued for it ended up on the cutting room floor due to space constraints.

⁶⁸ A similar point is made by Craig (1990) who takes a view similar to mine. The difference, as we will see, is that I do not think the primordial event has tensed properties because it cannot be individuated into past and present events to bear such properties. Also, in defence of this point, there many philosophers who think that monism of this sort is compatible with (or even suggested by) contemporary science. For a small sample of such views, see Horgan and Potrč (2008), Schaffer (2009) and (2010), Le Bihan (2016) and (2018), Perry (2017), and Builes (2021).

It might be objected that this proposal, of order and difference without division, is either incoherent or self-contradictory. However, I can show that it is not. This can be seen by showing that an analogous thing can plausibly be said about space and thus can analogously be said of time.

Order Without Division: Qualitative Structure & The Primordial Event

There can be order in space without division *or* internal relations. Take a complex painting like *The Garden of Earthly Delights* by Hieronymus Bosch. Now suppose that it is indivisible, that it does not have any parts. You can inspect it in all of its multifaceted brilliance, you can identify the seeming similarities and differences within its internal structure, you can speak of it as if it were broken up into units of being. However, nothing in its broad heterogeneity need imply that it is metaphysically divisible. All that it does have is a kind of *qualitative structure* such that its different aspects come together in a particular way to form it as *that* whole. To see this just take another painting and assume it to be indivisible. Both paintings are an indivisible spatial whole without any parts. This does not mean they are the same painting. Rather, what makes them distinct is their qualitative structure. This is what I mean by order without division. For sure, we can imagine that each person within the painting is its own independent whole, or that we can perform an operation of infinite subdivision upon the whole, but these operations really are just our imagination. At minimum, the possibility of performing these operations of division do not entail the actuality of their supposed objects. These objects have to be argued for. And if my arguments have succeeded, then any such argument will fail.

Thus, the same that can be said coherently of events and time once we project the idea into duration. The primordial event has a particular qualitative structure and internal differentiation that orders our universe within duration as *this* universe, the world we now inhabit. The things we call events are in fact real qualitatively differentiated phases of this duration (just as there are real aspects of spatial difference in an undivided painting), but not divisions within it. The term ‘phase’ is supposed to evoke the temporal but also the fact that time has profound and varied internal heterogeneity even though it has no joints carved within it. We can imagine this metaphorically as any continuous geometrical function in mathematics. The sine function has the properties of being whole, undivided, and having no discontinuities but also of having a particular character as opposed to say, a cosine function. The logic of this was already at play in the overdetermination section on states and events proposals. There I asked us to imagine an event with fictitious divisions within it, states. Then I proceeded to show that nothing, including the qualitative order of the event, is lost once we extract those fictions from it. The same is true here, just on a universal scale.

However, do not be fooled by the language. I am not even entitled to say these phases have ontological relations of “before” and “after” to each other because this implies their being an entity such that it can be related to other entities in that way. No such entities exist and indeed phases cannot exist. Thus, the terminology may be slightly misleading. However, make no mistake, the order I need *is* secured by the qualitative character of the world, in spite of its inability to divide. Logical problems with this only come about once we try to label such phases

with a variable, as if they were independent entities and not just an aspect of the primordial event. The facts of succession in experience and internal differentiation do not require and indeed cannot bear any such logical notation (as will be seen in the next section). The word 'phase' is just a way of *showing* you that the primordial event has both of these things, it has a purely imaginative function for the reader.

A further point to note is that sometimes views that posit more than simply the present, such as eternalism, are characterised as "static" worldviews. They are called static because they are views which construct the world-line out of unchanging states or timeless relations. However, this is entirely mistaken in the case at hand. It is precisely the opposite within the primordial event. It is undergoing a constant state of change such that at no 'point' or 'interval' along it can, in principle, be individuated against anything else in the universe. Thus, during any 'phase' of the universe, the world is in a constant state of becoming. The things we took to be events are just sharp qualitative transitions within universal becoming. To see this, imagine that some brief phase of the event is bounded by two states. Within this interval, there is nothing but unbroken and constant interpenetrating change. Such is the same for the rest of the world-line.

Transcendental Duration: The Perception of Change & The Primordial Event

At four points in this essay, I have referred to the experience of change and the necessity of some real succession or particular structure in the world that is required to guarantee it. I first discussed this in the context of the states-only view and showed that some real duration is necessary for the perception of change. Next, I discussed it in the context of the problem of the many which avoided the previous problem owing to the fact that common sense events have some duration. Finally, it came up twice in the discussion of atomic events. Both Hume and James thought that succession was necessary to explain the perception of change and that our theory of time must account for it. Hume was making essentially the same point as I did in the states-only chapter, though it later proved that Hume's own account could not satisfy this requirement. The solution to mine and Hume's problematic was that a real duration was necessary to posit in the world-line. James' view, however, is different.

James, as well as thinking that succession in time is necessary, further thinks that any kind of eternalism or monism about time cannot account for the perception of change because everything is already given as absolute and unchanging. This section will assess whether this argument is sound against the primordial event and conclude that it is not. I begin by affirming that the experience of change (succession) is a necessary explanandum of any metaphysical theory. Following this, I show how, against James, the primordial event can account for it. Further, I show how the same view can refute F.H. Bradley's anti-relational argument against the possibility of change. Doing this has the added benefit of showing how there is particular way of analysing change that will always be inadequate to reality. Here goes.

If we attend to perception properly, we cannot deny one thing. Namely, that: *there is change*. When Rowan crosses the road, we can be mistaken, even fundamentally, about whether any seeming *thing* (Rowan and the road) in our experience maps to reality, but we

cannot be mistaken that there is change. And for there to be a change in experience, there must be some corresponding change in reality, unless you hold a two-world theory where perception does not co-vary with reality. This is true even if the things we think are changing, such as the road or I, are entirely illusory; something is still changing. Even if it merely amounts to a change in our biological constitution or even merely within experience. Since we are beings embedded in and amongst the world, there is still change. If we were not embedded in the world, it is hard to see how there would be perception or experience at all to begin with. Further, even those who do think that experience is not strictly in this world, almost all ought to think it (even minimally) covaries with the body in such a way that some successive mental change corresponds to *some* (even non-brain-bound) successive physical change. In all cases, there still must be change. Importantly, it must also occur over some duration. This is because a change without duration, without some dimension in which to change, is a contradiction. Thus, there must be some real *succession* not merely in experience, but in the world. (See Chapter II for the full statement of this argument.)

James would argue that my view put forward here runs afoul of this datum of our experience. To quote him again on this point, he writes:

[The primordial event] contradicts the character of reality as perceptually experienced. Of our world, change seems an essential ingredient. There is history. There are novelties, struggles, losses, gains. But the world of the Absolute is represented as unchanging, eternal, or 'out of time,' and is foreign to our powers either of apprehension or of appreciation.

The argument here seems to be that because the structure of my event is given 'all at once' (because it cannot be divided), it cannot account for the perception of change. As already noted, this is because for there to be the perception of change, there must be some corresponding change in reality that has not already occurred. However, if everything is 'given at once,' reality does not change. Therefore, the perception of change must be an illusion.

This argument only works against some views, such as the states-only view, and some monistic doctrines (like F.H. Bradley's, who was one of the targets of James' attack) that abolishes duration entirely, that abolishes the dimension of time from reality. On such views, the perception of change could not be accounted for because everything must co-exist and co-occur, making succession impossible. As James' rightly points out, there must be a real succession change in the world that occur over some duration corresponding to the successive perception of it. Again, this is true regardless of the mechanisms by which one thinks perception is presented to the subject. But on the present view, there is still real duration, it is just not divided into intervals, states, or atomic events. It is a pure, undivided continuity unfolding over the total duration. This structure being given 'all at once' (because there is only one) is thus entirely compatible with the perception of change. This is because the perception of change is always correlative with and constituted by some real change in the world, because the world is in a constant state of change, and the perception of it is just one aspect of this same world.

To see this, imagine a universe whose entire content is made up by a single event lasting roughly 10 seconds. It is just a single room with a ball being thrown by one person on one end to another person on the other end. The entire structure of this world's primordial event being given in advance is perfectly compatible with each person perceiving a qualitative succession of change, as, at each supposed 'moment' they are present within some dynamically unfolding 'phase' of duration. By phase here I mean the real qualitatively differentiated structure of time. This succession will just not be divisible (and thus will have no parts). Indeed, since temporal monism abolishes the temporal individuality of each thing, each person, and their perception of change (assuming it is of this world) just are a particular configuration or phase of the entire universe's becoming. Change *and* its perception *just are* the existence of some phase of the primordial event, some local irregularity in the primordial event.

The difficulty in seeing how this is possible has always been in the application of states or intervals to time. Both Hume and James assume that a succession cannot occur without having the logical structure of ordered successive units that entirely exclude each other. This looks like A, B, C, D in Hume's case or AB, BC, CD, in James' case. However, once we get rid of this logical requirement of placing change or time into determinate units and instead posit a sub-representational becoming with internally differentiated but undivided phases, the paradoxes immediately dissolve.

We can see this exact dialectic unfold in the case of F.H. Bradley's argument against the reality of change. His argument illegitimately argues that the perception of change is an illusion because change is a relation between events. However, events, on my view, are fictitious carvings up of absolute temporal becoming.

Bradley's argument against the reality of change goes as follows.⁶⁹ For there to be change, it must either occur over either no duration, which is impossible, or it must occur over some duration and thus rely on relations, which are unintelligible. The former should already be familiar, as Hume and I have marshalled exactly the same argument against the states-only view. To rehearse it again: change requires succession. That is, for something, A, to change, it must become something else, B, such that A is before B. But if the change occurs over no duration, it cannot be change at all, because the terms co-exist, making their succession impossible. We saw that this was due to the stripping of duration from reality, which leaves no dimension for succession to occur across. "This is surely a flat contradiction."⁷⁰ Therefore, if change occurs over no duration, there can be no change. Here I agree with Bradley.

The latter, that change occurs over some duration, is unintelligible because any account of change must include relations, and they lead to an infinite vicious regress. If a change happens that has real duration, the changes must genuinely succeed each other and thus have a before and after. For there to be a relation of before and after, we must say that there is some event which can relate to some other event as "before." But then, if any x changes at all to any y, there must be some relation of order R, in the form of "before" and "after", relating them. This is because there is a difference between the mere existence of events x, y, and R, and the

⁶⁹ Bradley, Francis Herbert. *Appearance and Reality*. London: Swan Sonnenschein & Co., 1893. 45-49.

⁷⁰ Bradley, *Appearance and Reality*, 47.

relational fact xRy that denotes their succession. Bradley then argues that something must unify the relation to the relata. But this leads to an infinite regress because then something must unify the unifier relating the relation to the relata. Perhaps it would be replied that the succession xy , is brute and that no relation needs to be appealed to. But if you say this, you are rejecting that they succeeded each other, because then it is false that y is after x . If they do not succeed each other, there can be no change.

Perhaps then it would be replied that we can reject Bradley's premise that some further relation must relate the relation to the relata and say instead that the relation between the relational fact's constituents is brute and requires no further connection to the relata. However, we can see that this is not right because it would entail that the mere existence of the parts entails the existence of the whole. That is, since there is nothing unifying the constituents of the relational fact xRy , it is merely the sum of its parts. However, event x , event y , and the relation of order can all exist without the relational fact xRy existing. Most simply in the case of its reverse, yRx . Thus, some further relation must unify a given relational fact about change, and this leads to regress. If we say that the fact is somehow unified without such a relation, we are forced to hold the view that any given relational fact is both the sum of its parts (because that's all it can be analysed into), and not the sum of its parts (because it has some non-relational unifier that is a further part), which is contradictory.⁷¹ Assuming contradictions do not have any reality, it is impossible that change can occur over some duration, because change is contradictory.

Taken together, whether or not change occurs over some duration, it is impossible. Where does the argument go wrong? Change certainly does occur. As I note earlier, if there is one thing that cannot be denied it is that *there is change* or, at minimum, *we seem to perceive change*. And unless we think that perception occurs in another world than ours, the latter entails the former. Thus, we are going to have to provide an account of change within duration that is not defeated by Bradley's argument.

However, the solution is rather simple: the analysis of change Bradley gives is false. It is false for the simple reason that there are no events or states x and y that relate to each other in the first place. This is because, for there to be such events, the universe must be divisible in time, which, according to event monism, it is not. If there were such events, I think his argument succeeds, and is another point against the existence of events. However, his account does not exclude the possibility that there is one indivisible event, with a real duration, that is qualitatively heterogeneous.⁷² As I noted earlier, the fact of succession does not have to be

⁷¹ The relevant argument against the brute facts view is laid out cogently by Vallicella (2002), pages 11 and 20-21.

⁷² Granted, he does argue (like Hume) that you could not have a duration that is not divisible because you can always sub-divide it. He writes that "If the one time, necessary for change, means a single duration, that is self-contradictory, for no duration is single. The would-be unit falls asunder into endless plurality, in which it disappears. The pieces of duration, each containing a before and an after, are divided against themselves, and become mere relations of the illusory. (Page 46.)" However, my contention is that such subdivision is impossible. One wonders what the contradiction is here. Must a duration be resolved into a plurality of *smaller* parts? We both agree time must be indivisible, but why must the unit be a state rather than an event? There seems to be no conceptual abhorrence whatsoever in entertaining this (contra both Hume and now Bradley). Is the contradiction that something could both be internally heterogeneous but cannot be divided? But surely

relegated to the realm of contradiction and illusion, because change and the perception of it can be perfectly well explained as being an aspect or phase of universal becoming in the primordial event.

This is an improvement over Bradley's view in two ways. First, it accounts for the fact that there is change, which he argues is impossible. This is something I do not think can be denied, something that every metaphysics must account for. Second, it does not have to relegate the perception of change to the metaphysically suspect category of 'appearance', where no satisfying explanation or structural isomorphism can be posited to explain the relationship between the appearance of successive change and reality, which does not successively change. On my account, perception is an aspect of reality like anything else and the fact that there is constant change in perception is supported by the fact that reality (including our perception of it) is (in) a state of constant change. What we call change is just the universe's duration and our place in it.

Bradley's own Absolute has something of these two features. Indeed, he repeatedly characterises it as such. For example, he writes "The Absolute stands above, and not below, its internal distinctions. It does not eject them, but it includes them as elements in its fullness. (Page 472-73.)" Once we resist the urge to treat duration as a geometrical line segment and accept my argument that it is simple and indivisible, it stops becoming mysterious that we cannot subdivide it infinitely and stops forcing us to accept timeless Absolutes.

Chapter VIII: The Eternal Present

...the more we accustom ourselves to think and to perceive all things *sub specie durationis*, the more we plunge into real duration. And the more we immerse ourselves in it, the more we set ourselves back in the direction of the principle, though it be transcendent, in which we participate and whose eternity is not to be an eternity of immutability, but an eternity of life...

– Henri Bergson, *The Creative Mind*

This chapter will situate my own view as it relates to the current debates on time. First, as it relates to presentism, eternalism, and the to the occurrence/existence distinction. Second, as it relates to the A- and B-theories of time. Onto the first.

Occurrence/Existence & The Primordial Event

As I note in Part I, eternalists think that all of reality, what we normally consider the past, present, and future events making up the universe, exist. However, they do not all *occur* at the same time. Rather, there is an objective, tenseless, order of events given by a before/after relation which index each event to different times. Each event occurs in this order, and our experience of these events just is the occurrence of us within them. However, within the primordial event, there can be no distinction between occurrence and existence, because there are no distinctions between events that mean we can index their occurrence to some time. This is because there is only one event, and therefore no time indexes, in existence. We can see this with an argument.

Either some actually occurring event exists, or nothing does. If there were no actually occurring event no experience would be possible at all. That is, whatever *this* is, would not be occurring. But if some actually occurring event exists, there are no distinctions between it and the rest of the universe. This is because I have just determined that all that exists is a single event and that the boundaries drawn within it are always arbitrary derivations from a more basic duration. Therefore, the entire history of the universe must also exist and always be occurring. For the primordial event, existence is occurrence, because it cannot be divided in a way that can index occurrence to anything but its entire being. What we take to be the distinction is the pragmatic carving up of qualitative becoming. As I argued above, I do not think this poses a problem for the experience of change or its possibility in the first place, because it has a real duration.

This means that even though my proposal sounds like a kind of eternalism, it nonetheless rejects the occurrence/existence distinction essential to it. This is because that which exists, the primordial event, is also occurring. This strict connection between existence and occurrence aligns my view with the presentist, rather than the eternalist.

Presentism is the view that only the present exists. ‘Present’ could mean either that there is a single time that exists and at which everything occurs, or that the present just is the collection of occurring entities. In both cases, what constitutes ‘the present’ is constantly changing. And, importantly, on either construal, only that which is occurring, exists (at least for concrete entities). Thus, they reject the occurrence/existence distinction as well. However, there is a further distinction within versions of presentism that is important here. They differ essentially with respect to whether they grant the present some duration or not. In my terminology they would either be state-presentists, that is, those who conceive of the world-line as a single present state, A. Or event-presentists, those who conceive the world-line as a single present event, AB. State-presentism runs into essentially the same problems that the states-only views discussed in Part I does. Thus, it cannot be accepted. On the other hand, event-presentism has some real merit to it.⁷³ It says that all that exists is some duration that is presently occurring. This is exactly what I say too.

Thus, in a strange turn, I have to come to agree with the presentist, specifically the event-presentist: the only thing that exists is that which has some duration and is presently occurring. However, in this case, something like eternity (the entire history of the universe) is occurring, but it is occurring presently. Thus, the eternal is present, and the present is eternal. *The primordial event is the eternal present.* The primordial event is an *eternity of duration*. This means that both presentism and eternalism have some truth in them.

Presentists (both state and event) are right to restrict existence to occurrence in the present but failed to appreciate its breadth by arbitrarily drawing a line between what counts as that present and what no longer exists. Therefore, both presentists agree with temporal monism that the world-line must be indivisible, but state-presentists disagree that there is duration at all and event-presentists disagree that it is so wide. Indeed, that is all my view is: event-presentism inflated to capture the entire history of the universe.⁷⁴ Eternalists are right to extend existence beyond a mere state, chronon, or common-sense event and instead to the entire universe in duration, but are wrong to suppose that the universe could be structured in a discrete quantitative way that lets us index the occurrence of events to some time. The primordial duration of the universe is prior to the metrical concepts we stretch over it. Therefore, eternalists disagree with event monism that the world-line is indivisible, as they accept that there are events. However, they agree that the world-line is constructed out of a real duration, constitutive of the history of the universe. By this route, the debate is made soluble by combining both views into one. Albeit, in a way neither side would likely accept!⁷⁵

⁷³ For a discussion of these views see Appendix I on the problems with presentism. It is not essential to my argument to outline them here, but I do reject both views, mostly for reasons already gone over in Part I.

⁷⁴ This is what Hestevold (2008) calls ‘unlimitedly thick, time-free presentism’ (see pages 331-333), which they reject. They reject it because we can generate an argument where events from (what we ordinarily call) the past and the future both exist and do not exist. Since I do not think there are such events, the argument does not touch my own view. Furthermore, I think that the view they accept, ‘limited time-free thick presentism’, will have individuation problems with regards to what is rightly considered the present. Unfortunately, I do not have space to get into these problems, but hopefully the above discussion of event-presentism sufficiently covers it.

⁷⁵ Interesting to note that Craig (2000) accepts a view similar though (seemingly) not identical to this. He is a presentist who thinks that the present is a mode of being that does involve metrical ideas. Namely, ideas that the present has some extent, in the quantitative sense, to it. I agree in the sense that I think our quantitative

Superseding the A-and B-Theories

I will now situate my views within the A-B theory debate. The idea of an A and B theory of time comes from McTaggart's classic paper "The Unreality of Time."⁷⁶ A proponent of a B-theory of time thinks that all that exists is a series of events ordered by a before/after relation. On this view there is no privileged time that counts as the past, present, or future, there are only locations within this order. This differs in no essential way from how I have been discussing eternalism. However, the key to it being a B-theory, rather than the A-theory (or merely eternalism), is that no events are ever 'tensed' as past, present, or future (even though there is an occurrence/existence distinction). A proponent of an A-theory of time thinks that events are still ordered, but that they also have 'tensed' properties. What this means is that any given event on the world-line, say, CD, will at some point have the property 'future', then 'present', and then 'past.' The impetus for believing this is, first of all, that it seems to capture common-sense. It seems we ordinarily think that some event will happen, then it does, then it becomes past. Further, proponents of an A-theory think that it accounts for ineliminable aspects of our experience that the B-theory cannot. Before getting into this, I will outline McTaggart's argument.

McTaggart's argument is that *neither* the A nor B series is sufficient to capture time. Therefore, time is unreal. His argument runs as follows. First, it is an essential feature of time that something changes. This is because if nothing changed, some time could not really have been said to 'pass', because any given moment is completely unchanging. Therefore, a universe in which nothing changes, would be a universe that is timeless. However, if the B-theory is true, each event exists, exactly as it is, ordered into a series of before/after relations. Therefore, in this universe, nothing actually changes, and time is therefore unreal. Perhaps it could be objected that "the change consisted in the fact that an event ceased to be an event, while another event began to be an event?"⁷⁷ But this is impossible according to the B-theory because an event never ceases to exist, its relations (exist as) timeless. Therefore, an actual change must

understanding of time comes later than a more basic duration that makes that understanding possible. He writes that "there is no minimal temporal interval which is now" because once a finite duration is specified, it can be arbitrarily divided to be made smaller and smaller, and that this is merely a potential, not actual, divisibility. Agreed. However, it seems like this process could not only get arbitrarily smaller through division, but also bigger through the inclusion of some further duration. That is to say: what warranted us in selecting what counts as the 'present' duration in the first place? Carry this process on indefinitely and you get the primordial event, the only non-arbitrary specification of duration: all of it. Craig would reply that I am still applying metrical concepts to duration, which is where the problems arose in the first place. However, it is not clear on his view how he even specifies what the present is supposed to be. If you do specify it, you are applying metrical concepts to it but if you do not then the present just is duration. But what is 'duration' supposed to include for him? Given he's a presentist, it does not seem to be what I am advocating. However, both of us do seem to think that there is something called duration that is prior to any conceptualisation of time, and I agree with all of his arguments against other views. I think my view is superior if only because it actually works out the consequences of taking duration to be primary by not arbitrarily individuating some extent of time as the present without explaining what, how, or why.

⁷⁶ McTaggart, J. Ellis. "The Unreality of Time." *Mind* 17, no. 68 (1908): 457-474.

⁷⁷ McTaggart, "The Unreality of Time", 459.

somehow happen to the existing event such that we can say it is different at one time than it is at another. This is where the A-theory comes in.

Suppose the universe is an ordered set of B-relations but each event is also described by the properties ‘past’, ‘present’, and ‘future’ at a given time. These are A-relations. If these A-relations can unproblematically be applied to the series of events, the A-theory of time might be true. On this view there can be the kind of change McTaggart wants because at one time an event is present and at another time it is past, which means that events themselves do actually change. Therefore, we can say that some change occurs between C and D because C was once present and now is past and D was once future and is now present. However, there is a problem. First, A-relations must be something outside the B-series. This is because they are changing relations and as we determined, the B-series is unchanging. Therefore, they must apply to the events in some way independently of their order. The problem is that once this is admitted, we must also admit that any given event has contradictory properties. This is because each event must have the properties past, present, and future, which are each mutually incompatible. And assuming contradictions can have no reality, an A-series cannot exist.

The obvious reply at this point is to say that each event does not instantiate each of these properties simultaneously. Rather, each event instantiates them at different times. However, this begs the question. To say that each event instantiates the A-relations at different times is to invoke precisely the thing that is trying to be explained: time. Nor does it help to say that any given event has the mutually compatible properties ‘is present’, ‘has been future’, and ‘will be past’, because in order for these to be true of the original A-series, there must be another A-series that is equally as contradictory and therefore in need of explanation as the first. For example, suppose CD has these properties. To say truly that it ‘was future’ means that it was, or more precisely, *is, ((future) in the past)*. However, this same event *is also ((past) in the past)* which is another contradiction. Thus, the second-order predicates adduced to prevent contradiction only proliferate it. The attempt to avoid contradiction in the A-series this way leads to an infinite regress of contradictions, which is unacceptable. Therefore, an A-series cannot exist.

Before discussing my own view, I will offer some brief remarks on what people often take to be the difference between the two theories, supposing they can be formulated consistently. Proponents of the A-theory often argue that temporal passage is an obvious aspect of experience and common-sense, but also that there are certain truths that are irreducibly tensed. They then argue that only the A-theory is sufficient to the task of explaining such features of the world. I agree that the A-theory seems to best capture much of common-sense and the many seemingly tensed truths, but I disagree that only the A-theory is able to adequately capture the experience of objective temporal passage. In fact, I think that many versions of the A- and B-theories are experientially equivalent.

What my arguments have shown so far is that what is necessary for the experience of the passage of time is not that existing events have different properties at different times that only A-relations can provide (‘change’ in the sense that McTaggart demands). Rather, it is only necessary that there is some qualitative structure to the world-line in time that can be

successively experienced.⁷⁸ And all that is necessary for qualitative structure and succession is that the universe has some real duration. Many A- and B-theories accept that there is, at minimum, an ordered series of events. What I am arguing is that the events simply existing in this order is sufficient to explain the experience of change because the experience of change just is a part of the ordered occurrence of those events. That is, assuming that this order is an order of *events*, rather than *states*. This means that ‘thick’ A-theories, such as event-presentism, A-theoretic growing block, and the moving spotlight view are all experientially equivalent to B-theories like eternalism and B-theoretic growing block. (Namely, any non-state-presentist A-theories, which make experience impossible.) However, in my view (even without my arguments against presentism in Appendix I) this removes much (but not all) of the impetus to endorse an A-theory in the first place. Especially since I am not that concerned with common-sense. But I digress.

I won’t discuss the many other problems one might have with this argument, nor the many interesting avenues and contours McTaggart explores in his paper. I will only discuss how I purport to solve these particular problems with my own view, as they have had an outsized and lasting effect on the philosophy of time. Thus, doing so will situate my views within these debates. The primary difference between the A- and B-theories and my own is that both sides assume that time is either divisible or made up of events and/or states that differ from one another numerically. This is even true for presentists who think that there is a real succession of numerically different states, even though only one of them can exist at a time (what I call *perishing state-presentism* in Appendix I). McTaggart writes at the beginning of his paper when he is laying out his assumptions that:

I shall speak of the series of positions running from the far past through the near past to the present, and then from the present to the near future and the far future, as the A series. The series of positions which runs from earlier to later I shall call the B series. The contents of a position in time are called events. The contents of a single position are admitted to be properly called a plurality of events [because many things will be happening at any given time]. (I believe, however, that they can *as truly*, though not *more truly*, be called a single event...) A position in time is called a moment. The first question which we must consider is whether it is essential to the reality of time that its events should form an A series as well as a B series.⁷⁹

His description should be reminiscent of the world-line that I have been discussing. But in formulating it as such, McTaggart simply presupposes that there are such things as events and states that in the first place could be constitutive of it. However, that is precisely what I have called into question. McTaggart’s direction of explanation is the reverse of mine. He goes from events and states as given, to time or order as ‘merely subjective’ or (falsely) inferred from experience. But he (and the whole debate) has it the wrong way around.

⁷⁸ This is also the view expressed in both Williams (1998) and Deng (2010) (which is a critique of the former) who writes that “the right way to resolve this tension is by reserving the label ‘passage’ for the process that is included in the B-theory simply by virtue of its including succession. (Page 747.)” I wholeheartedly agree.

⁷⁹ McTaggart, “The Unreality of Time”, 458.

My argument has been that duration is given (to account for the perception of change) and that we would have to go from there to events – and further that there are no events to be found. Thus, according to temporal monism, what is essential to time is that it be an indivisible whole and have a real duration, rather than that it forms an A- or even a B-series at all. Such a series cannot even be constructed in the first place, because all there is, is the primordial event and these series require a plurality of events, not just one.

Therefore, temporal monism avoids his argument against the reality of time for the A-series because it does not presuppose that temporal entities, whether they be states or events, can be carved out of duration in the first place such that a contradiction can be elicited from mutually incompatible temporal properties. The primordial event is wholly present and therefore doesn't have such temporal properties, it just exists. It also avoids his argument against the B-series because change in the sense that he demands it is not necessary for time (though this is merely terminological), nor (more importantly) for the experience of passage. The only thing necessary for time is that there be some real duration and therefore some qualitative structure that constitutes temporal becoming. This is precisely what the primordial event has. This thereby circumvents the entire framework of A- and B-theories of time, as well as the debate about tense, by rejecting a supposition they both hold. Namely, that there is a plurality of events that constitute the world in time. By denying that there are such events, everything is present, but still ordered such that passage is experienced in the way that it is. Indeed, on this view, passage itself, the temporal becoming of duration, is the primary stuff of the universe.

Chapter IX: Concepts of Qualitative Cartography

But if we are all one, can there be anything to say, anything to refer to? ...Rather than moving from anywhere to anywhere, then, let us just go along with “thisness,” relying on the rightness of whatever is before us as the present “this”...For courses have never had any sealed boundaries between them, and words have never had any constant range. It is by establishing definitions of what is “this,” what is “right,” that boundaries are made.

– Zhuang Zhou, *Zhuangzi*

In this chapter, I close the essay by outlining an alternative vision for conceiving of the entities in time based not on ontological commitment to mereological sums or common-sense entities, but on pragmatic commitment to *mapping* the real but undivided features of reality.

The last thing I would like to discuss is regarding the individuation of entities in time and of explaining our practices regarding them. First, suppose a red ball is painted green. If my view is correct, there is no determinate time, and therefore no fact of the matter, when the red ball actually became a green ball. There is no tensed or tenseless fact of the matter about either the ball being red or it being green, at any time whatsoever because, there are no times whatsoever. There is only an eternally present duration that admits of no division. Further, there is no time when the ball itself came into being from whatever elements it (seemed to be) before that, because there can be no moment of its conception, because there are no moments.

Remember, there is no order in the sense of before/after relations within the primordial event, it just has a qualitative structure without division. The problem is that, once we admit that the world is in a constant state of absolute becoming with no determinate breaks, the entities seemingly within it slip away. With no real resting points in time, there is no longer any beginnings and endings to any particular entity. The problem can be stated more precisely as follows.

For something to come-to-be it must come-to-be at some time t_1 and for something to cease-to-be, it must cease-to-be at some time t_2 . Similarly, for some event to have truly lasted some determinate duration of time D , there has to be some real interval, D , we can point to. But there are no times, and there are no intervals other than the entire history of the universe. This problem generalises to all entities that seem to be in time, objects, substances, organisms, and persons included. The conclusion we are forced into this: nothing can be created or destroyed, because if something were created or destroyed, it would have to be created or destroyed at some time. Therefore, if something does have existence that is independent of or over and above the basic qualitative structure of the primordial event, it must exist eternally and presently, i.e., throughout the entire primordial event.

This means that the options for allowing the existence of other entities within the primordial event are as follows. We could either say that all entities exist as wholly present throughout the entire history of duration (because it is one), or we could say that they do not exist at all and that what we ordinarily take to be entities are merely complex or interesting aspects of universal becoming in the primordial event, that are not truly individual at all, even though they seem to be. Note that the former view is not a kind of Platonism where there are eternal abstract objects that are variously concretely instantiated in reality. Rather, it is the view that all concrete objects actually exist eternally and are neither created nor destroyed. I think there is a strong case for the latter view. I give three reasons.

First, it seems absurd to say that the entity we call Rowan exists omni-presently and omni-occurently, even since the early phases of the primordial event's duration and long into the later phases. This would mean that I existed millions of (what we call) years before the when we ordinarily think that I was born, namely, (what we call) the year 2000. In what sense am I existing alongside dinosaurs? In what sense are dinosaurs existing alongside us? I cannot have been in some merely nascent or potential existence because when we ordinarily think dinosaurs were around is just as 'present' as 'this' phase is. That is to say, they are the same eternal present. However, absurdity is no real barrier to accepting a theory. This conclusion could still be accepted.

The real reason we should reject this view is because it is (1) entirely uninformative about the seeming world and (2) it is post-hoc. First, saying that there is an entity called Rowan that exists eternally reveals absolutely nothing about the actual, particular, seemingly fine-grained, qualitative structure of the world. It says nothing about the finite aspects we actually mean to describe when we talk as if there were entities like Rowan or concerts. On no normal or useful construal of Rowan is he an object that has concretely existed throughout the entire history of the universe, was never created, and can never be destroyed. Further, since all concrete objects would be omni-present and omni-occurent within the primordial event, it explains none of our actual behaviour, which is always directed towards seemingly finite, unstable, and precarious existents, not things that are eternally present alongside duration. Such objects are uninformative.

Second, the initial justification for postulating objects as existents was precisely to capture these finite, perishing aspects of reality. Thus, given that in this theory, the entities postulated no longer line up with the initial targets of our explanations, accepting this conclusion would also be post-hoc. These dual problems of the theory being uninformative and post-hoc are obviously entwined. Both of them come from the fact that postulating such entities would be entirely against the spirit they were first postulated and done so without their initial justification.

Third, since most of my arguments against the divisibility (but for the substantiality) of time apply equally to the divisibility of space. We have good reason to reject the existence of spatial entities as well as temporal ones. Both the argument from the vagueness of composition and Unger's problem of the many equally apply to them. This means not only that all entities will be omnipresent, per my last objection, but that there will be an infinite number of them. Therefore, we should reject the existence of them entirely. However, these objections may only

be convincing if I sketch a compelling picture of what I think an alternative could look like. I intend to do just that.

I will give a brief birds-eye overview of my proposal here and then go on to explain what I mean by each aspect afterwards. Here goes. Instead of treating our concepts and language that purport to describe the world as joint carving, we should instead treat them like maps that model the world in better or worse ways, for different purposes. I call this process *qualitative cartography* and the relationship of better or worse (for some purpose) *fit-to-world*. By doing qualitative cartography, we are picking out what are actual *meta-stabilities* in the particular character of the primordial event. The basic idea is that we should take ourselves not to be butchers that carve nature's joints, because nature has no joints. Rather, we should take ourselves to be cartographers that produce maps of the world bearing specific information about a particular aspect of it, for some purpose.⁸⁰ I use the metaphor of cartography for two reasons.

First, because maps are ontologically innocent. They describe without dividing. Just because we draw boundaries and divisions into our maps, does not mean there is really something out there corresponding to them precisely. Rather, they convey information to the reader about the qualitative character of the landscape, using divisions as informational aids. For example, among other things, a map might depict changes in altitude, whether there are hills or valleys, changes in the environment, like whether a place is arid or snowy, and socio-political aspects, such as towns, cities, and borders. But just because a map is marked with a mountain or city, does not mean there is a fact of the matter in the world about precisely where those boundaries are in space. Nor is the map any worse for the non-existence of those boundaries, which brings me to my second point.

In spite of their ontological innocence, maps can still be truly informative and about the world in a meaningful way, even while representing the world as carved in ways it cannot really be. Indeed, they are *rooted in reality* in a way that they are obviously better or worse, depending on how well it depicts the character of the particular aspect it is supposed to be mapping. For example, a map of (the region) Taranaki with no mountain would be a shocking omission and a failure to capture the character of the area, even if there is no such thing as the mereological or common-sense entity Mount Taranaki. They can also be better or worse fit to some purpose. For example, a map of the entire world has a different purpose to an altitude map of Mount Taranaki. One could not climb to the peak of Taranaki with the map of the world, but nor could they sail around the world with the altitude map. We can tell whether a particular map is rooted in reality or fit-for-purpose by simply going out in the world and looking at it or trying to complete some end and seeing how well we are served by it. A map has a better *fit-to-world* insofar as it is commensurate with either our observations of its subject matter or our aims, in the broadest possible sense. We do not need there to be any mereological

⁸⁰ The idea is not far dissimilar to Builes' (2021, 16-20) suggestion that the universe just is a single property, the "World Quality", and that we can speak of 'aspects' of that quality as 'higher-order properties' of it, without also reifying them into entities in their own right. Something similar to this is what I suggested earlier with my proposed use of 'phase' and will suggest now with my use of 'meta-stability.' Ultimately, I develop the view slightly differently, but there are similarities.

sums picked out by a map that makes it ‘true’ or useful. In fact, I fail to see what difference this would make to a map at all.

To see this, imagine two worlds. In one we have the region Taranaki, where Mount Taranaki is an entity in its own right. It is something over and above the particular character of this aspect of the world. In the other, the particular character of the world is the same, but there is no object, Mount Taranaki, over and above it. Thus, the world is still arranged (what we call) “Mount Taranaki-wise”, but there is no entity Mount Taranaki. Now suppose that in each world an identical cartographer wishes to make a map of the region. Suppose further that the only difference between them is that they are aware of the ontological situation in their world. The first cartographer knows that in their world there is an object Mount Taranaki, and the second knows that in their world there is no Mount Taranaki. What would be different about the maps that they make?

I cannot see that anything would be different. Indeed, both cartographers would produce an identical map, in spite the lack of entities in the second world. Both would label the centre of the region ‘Mount Taranaki’, even the second who believes there is no such entity. Does this mean the former map is more accurate or true than the latter? I see no reason to think so. Both maps succeed in representing their particular situation to the best of their ability, both convey real information about their world, and both maps rise and fall, with regards to their ‘truth’, insofar as they adhere to observations made about the changing state of the environment they inhabit. Thus, the one is only as good as the other, completely independent of the mereological or common-sense constitution of the world.

What I call *qualitative cartography* is just a cartography generalised to all forms of inquiry. We should reconceive *all* of our attempts to describe and capture the world to have the same structure as the cartography I describe. That is to say, all inquiry is ontologically innocent, but also has a meaningful fit-to-world. Whether it is through the progressive elucidation of language, talk of objects, events, and processes, the positing of unobservables to explain and predict the qualitative structure of future becoming, or the mathematical models we use to describe the temporal unfolding of the universe and all of its supposed denizens; each of these forms of inquiry should be taken not to be joint carving, but cartographic. For example, the map that captures Rowan best will be the one that describes the aspect of the primordial event we ordinarily call 2000 and will be stretched over a duration that goes at least as far as what we ordinarily call 2022, including all of the events we might ordinarily attribute to his life during that time. Just as we saw with the story of the cartographers, whether or not any of these entities posited are in the world is beside the point. Their validity is completely independent of that question. All they are held accountable to is the particular structure of the primordial event.

This process is commensurate with the conclusions I have reached regarding time, events, and objects by being metaphysically non-committal. Yet it is also properly rooted in the world in a way that explains our practices as practices that are *truly about* world, even though the divisions we necessarily (for purposes of inquiry and action) carve within it are not really there. Further, it seems preferable to the only alternative: the omni-present, omni-occurrent, infinitely proliferated, arbitrary, and uninformative objects discussed earlier. However, I need to further defend my second claim here. You might still wonder: how can

something be about something else if there is no such thing? How can something be about Rowan, Mount Taranaki, or the movement of the planets, if there are no such things as persons, mountains, or planets? This brings me to my next point.

Even though we are in a constant state of absolute temporal becoming, even though there is no real rest within duration and therefore nothing that could come-to-be or perish, there is real stability to some of the qualitative features of the world. I think that this idea of stability can explain why we ordinarily think there are independent and determinate entities. My account will not ultimately be describing individual entities because stable qualities are not really independent entities at all, they are aspects of a whole. However, it will still be a real description of the world, and it does explain the seeming individualities in it. It does this in the same way a map describes a particular landscape without committing us to the existence of the entities it uses to describe that landscape.

I call some aspect of the primordial event *meta-stable* if it retains qualitative continuity over some duration. I call it *meta-stable* because nothing can be absolutely stable, as everything is in a constant state of becoming. However, within absolute becoming, some aspects can be relatively more stable than others (at least to our own human way of seeing the world), making them meta-stable. They are meta-stable if they exhibit qualitative continuity, by which I mean they have a kind of intelligible unity that persists over time, even though they are not individuals. Some aspect of reality has this intelligible unity if the way it unfolds in time is uniform, controlled, predictable, or displays a kind of qualitative logic we can develop to understand it. What we ordinarily call objects just are these kinds of meta-stability. For example, things like rocks, chairs, mountains, cars, and planets are aspects of reality that have virtually the same character, even over a very long duration. The reason that we take them to be individuals is precisely because of this lasting stability, precisely because they are not centres of excessive change.

Humans, plants, and animals display a kind of qualitative logic to them, owing to their predictable growth and development. Thus, in spite of the fact that they are less stable than something like a rock, owing to their constantly evolving behaviours and developments over time, they are relatively predictable in those changes as they follow a kind of trajectory that (partially) makes sense to us. In both the cases of objects and organisms, the reason we pick these things out as we do and not in other ways, and come up with names for them, is because the world exhibits its particular character of differentiation and features more and less stable spatiotemporal uniformities within it, meta-stabilities.

Some aspect will not be meta-stable insofar as it does not retain qualitative continuity over some duration. Things we often call processes or events have this character. For example, a concert occurs covers a lesser duration than a rock or chair, but it also has considerable internal heterogeneity making it much less meta-stable. The same is true of things like explosions or car crashes. We do not always take things like this to be individuals because they mark sharp and brief qualitative transitions in what was until then a relatively stable aspect of reality. This also includes what we might think of as senseless groupings, like the combination of Mount Taranaki and the Eiffel Tower, which nothing about the world suggests that some qualitative connection should be drawn. The reason we do not have a word for these kinds of

entities is owing to the fact that the world exhibits no meaningful spatial or temporal qualitative continuity between them, even though they are both themselves meta-stable features of the world, we have no purpose for such a bundling. (Though of course, such a grouping is not ruled out in principle.)

The important thing to note is that talk of these meta-stabilities extends along a continuum. There is no difference in kind between each aspect, for example, between what we call objects and what we call events, they are each captured under the same construct: mapping. It is just that, by artificially isolating some aspects of reality, we can compare them relative to each other's stability and qualitative continuity in a way that actually captures the particular character of the world.

The fact that the world exhibits these relatively uniform regions of qualitative stability, combined with the fact that there is a similar biological and cultural substate to how we characterise the world, makes possible effective communication and the intersubjective validity of claims about it, without a direct correspondence between the entities we suppose ourselves to be picking out and the world in its undivided state. Thus, even though it is becoming all the way down, that becoming takes a particular character that ebbs and flows into seeming individualities that are either natural (exhibit qualitative continuity, at least to us) or useful (help guide action).

Concepts and language are simply maps that purport to describe some phenomena that seems to repeat itself similarly and often enough that it is more useful than not to be able to talk about it. We then pick out these meta-stabilities as if they were independent entities so that we can effectively navigate and communicate about the world, just like how markings on a map help us navigate the world, without having committed ourselves to the corresponding existence of those markings in the world. There is, of course, a genuine sense in which there are mountains and persons out there in the world and that those terms point to some reality. But it is the qualitative becoming occurring over some duration that is the reality, not the pragmatic entities we decompose this becoming into after the fact. To take our constructed and determinate entities for the reality, rather than the particular character of the primordial event in its stabilities and instabilities, is to mistake the map for the territory.

Just to have one more try at making this clear, think back to Bosch's *The Garden of Earthly Delights*. Once again, it is easier to imagine what these things will mean in the spatial case and then extending the ideas in time. Remember that we were to imagine that it is indivisible, it has no real parts, even though it has spatial extension and therefore a particular qualitative character. See the picture below:



Even on the supposition that this painting as an entity is, in fact, indivisible, an accurate mapping of it might start by pointing out a few things about its qualitative structure. First it might point out that there are three meta-stable ‘regions’ that exhibit a character that is continuous within themselves, while at the same time being particularly discontinuous with each other, owing to the sharp qualitative transition between them. This marks them off as an appropriate point of departure for a cartographic description of all or some of its structure that wields divisions without positing them.

Indeed, anyone that looks at the painting will see that saying something like “the painting is broken up into three regions” is going to have a very close fit-to-world due to the fact that it captures the relevant meta-stabilities actually present in it. Nothing about the fact that we are supposing it to be indivisible can stop us from seeing that. The same is true for the meta-stabilities within the middle section that appear to be individual entities. For example, the statement “there are many humanoids in the centre region of the painting” is going to have a strong fit-to-world, owing to the fact that the meta-stable qualities relevant to our use of the concept ‘humanoids’ are actually present in droves within the centre region of the painting. Indeed, without ever supposing there are divisions within it, running afoul of my earlier conclusions, I can formulate a genuinely illuminating description of it by taking advantage of our accumulated conceptual distinctions that capture similarities within qualitative reality.

All we need to do to extend this account to the primordial event is to suppose now that the painting is extended in time, and that it is therefore undergoing constant internal change. The added difficulty of doing so is in no way conceptual. Precisely the same tools are sufficient to describe its now four-dimensional qualitative structure. The difficulty is rather only to do with complexity. The region requiring mapping is extended in a whole extra dimension and will therefore be exponentially harder to capture. However, this is precisely what, for example, theories of cosmology are attempting. They are trying to describe, in the broadest possible sense, the temporal unfolding of the entire spatiotemporal universe. And for me, that just means they are mapping, to their best approximation, the qualitative structure of the entire primordial

event, the entire undivided history of the universe. Checking that such theories line up with observations made is simply them checking that the fit-to-world is right, in the same way that we must go out and look at some landscape when drawing up a map of it.

The same is true for historians who study a particular aspect of history. They might begin by pragmatically lifting some phase of history that exhibits some particularly sharp (and therefore interesting) qualitative transitions and attempts to map the structure of its unfolding in duration. For example, we might take the fall of the Roman Empire or the Thirty Years War. Checking these descriptions against historical data is once again the process of securing a tighter fit-to-world on that particular phase of duration. Once again, ‘the events making up history’ whether it is in the cosmological or historical sense, need not actually exist for such descriptions to be faithful to the structure of the world, they need only to capture the real meta-stabilities within the particular character of our world. It should be noted that I am not saying this is what cosmologists or historians *think* they are doing, I have no idea what they think the ontological consequences of their practice is. Rather, it is my attempt to explain the seeming intelligibility of what they are doing, in light of my own view that their objects of study do not actually exist.

This is in stark contrast to how uninformative the alternative would be. As noted earlier, the alternative that saves independent entities within the primordial event would have to suppose that they, like the primordial event itself, are eternally present, because there are no times for them to come to be or be destroyed. Therefore, the cosmologist would have to say the big bang exists and is occurring now. The historian would have to say that the French revolution exists and is occurring now. I would have to say that the object Rowan is wholly present now and alongside the pterodactyls. In all of these cases, what would have to be said is entirely uninformative. They do not illuminate anything interesting about the real qualitative structure of the world, as there is no relevant cartographic relationship between (what we call) pterodactyls and (what we call) Rowan. Nor does it help us navigate the worlds’ actual structure. Unless we learn about some actual or expected qualitative relationship between the phase of duration we call the French revolution and other phases, no lessons are learnt. And it does not explain why we seem to experience things that come to be and go away. The idea of the big bang loses whatever meaning it could have by supposing it to be occurring at all phases of duration.

The mereological or common-sense explanation of the world is riddled with paradox and ultimately uninformative. It must be overcome in favour of cartographic explanations. Once we reject the idea that what we see, talk about, and model is made up of things and events with perfectly determinate boundaries and lifespans, and accept that our real role as inquirers is to map, interpret, and categorise an absolutely fluid but qualitatively structured becoming into groupings of vague but useful chunks to elucidate and then synthesise into a wider and wider schematic map of the universe, these difficulties all slip away.⁸¹

⁸¹ Further, if qualitative cartography can successfully explain our theorising without the postulation of entities that correspond to its maps, then we have no explanatory need to postulate them. And according to the razor developed in Chapter III, if we have no reason to postulate those entities, we should not do so.

Appendix I: Whither Presentism?

Another possibility for construing the world-line that I did not have space to discuss in the main text is presentism. It is not essential to my argument that I reject the presentist construal because everything that is needed to respond to it is already in the main text. However, I think that it is important to have it recorded here how that rejection would go, and that I do have the arguments.

There are two ways that we could carve up the world-line in a presentist way. One would be a states-only presentism, what I will be calling *state-presentism*. The other is an events-only presentism, what I will be calling *event-presentism*. While there are different ways you could fill in the details of both approaches, they differ essentially with regard to whether they grant the present some duration.

State-Presentism

On the first proposal, state-presentism, instead of there being one indivisible event with a substantial duration constitutive of the entire history of the universe, there could be either one indivisible state of no duration and its many objects, or one (duration-less) time where many events and objects are co-occurring. Once again, what view one takes will depend on one's views on whether time exists over and above events. However, both views are states-only views regarding events. On the world line, this would simply be state A. Unlike the eternalist, the presentist rejects the occurrence/existence distinction, at least for concrete entities. Instead, they hold that only that which occurs, also exists, while that which is not occurring, does not exist. Thus, B, C, and D would not exist unless they were occurring *now*. However, it may not be completely clear why this view should be interpreted as a states-only view. Is this presentism only committed to states? To exclusively duration-less events?

In short, yes. In fact, they are committed to exclusively duration-less entities, not just duration-less events. Consider this variation on an argument from Merricks (1995). Any state-presentism (per my definition) is committed to a duration-less present. Since the only things that exist are the things that exist at this duration-less present, the things that exist must also have no duration. Why would we think this though? For the simple reason that no existing entity can have parts or aspects that do not exist. Any question of whether some entity has some other entity as one of its parts is surely settled by the fact that the entity that is supposed to be a part of the other entity does not exist. Once this is recognised, we can construct a reductio of the conjunction of these two positions.

Suppose presentism were true and there exists some entities with a real duration. If presentism is true, any entity that is not in the present does not exist. If there exist entities with duration, there must exist entities (the past and future parts/aspects of those entities) that do not exist at the present time because these entities are temporally extended. All real parts must

exist. Therefore, if presentism is true, entities with duration both have and do not have real parts. Contradiction. Therefore, state-presentism is incompatible with any entities having real duration. That is to say, either presentism is true, *or* entities have a real duration.⁸² State-presentism can only be committed to states, not events.

Now recall my argument from Part I earlier. There I argued that the existence of some duration is a necessary condition for the perception of change if that change is immanent in the world. Given that I think, and you should think, that it is immanent in this world, does the fact that presentism does not admit the existence of anything with a real duration preclude it from being true?

In short, yes. The argument will go as follows. There is the perception of change. The perception of change necessarily occurs over some duration. A change in perception is a change in the world. Therefore, there is a real duration (or the perception of change is illusory). State-presentism is incompatible with the existence of a real duration. Therefore, state presentism is false (or the perception of change is illusory).⁸³ However, I should not be hasty here. I should look at what presentists actually say about the nature of change, and our perception of it. Perhaps what they say in this regard will get around the problem. We must ask: how does state-presentism account for the perception of change?

One way the state-presentist could explain such change is similar to how the other states-only views would describe it. They could say that the experience of change is made possible by the constant passing of duration-less states. If you watch me cross the road from A to B, that experience will be constituted by a succession of individual states coming into existence in the present, and going out of existence, to be replaced by the next state. Thus, on this view reality is, at bottom, like a film. The present must be constituted by one configuration of entities, and then another after that configuration has elapsed, and another at the next, and so on. However, only a single duration-less ‘still’ can exist at a time, so it is more like the mere (present) projection of a film, rather than the reel of film plus the projection. (The latter metaphor captures an eternalist states-only view which maintains the occurrence (projection)/existence (reel) distinction.) The still that is being projected is the present, and the content of that still is analogous to the entities occurring at the present time. As with the present projection of a film, this present is constantly changing.

Since only the present exists for the presentist there is no actual change and motion of the presently existing entities at a given individual present, because the only thing that exists, the present, does not itself change or move, just as any given individual still of a film does not change or move. (This is because it has no duration, not because it has no temporal parts. A present with duration could change.) Each present event or object comes in to, and goes out of, being. However, the constant succession of differing states going through this process is

⁸² It will not do to reply that present objects have the property of “having-been” a way they are not now, in the present. For example, a red ball might have the property of “having-been-green”, which is a present property. I do not deny this. That individuals change and endure is fine for a presentist and is not the point at issue. It is just that such properties do not mean the entity itself will have a real duration. It just means that they have further duration-less properties in the duration-less present.

⁸³ Variations on this argument, or at least arguments that are raising analogous problems, and two different responses to it, can be found in Craig (2000) and Pezet (2020).

supposed to explain our experience of change and motion and the continuity between evolving states of the world, even though the states themselves neither change nor move.

According to the account given so far, the state-presentist must accept three theses. They are as follows:

- (1) All that exists is a single time, the present, and the entities occurring at it (or just those entities).
- (2) The present, and the entities occurring at it (or just those entities), must have no duration.
- (3) Change is the succession of different (but themselves unchanging) duration-less presents.

For simplicity's sake, from now on I call (1), the particular collection of entities at the present time, the 'configuration' of the present. The problem with these theses is that they (1) make any change impossible and thus (2) cannot actually account for the immanent reality of change. I will bring these conclusions over a few piecemeal steps. However, most of the reasoning should be familiar by now.

If there is a succession of different present configurations, posited by (3), according to (1), none of these configurations can co-exist with one another. This is because if any one state occurs after the present state, it must occur at a different time to the present. But no other times exist. Therefore, the states that will come to be cannot exist, unless they exist in the present. The existence of one time and the entities occurring at it must therefore exclude the existence of *any* other times and their respective entities. This is just a restatement of the view, the presentist should agree with me here.

However, if change is supposed to be the succession of different duration-less present configurations, we have a problem owing to the fact that these presents are duration-less. Take the event of my crossing the road from A to B. Suppose that it appears to occur over some finite duration (even though we know there could be no actual duration). On this view, for you to have perceived my crossing the road as occurring over a finite duration, an infinite number of *successive* individual states must have passed into and out of being. This process must have occurred one-by-one, where each individual configuration follows from the last. This is because only *one* configuration of the present can presently exist and all that exists is that present. If this process did not occur one-by-one, there could be no apparent change at all because the present configuration would never be replaced by a different one. The world would be a single, static, unchanging configuration, never to be replaced by another configuration. This would make all perception an illusion in the otherworldly sense because our perception of events occurs over some duration and in such a world there is no duration. Thus, to make the perception of change possible, we must accept that present configurations replace each other one-by-one.

But here is the problem. Since states are duration-less, for any finite duration to have passed, an infinite number of states must have come to be and passed. However, as noted in Part I, it is impossible for a process with an infinite number of individual steps ever to

terminate. This is a growing infinite, like the process of counting all of the natural numbers, which is impossible in actuality. The perception, given it must be constituted by the one-by-one coming-into-being of an infinite number of entities, could never be completed. Once again, this is not due to a lack of time, a physical difficulty, it is due to metaphysical impossibility.

In fact, it is worse than this. It not only makes changes of a seemingly finite duration impossible, but it also makes even a single change of configuration impossible. For there to even be a single successive change of the present time from the present configuration to the next, there must be a next configuration of entities that follows from the present configuration. However, *there is no next configuration*. For a next configuration to come to be, an infinite number of intermediate configurations must become present before that, which is impossible. Once again, I am not saying this is because it would require an infinite amount of time to pass through these configurations, but because there simply cannot be a 'next' configuration, because for any given configuration, there are always infinitely many configurations that precede it that would have to come into being first. And indeed, we saw the same thing in the case of Zeno's paradox in the first part. It was not necessarily damning that we could not finish our crossing the road, *it was that it was always impossible to even start*.

In the first part my invocation of Zeno's paradox perhaps was not convincing due to the perceived victory of the methods of modern mathematics over its reasoning. However, putting aside whether this is the case, this route of reply is not open to the state-presentist (in a way that it might have seemed to be open to the state-eternalist). This is because, as I have taken pains to show, they must explicitly posit a premise like (3). Namely, one that says change is constituted by a one-by-one process of new configurations coming-to-be. They must do this because of (1), because only *one* configuration, the present configuration, can exist at the present time. Denial amounts to a denial of presentism. However, affirming it means that presentism is straightforwardly defeated by Zeno's paradoxes. Is there a way out of this predicament?

Denying (2), that the present must have no duration, would be a denial of state-presentism in favour of event-presentism, which I will discuss soon. And again, denying (1), that all that exists is the present time and configuration (or just that configuration), would also be to deny the truth of presentism because then more than just the present time or configuration could exist. Suppose finally that we deny (3), that change consists in the step by step coming into being of different configurations of the present time. If there are no changes in configuration, no new states, then all that could possibly exist is the one present configuration, which never changes. But surely this is false. To think that this is true would be to think that there is a single duration-less state of existence and that all change must be an illusion in the otherworldly sense. Usually, the whole point of advocating presentism is that it accounts for the actual becoming of the world. If only one configuration of the present can exist, this becoming would be an illusion.

However, presentists will object here. I have surely mischaracterised their position. They will say that of course, there are not a discrete sequence of configurations that succeed

one another. Instead, they will say that *it is of the nature of the present, that it is changing*.⁸⁴ On this view, (3) that mischaracterises the position as one that posits a single time that is occupied by successive discrete configurations. However, what is really happening is that there is a single time at which a constant flux is occurring, where events and objects are constantly coming into and going out of being, dynamically changing in a way that is not problematic because it is not tied to static theoretical formalisms like ‘states.’ My presentation presupposes that presentists are thinking in terms of order and events in a way that they are not, in a way that their eternalist opponents do, leading their position to appear weaker than it really is. The present is not the succession of discrete configurations, but rather a flux of change that is undivided but constantly differing from itself.

The problem with this complaint is as follows. Either entities in the present have duration, or they do not. If entities have duration, state-presentism cannot be true as it leads to contradiction because then there would exist entities that exist somewhere that is not in the present. Therefore, all entities that exist for a state-presentism must have no duration. However, this means I have not misconstrued the position, because everything that exists and can exist is like a state in the relevant respect: they must have no duration. Thus, an account such as (3) must be given to account for how things appear to us to be changing successively. If a state-presentist does not accept (3), that there are different successive configurations of the present, then the perception of change must be an illusion in the otherworldly sense. This is because our perception of change is successive and denying (3) denies that there is an objective succession at all. For example, it would not be the case that when I cross the road, I start from one side and then get to the other, which we undeniably perceive, because these states of affairs must actually succeed each other in time and there can be no such succession in a single state.

Perhaps the presentist replies that I am still mischaracterising their position. Just because the present changes does not mean it has to exist as extended in time or as a discrete sequence of entities that are constantly coming-into-being. I am begging the question in favour of my own view to presume as such, because it rules out another possibility. Namely, that the present itself could *endure*. Something that endures is something that can exist at two different times with different properties but be ‘wholly present’ at each of those times. Perhaps this is what is meant by the idea that it is of the nature of the present that it is changing: that it is of the nature of all that exists, the present, to constantly be changing, without thereby implying the coming into being of more temporal entities or the division of that change into smaller entities, but only the *reconfiguration* of a single entity and the spatial objects that make it up.

Thus, suppose that there is exactly one event that exists: a single state called *the present* that is constantly changing, but endures through those changes. This would mean that change does not require the coming-into-being of an infinite number of entities nor does it require that there is a duration for this present to exist over. Instead, change would just be the constant *reconfiguration* of the same entity, the present, which only exists at any given time instantaneously. On this view reality is not exactly like a film because there are no separate

⁸⁴ Thanks to Dan Simpson Becklloyd and Cei Maslen for independently pushing me on this point.

stills. Changes would all be dynamic changes *within* the same entity not the coming-into-being of new ones.

I do not think this would help either. Consider the following argument, which captures both proposed presentist views: one where the present is the succession of different entities (different states), and this one, where the present is a single entity changing (same state). I call the former *perishing state-presentism* and the latter *enduring state-presentism*.

Either the way the present changes makes it a different entity to what it was prior to the change (perishing present), or it does not (enduring present). If the present is a different entity every time it changes, then my original characterisation is correct, and it cannot account for change because of Zeno's paradox. If the present is the same entity enduring *through its changes* (as this objection suggests), then it must actually exist over some duration, contradicting presentism. The presentist will object here. Endurance does not entail the existence of an entity over some duration. Rather, an enduring entity changes without loss of identity and *without existing over some duration*. For example, a tree whose branch you cut off is still that tree and it is wholly present before and after it is cut off. It never exists over some duration it only exists wholly and presently at different times, with different properties. Therefore, the tree exists in both cases, without duration and therefore unproblematically in the present. The same could be said for this enduring present which, it has been proposed, is itself an entity. It exists, has no duration, and has properties that are constantly changing, all while being wholly present. However, the problem with this is that the proposal that this present undergoes an actual successive changing of properties covertly presupposes a real duration.

Take the tree first. The tree could not have gone from having that branch to not having it afterwards without there being some real duration. This is because, the tree's having that branch then the tree's not having that branch must have succeeded one another over some finite non-zero temporal interval. If there was no finite non-zero temporal interval between the tree's having these properties, that is, if they were not temporally posterior to one another at all, then there could be no temporal distance between the point at which the tree has the branch and the point at which it does not have the same branch. These points would then occur simultaneously. This means that the tree would have both mutually exclusive properties at the same time, generating a contradiction. These properties cannot be time-indexed to avoid contradiction either because for the endurantist, properties are brute constituents of entities and are not tied to times.

It could be replied that there does not need to be a non-zero temporal interval between the having of these properties for the tree. It is enough that the loss of the branch occurs instantaneously after the instantaneous (last) occurrence of it having the branch. However, this will not do. The idea that some occurrence can actually happen after another for a states-only view I have already shown to be illusory. This is due to the fact that any such ordering presupposes a dimension whose occupation has been ruled out a priori by the theoretical assumptions of any states-only view. It would be like saying that even though two two-dimensional books stacked tightly beside each other are spatially co-located, one is still actually next to the other.

It could also be replied that I am smuggling non-presentist assumptions back into the evaluation of this presentism. However, the changing of the trees properties does not presuppose anything like states or times. It only presupposes that there are enduring objects that dynamically change their properties, and that the entity having those properties, when they exist with them, exist without duration. All of these things the presentist should accept. My argument is that if there is no real duration over which such an object can change its properties, no such dynamic change can occur at all.

Precisely the same thing happens with the present state itself, because of its temporal width. Since the present is duration-less, any change it undergoes would have to be constituted by it having different properties. However, in all instances where it has different properties it would be duration-less. Therefore, all changes of properties elapse instantaneously, because no number of duration-less changes could ever add up to a change that occurs over a non-instantaneous duration. Thus, any change that it undergoes could not actually occur successively. Rather, any and all changes would all be instantaneous and therefore all occur at the same time. But if they all occur at the same time, then they all exist, because they are all present and presentism is the thesis that all and only present things, things occurring simultaneously, exist. But this means that our entity, the present, has all of its supposed transformations immanent to itself in the present. Further, since all of these property transformations will be mutually exclusive, as in the case of the tree, it will also be a contradictory entity.

Worse than this, as with all states-only views, this also makes the perception of change illusory. If any change of the present only occurs instantaneously rather than over some duration, then our perception of change, which must occur over some duration, must be an illusion, that is, be in another world. Thus, maintaining state-presentism means one will have to relegate perception to another world, which is especially unattractive for a presentist, due to the fact that their often purports to be the only view able to explain our perception of objective passage. Presentist passage ends up being merely subjective and of another world. Enduring state-presentism should be rejected.

I will now summarise the preceding dialectic just to make it clear. My argument against state-presentism goes as follows. State-presentism is incompatible with the existence of any entity with some duration. This is because, for state-presentism, everything that exists, exists in the present and anything with duration would exist partially outside the present. Therefore, everything that exists in the present is duration-less. This means there can only exist one duration-less 'configuration' of the present, which is the collection of all that entities that presently exist. There are two ways you could explain change on this view.

The first is that the present is perpetually perishing and that for every change there is a succession of new entities that come to be. However, the only way to explain change on this view is by saying that there is always a 'next' configuration of the present which is the coming-into-being of another individual configuration. However, since the present is duration-less, for the appearance of the passing of any finite duration, an infinite number of present configuration's would have to come-to-be before there can be a 'next' configuration. It is impossible that an infinite number of present configurations could ever come-to-be. Therefore,

change would be impossible. This means either that (1) presentism is false (because change is immanent) or (2) presentism is true, but the perception of change is illusory (that is, transcendent in the objectionable sense, outlined in Chapter II). Given that most state-presentists think that presentism is the position that takes temporal passage seriously, but we would have to come up with a two-world ontology to accommodate it, (2) is not an attractive position. We should instead affirm the perception of change is immanent and real, rejecting perishing state-presentism.

The second way of explaining change is to say that there is a single temporal entity, the present, which is an entity in its own right that endures through all change. On this view, change is just the constant reconfiguration of this entity. Here, it is of the nature of the present that it is changing. However, the problem with this view is that any changes of a seemingly non-zero duration must actually occur instantaneously. This means (1) that our successive, non-instantaneous perception must be an illusion in the otherworldly sense and (2) that the entity, the present, has contradictory properties at the same time. Once again, given that state-presentists ordinarily think that their view best accounts for temporal passage, relegating the perception of change to another world is not an attractive option. And nor is the acceptance of real contradictions. Both of these problems are grounds for rejecting enduring state-presentism.

Neither version of state-presentism can satisfactorily explain change. Thus, if one wishes to be a presentist, they ought to drop the requirement that the present cannot have a duration. This is event-presentism, which I now move on to.

Event-Presentism

On the second proposal, event-presentism, instead of there being one indivisible event with no duration, there could be one indivisible event with some finite duration. This is an events-only view. On the world-line, all that exists would be a single event, AB. This view differs from the other kind of presentism because it is a kind of ‘thick’ presentism, one that takes the present to have some duration, rather than being duration-less. This present could either be common-sense or atomic. On common-sense event-presentism, the present would be individuated by the particular events that are currently occurring at a given time. For example, going to a concert will be present and thus existent insofar as it is occurring. Therefore, it will exist (say) beginning at the first song, and it will pass out of existence once the band has left the stage. Within this event there will be considerable change unlike on states-only views where there would be no change. On atomic presentism, the present will be individuated either by some uniform temporal atom (like Hume), our specious present (like in the case of James), or some other such minimum.

Importantly, in both cases, this present could either be *elastic* or *rigid*. The only rigid version of either alternative that I am aware of is Hume’s. By rigid I mean that the present is of a uniform temporal extent. I do not think his account works for the reasons given above. In short, Hume’s collapses into a states-only view because the only non-arbitrary uniform duration of time is duration-less, which makes succession impossible. However, the idea of a minimum non-zero temporal extent, “chronon”, “temporal atom”, or “temporal quanta”, is not unheard

of. While I think they have some serious conceptual difficulties (especially as a version of presentism), I do not think they are incoherent. However, if such postulates are motivated by scientific theory, I am tempted to chalk their appearance up to the inadequacy of our measurements, rather than our direct access to the metaphysical structure of time. No doubt there are metaphysical reasons why we might pursue such a proposal. For one, it avoids my use of Zeno's paradoxes. But given my own predispositions against it, its relative lack of defenders, and length of my discussion, I will omit further discussion of rigid event-presentism.⁸⁵

The rest of the views are elastic. By elastic I mean that the thickness of any present event (the present duration) is of a variable duration. Since the elasticity of the boundaries is what I take to be the source of the view's challenges, I will discuss the objections as they apply to all elastic views of event-presentism.

I have both an epistemological and metaphysical objection to the idea. They will be similar to the objections levelled against common-sense events in general, and James' view of elastic atomic events, so I will keep the discussion short. First, the boundaries marking the beginning and ending of the present, common-sense or atomic, can only ever be arbitrarily drawn. We can always ask of some purported present, "Why is it not an arbitrarily minute duration longer or shorter than you are supposing it to be?" And this is always an appropriate question because of its supposed elasticity. It is not enough to say that even though there are many possible choices for individuation of the actual present, there is nonetheless a single determinate boundary out there, because there are an infinite number of possible alternatives that we would be just as warranted to select as the single present. And these alternatives stretch out from the given event to arbitrarily large or small durations. Even if we did arbitrarily select some boundary as *the* present, we could never know that we know it, because there are an infinite number of minutely similar other thick presents that also seem to currently exist. Thus, contrary to what is normally thought about presentism, if the present is thick, we can never have knowledge of it. We can only assume that such a present exists.

However, suppose one grants that the boundary is unknowable but (rightly) replies that epistemological difficulties do not equal metaphysical difficulties. It is a necessary posit of the theory that there is some boundary, regardless of whether we can know it. However, supposing that some boundary is actually drawn, there are two problems. First, if we allow that present events could overlap, Unger's problem of the many comes back. Any event we are willing to admit into our ontology forces us to admit an infinite number of other actual but minutely different events that almost entirely overlap. But then, every time an event comes into being or there is even some minute change, an infinite number of events would have to successively occur in some finite duration. But assuming that change is not already existing and complete like eternalism does, it is impossible that an infinite number of events occur over any finite duration. Therefore, change would be impossible. But change is possible. From here, the event-presentist must deny that the events overlap, thereby blocking the problem of the many.

⁸⁵ For mostly critical discussions of such views see Čapek (1961) 231-38, Craig (2000) 174-78, Dainton (2010) 294-97, and Pezet (2020) 406-411.

However, if we say that the events cannot overlap then the connection between the present event and the immediate past is thrown into question and the epistemological difficulties become metaphysical. Take the events AB and BC that are supposed to succeed one another. Now imagine that AB has gone out of existence, and the duration which is of an arbitrarily small temporal distance after the end of AB is occurring. Why should we think that point B is the point where AB is abolished forever? Why cannot the event have gone out of existence slightly before or after where it did? If we allow that the present events are elastic, why shouldn't they have a greater or lesser temporal extent? If there is no principled answer to these 'why' questions (like there being the presence of some essence), then there is no principled response to my rejection of these arbitrary boundaries.

Further, their transcendental necessity can no longer be appealed to by saying that experience would not be possible if the present were not individuated into these thick present events, because I have already shown that temporal monism, an indivisible world-line with a real duration, constitutive of the entire history of the universe, can also account for change and experience just as well as event-presentism can. Thus, the choice is between a theory that explains less (by only accepting some aspects of reality as existing) and posits arbitrary entities (boundaries and events) and one that explains more (by counting all phases of existence in time as real) and is nowhere arbitrary (because it refuses to divide time at all). I go with the latter (though have by no means refuted the former).

This discussion should have hopefully revealed that my own view is not so different from event-presentism at all, it just counts as real a much, much, thicker present. But we both agree that all there is, is an indivisible present with a real duration.

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